Lectures on Optics

"To Understand the universe we must understand the light"

Friday 7$^{\text{th}}$ August, 2026

LECTURES ON OPTICS

Jafert Serrano

Professor: Rafael Ángel Torres Amaris

Universidad Industrial de Santander
Sciences faculty
Physics school
Group of optics and signals treatment

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Abstract

The field of optics is a captivating and continually evolving branch of science that plays an indispensable role in our modern world. "Advances in Optics: Exploring the Boundaries of Light and Vision" presents a comprehensive overview of the latest developments, theories, and applications in optics, spanning from fundamental principles to cutting-edge technologies.

This book delves into the fundamental properties of light, offering a clear and concise explanation of wave-particle duality, the nature of electromagnetic waves, and the behavior of light in various media. It explores the principles of geometric optics, including reflection, refraction, and the formation of images, while also addressing wave optics phenomena such as interference, diffraction, and polarization.

In addition to the foundational aspects, "Advances in Optics" goes beyond traditional optics to cover emerging topics and technologies. Readers will find detailed discussions on advanced optical materials, including metamaterials and photonic crystals, and their revolutionary applications in the manipulation of light for imaging, communication, and sensing. Quantum optics, an exciting frontier in optical science, is also explored, with a focus on quantum entanglement, quantum cryptography, and quantum computing.

Moreover, this book highlights the practical applications of optics in various fields, including medicine, telecommunications, astronomy, and environmental monitoring. It provides insights into state-of-the-art optical instruments and techniques, such as lasers, optical fibers, and adaptive optics systems, and their pivotal roles in advancing technology and scientific discovery.

"Advances in Optics" is designed to cater to a broad readership, from students and educators seeking a comprehensive textbook to researchers and professionals looking for the latest developments in the field. Its accessible style, accompanied by illustrative examples and diagrams, ensures that both newcomers and experts can appreciate the wonders of optics and its ever-expanding frontiers.

As we venture further into the 21st century, the exploration of light and vision promises to yield groundbreaking discoveries and transformative technologies. "Advances in Optics" serves as an invaluable resource for anyone curious about the principles, innovations, and limitless potential of optics in shaping our future.

Chapter 1

The loving Story of Optics

“The light is the God’s shadow”
– Saint Agustin

When we study physics beyond doing long calculations, formulate complicated theories or doing extremely elaborated experiments in an advanced laboratory there is a really important fact many people ignore, the *story of physics* if we want to study physics we must have understand where was born the physics and more important, how was it done. Here we are going to check the story of physics from the geometrical optics, to corpuscular optics to wave optics, to electromagnetic optics and the most recent the quantum optics.

1.1 Optics from Euclid to Huygens

Here I'm going to copy and paste one of the best articles I've read about the story of optics (Herzberger, 1966).

If I were endowed with dictatorial powers, I would require everyone receiving a degree in a scientific subject to know its history and to have read the classical papers relating to it. Historical knowledge is important because it stimulates creative thinking. The man who first struggled with an idea, trying to find a law, looked at the situation with different eyes than do we who accept the law as a matter of course. He considered alternatives to the law, and different interpretations, and some of these alternatives and interpretations may still be stimulating and worth thinking about.

I was fortunate during the eight years of my affiliation with the Zeiss works at the beginning of my career to have access to their extensive library; it contained the originals, or at least translations, of nearly all the writings in optics from Euclid to contemporary times. Over thirty years ago, I wrote a historical survey describing the content of important papers on geometrical optics. This task was immensely facilitated by the extensive bibliography in the third edition of Czapski-Eppenstein,I to which I had contributed my small part as a work student at the firm of Zeiss. Lately, I have looked through most of these papers again, as far as they were available to me, and have discovered many important details that I had not understood at that time. I should like to share these with you. I emphasize that my interest is not basically historical. I am not interested so much in questions of priority and originality of documents. My object is to entice you to read and to enjoy the old scriptures, and maybe take along an idea or two that might be of interest.

1.1.1 Ancient times

I am sure that the Egyptians and, even more, the Babylonians, with their great interest in astronomy,must have had some knowledge of optical laws. But the first important mention of one of the phenomena of refraction I found was in Plato's Republic.' Plato(427-347 B.C.) quotes Socrates as saying (Jowett's translation): " . . . the same object appears straight when looked at out of the water, and crooked when in the water; and the concave becomes convex, owing to the illusion about colours to which the sight is liable. Thus every sort of confusion is revealed within us; and this is that weakness of the human mind on which the art of conjuring and of deceiving by light and shadow and other ingenious devices imposes, having an effect upon us like magic.¨ Our senses give an erroneous picture of reality; they are untrustworthy;alone can lead us to truth pure thinking.

This idea persisted for two millennia under the influence of the misinterpreted teachings of Aristotle.

How we see is a question that long occupied the Greek philosophers, and, through them, the Neo-Platonists and the Scholastics. (Indeed, it cannot be said that the question is settled, and in recent times Ronchi has set forth some cogent ideas.) Even before Plato there was speculation in the writings of the Greek philosophers about the process of seeing. Most of the Greek thinkers believed that the eye sends out rays to the object, and Aristotle (384-322 B.C.) seems to have been the first to question this belief. If, says Aristotle, the eye sends out fiery rays that stream from it as from a lantern, why can we not see in the dark? His explanation, that the eye contains a transparent substance and that seeing is associated with the motion of something between the object and the eye, seems quite modern.

After the conquest of Greece by Alexander, Alexandria (in Egypt) and Sicily became the strongholds of Greek culture. The first great textbook of optics was written by Euclid of Alexandria, who lived about 300 B.C. And is better known for his Elements of Geometry. This famous work is still must reading for every scientist and is available in several editions. When King Archelaos once complained about the difficulty of learning mathematics and asked whether there was not an easier way, Euclid's answer is supposed to have been: "There is no royal route to geometry¨. When one of his pupils asked him what of material value could be gained by learning geometry, he told a slave, "Give the man threepence and tell him to go away¨.

Under Euclid's name we have two books of interest in the present connection: Optics and Catoptrics. The first deals with the laws of vision and perspective and the second with reflection. An English translation of the first was published in the Journal of the Optical Society of America twenty years ago. 7 It is a striking fact that one of the definitions in this book contains an idea that could be used to develop many problems of diffraction theory.

I quote some of Euclid's definitions, slightly paraphrased from the Optical Society of America translation: "Let it be assumed that lines drawn directly from the eye pass through a space of great extent [not infinite]; that the form of this space is the mantle of a cone with its apex in the eye and its base at the limits of our vision; that those things upon which the cone falls are seen and those things upon which the cone does not fall are not seen; and that things seen through several cones simultaneously appear to be more distinct.¨ From this, Euclid concludes, among other things, that nothing can be seen in its entirety. The same objects located nearby are seen more distinctly than if they were at a distance. Every object that is visible has a certain limit of distance. If that is reached-that is, if the object lies entirely inside the mantle of the cone of vision-it is no longer seen. These are the axioms from which he derives the laws of perspective.

In Catoptrics the law of reflection is stated, namely, that incoming and outgoing rays form the same angle with the surface normal. From this he derives the exact position of the image behind a plane mirror and the principle that left- and right-hand sides are interchanged. He generalizes these laws for spherical mirrors, both concave and convex, describing virtual images as well as stating that concave mirrors can give real images. As the position of the image is formed by reflection at curved surfaces, he suggests the point of intersection of the ray with the axis (shades of my definition of diapoint!). He shows the existence of a focal point for spherical mirrors, although he errs with respect to its position. One of his theorems says that an eye at the center of a spherical mirror sees only itself. Another states that you can light a fire at the focus of a concave mirror. He states that straight lines appear convex in a convex mirror. One can see the same object through many plane mirrors if they are positioned along the sides of a regular polygon with two more sides than there are mirrors.

According to Roman historians, the greatest mathematician of ancient times, Archimedes (287-212 B.C.), is supposed to have used mirrors to destroy the Roman fleet besieging Syracuse. Like most scientists, I have doubted this story, but a personal experience made me doubt my doubts.

Two years ago I had an optical accident. My wife had me get a large bottle of distilled water, and I left it in my auto in front of our laboratory. Driving home that evening I smelled smoke and stopped at a garage. After thoroughly investigating all the mechanical details and finding them in order, the mechanics discovered that smoke was coming out of the upholstery. The cylindrical bottle had acted like a lens and the sun had made a hole in the upholstery and ignited the stuffing inside. After this personal experience, I am somewhat more reluctant to doubt the story of Archimedes' mirrors.

The next person I should like to discuss, Hero of Alexandria, lived sometime between about 150 B.C. and A.D. 250. We have two outside dates for this philosopher: he mentions Hipparchos, who flourished between 146 B.C. and 126 B.C., but the mathematician Pappus (ca. A.D. 300) follows Hero's treatment in his own writings on mechanics. Be this as it may, Hero's Catoptrics is one of the most interesting books on optics. He is not content with Euclid's statement that light travels in straight lines, nor is he content with the form of the reflection law. He looks for a reason for both laws and derives it from the assumption that light chooses the shortest possible path between two points. We know that this minimum principle is not quite correct: the path is, in general, only an extremum. It may or may not be significant that Hero's book illustrates the concept with a picture of a convex mirror, for which the minimum principle is correct-not a concave mirror, for which it can be wrong. We shall find the same trick worked later by Fermat, who extended Hero's law to derive the refraction law. However, Fermat was called on the carpet for this by Abbé Mersenne, one of his contemporaries.

Hero of Alexandria-who, by the way, in his Mechanics described the so-called seven simple machines known to us-not only discussed spherical mirrors in his Catoptrics but, as an early practical experimentalist, also showed what can be done with cylindrical mirrors. If you go into an amusement parlor and see yourself in mirrors of all kinds (I especially like those that elongate the height and diminish the girth), you know the proprietor has read Hero of Alexandria. If you see a play in a tanagra theater, where the actors appear on the stage, by the skillful use of mirrors, as beautifully dressed dwarfs, you know that Hero of Alexandria wrote the prescription. If you want to see around the corner or look at a person without letting him see you, Hero of Alexandria tells you how to do it with mirrors.

Ptolemy of Alexandria, who lived about A.D. 150, is best known for his Almagest, but he was also the first to make a serious attempt to study the law of refraction. He wrote a five-part volume entitled Optics, several copies of which have come down to us. This work, of which the first part has been lost (although we have a description of its contents), can be summarized as follows.

The first part deals with the relations between the eye and light, and the second with the conditions of visibility: size, form, color, and motion of bodies. Ptolemy explains why we usually see only one image of an object, although we have two eyes, and describes the conditions under which we see more than one image. The third part deals with the laws of reflection by plane and convex mirrors. Ptolemy gives probably the first attempt at an explanation of why the moon and the sun appear larger at the horizon than at the zenith a question that is still being discussed.

In the fifth part, Ptolemy tries to find a law for refraction. First he observes that for small incident angles the angles of incidence and of refraction are proportional to each other. He also remarks that, in transition from a thinner to a denser medium, the refracted ray comes earer to the perpendicular, the opposite being true for the reverse direction of travel. He uses an ingenious contraption to measure the incident and the refracted angles for the transition from air into water, air into glass, and water into glass. This instrument consists of a circular disk divided into 3600 with a colored marker in the center and two movable markers on the periphery. You put the instrument into water, line up the three markers, and read the two positions, one giving the incident angle and the other the refracted angle. Ptolemy gives refraction tables in 100 intervals. My friend, the late Hans Boegehold, studied hese tables carefully and found that the values for 500 and 600 incident angles are surprisingly accurate, but that the other angles were obviously calculated by a second-degree interpolation formula. (The first differences are exactly 1 30'.14) Ptolemy applies the same method for the transition of light from air into glass and from water into glass. He also discusses atmospheric refraction, the cause of which he assumes to be that the densities of air and of ether vary with height. He points out that only for a star in the zenith do the true and the apparent positions coincide.

1.1.2 The arabain period

After the fall of Greece, scientific progress ceased on the continent of Europe for nearly a century and a half, but Alexandria remained the center of scientific discovery. When the Arabs took over, they brought their scientific inheritance and knowledge of mathematics and the mathematical sciences into northern Africa and Spain. It is remarkable that, at a time when magic and mysticism and misinterpretation of Aristotelian doctrines prevented scientific progress over most of Europe, the Arabs held high the torch of science in their countries.

The next big contribution to our science was by Ibn al-Haitham, or Alhazen of Cairo, whose date is frequently set around 1100, although some sources tell us he lived from 965 to 1038. Alhazen's Optics" consists of seven parts. The first gives an anatomical description of the eye, which, he maintains, contains three liquids separated by four membranes. The names which he gives to them the watery medium, the crystalline medium, and the glassy medium are still used today. The phenomenon of binocular vision is explained by assuming that the pictures from the two eyes are transported to one visual nerve. Alhazen discusses the pupil of the eye and speaks of a pyramid of rays (rather than a cone) coming from the object to the eye.

The second part, like the corresponding book by Ptolemy, deals with the properties of bodies, and he distinguishes no fewer than twenty-two different properties.

The third part deals with optical illusions. He draws attention to the illusions arising from a false interpretation of a picture by the human mind and imagination. The fourth, fifth, and sixth parts deal with problems of reflection and mirrors. One of the problems studied in great detail has since been called after Alhazen: it consists in finding the point on a mirror where the incident and the reflected rays meet. Alhazen corrects errors of the Greeks about the position of the images in spherical, conical, and cylindrical mirrors. He also emphasizes the fact that the incident and reflected rays lie in the same plane.

Alhazen's seventh part deals with refraction. He proposes methods for measuring incident and refracted rays through different media, but he gives no tables. He knows that a glass sphere magnifies. He explains the apparently large size of the sun at the horizon by the assumption that we compare the sun with objects of known size lying along the line of sight. He investigates binocular vision and makes a good guess at the height of the atmosphere. He breaks with the idea of the Greeks that light rays originate at the eye and go to the objects, supporting his claim by the circumstance that sunlight can hurt the eye and that we can see afterimages.

Other manuscripts show that Alhazen used the pinhole to view solar eclipses and that he knew how the mage becomes blurred and finally becomes circular as the pinhole increases in size. He did not write as though this were a discovery of his own, and Aristotle had noted that the spot formed by the rays from the uneclipsed sun is circular when the rays pass through a rectangular hole in wickerwork if the hole is small, but becomes rectangular when the hole is large. However, Alhazen's is the first known unequivocal description of the camera obscura. Alhazen's works were of enormous influence in the development of optics.

1.1.3 The middle ages

The knowledge of the Arabs was transmitted to Europe via Spain and Sicily and set in motion the intellectual revival that culminated in the Renaissance. The first of the European scientists to play a role in optics was Vitello, a Polish mathematician who lived and studied in Italy about 1270. His book in ten parts is probably the most voluminous optical work ever written. It contains about five hundred closely printed folio pages. Unfortunately, he is more famous for his errors than for his contributions.

He reissued the tables of Ptolemy and added tables for the passage of light from water into air, from glass into air, and from glass into water. It is unfortunate that he never tells us how he arrived at his values, for they include impossible data, being in contradiction with the then unknown phenomenon of total reflection.

It was this error that later prevented Kepler from finding the correct form of the refraction law. Nevertheless, Vitello made two notable contributions. He explained the colors of the rainbow as originating from refraction and reflection by the water drops, and he made the proposal to collect the light of the sun into a point by using parabolic mirrors.

The Englishman Roger Bacon (1215-1294) discovered the spherical aberration of a mirror; he further stated that rays forming a cone around the axis in object space also form a cone around the axis in image space. From the magnifying property of a sphere, he speculated that it might be possible to invent instruments in which a man would look like a mountain and a small group like a big army. However, since he did not say how this could be achieved, it is doubtful whether he should be called (as he was by Sir John Herschel) the inventor of the telescope. His writings are so sketchy and in definite that it is impossible to attribute specific inventions to him or to reach any conclusions about what he actually knew.

1.1.4 The renaissance

Coming to the Renaissance, we must mention Leonardo da Vinci (1452-1519). His notebooks show that he studied the eye anatomically and made a model of it, although he supposed the rays to cross twice in it to form an upright image. 18 , 9 Immediately afterward he described the camera obscura but correctly showed the image inverted!20 He described, with sketches, a Rumford-type photometer and a machine for grinding concave mirrors. Unfortunately for his optical reputation and the development of optics, he never organized his notes into publishable form, so they were practically unknown until Venturi published them in 1797.

About half a century later came another Italian we should mention, Francesco Maurolico (1494-1577), who solved one of the problems that had bothered previous thinkers: that the image of the sun is round even if formed by a square pinhole. He improved our knowledge of vision by suggesting that the crystalline lens of the eye works like a double-convex lens. This enabled him to attribute myopia and hyperopia to an eye whose lens has a larger or smaller curvature than is proper for the length of the eyeball. He emphasized for the first time that the rays from an object point envelop a surface that is now called the caustic surface and that this surface is the place of the highest light concentration. And he explained the colors of the rainbow by multiple reflection in the water drops. All this information was concisely organized by 1567 into a work entitled Photismi de lumine , although it was not published for some thirty years (1611) after the author's death, and then only as a consequence of the interest in optics engendered by Galileo's telescopic discoveries.

In Italy, under the influence of the philosophy of the time, scientific progress and magic were mixed in a way that led to interesting reading. The development of mathematical methods, however, served to gradually separate speculation and imagination from the solid basis of observation.

One of the most colorful books of the period was the Magia naturalis of Giovanni Battista della Porta (ca. 1538 or 1543 -1615). In his youth, della Porta had written a four-part volume entitled De miraculis rerum naturalium (1558) and, subsequently, many books and plays, including De refractione (Fig. 6) in nine parts in 1593. The Magia naturalis, which was an expansion of the earlier De iraculis into twenty volumes, appeared in 1589. It became so famous that it was soon translated from Latin into the vernacular languages Italian, French, Spanish, and English. It is a strange medley. Only the seventeenth part concerns optics, but some of the others are also worth noting for our amusement if nothing else. I mention the titles of a few of the twenty parts, to give you an idea of their contents. The first is entitled, "The Causes of the Miraculous. It deals with astrology, sympathy and antipathy, and the influence of place and time on events. The second deals with the origin of animals. He quotes many authorities who agree that snakes are made from the marrow of men and the hair of women. The third deals with garden plants; the fourth, with how to become wealthy by good housekeeping; the fifth, with how to make precious stones artificially; and the ninth, with make-up and other ways of improving the beauty of women (as if this were necessary!).

Finally, the seventeenth part deals with focal mirrors and other mirrors and explains what one can do with them. This book exhibits progress in the understanding of optical phenomena. In it are discussed the same tricks that Hero had dealt with 1400 years earlier, namely, how to use mirrors to obtain desired effects. However, we find here the first exact theory of multiple mirrors, the first complete description of the camera obscura, with pinhole or lens, and a discussion of combinations of positive and negative lenses designed to improve vision. Porta compares the eye and its pupil to the camera obscura and its stop. (Leonardo had anticipated him here, but less definitely and accurately.) At the end of this century also falls the invention of the telescope and the microscope.

Eyeglasses had apparently been invented toward the end of the thirteenth century, but they seem not to have attracted the attention of writers on optics until the practical Leonardo discussed methods of making them. Some lens grinder-more likely, several independentlymust have discovered the principle of the Galilean telescope by accident, and certain passages in Leonardo's notebooks a hundred years before Galileo can be interpreted as instructions for making such a telescope. Be that as it may, a Dutchman, Zacharias Jansen (1588-1632), was said by his son to have made a telescope in 1604 according to the model of an Italian dated 1590, and soon telescopes became well known in Holland. Galileo Galilei (1564-1642) heard of them and, apparently independently, made an instrument that he exhibited in 1609, and in his lifetime made over a hundred. The Dutch instruments were so imperfect that a magnifying power of more than 3X was impractical. Galileo made instruments with a useful power of approximately 30 X, an order of magnitude greater. He was by no means the inventor of the telescope or even the first scientist to describe it; Vavilov reminds us that Leonardo may have anticipated him and that Porta described, three months after Galileo's demonstration, an instrument he had made independently. But Galileo is outstanding on two counts: he revolutionized man's ideas of the uni- verse by his applications of the telescope and, by improving it by an order of magnitude, Ronchi points out that Galileo is entitled to be considered the father of the art of fine optics.

Moreover, from telescope to microscope is a short optical step, and by 1624 Galileo was using lenses of short focal length and thus making precursors of the modern compound microscope. 9 A few decades thereafter, the Dutchman Antonj van Leeuwenhoek (1632-1723) improved the quality of simple lenses used for magnifiers or simple microscopes, and in a letter of 1674 he began the descriptions of the discoveries for which he became justly famous.24 Meanwhile, Robert Hooke (1635-1703) in England also improved the quality of lenses in order to make the compound microscope practical, and in 1665 the Royal Society published his Micrographia.

Aside from publications, which were in an embryonic state, the seventeenth century brought a lively exchange of ideas and thoughts between scientists, mostly friendly but sometimes polemical and frequently expressed in code to conceal them from the uninitiated. It is therefore necessary to study not only the books of these scientists but also their correspondence (cf. Leeuwenhoek's letters quoted by Dobell).

The same period witnessed the development of scientific knowledge to the system that is still important.Indeed, scientific development in the seventeenth century was greater than in any century before or since. Scientific giants like Kepler, Descartes, Pascal, Fermat, Newton, Leibniz, and Huygens appeared and left their marks in all fields of science, including optics. Let us sketch some of their contributions to geometrical optics.

Johannes Kepler (1571-1630) wrote two books on optics: Supplements to Vitelo in 1604 and Dioptrice in 1611. The object of the first is to find the law of refraction. Kepler nearly succeeds, though he does not look for a relation between the incident angle and the refracting angle but between the incident angle and the deviation. He asks himself what form a surface should have to refract a parallel beam of light exactly into a point image. He studies the hyperboloid, which, as we know, fulfills the condition; but he rejects it because it disagrees with the values that Vitello gave for the relation between the angles of the incident ray and the refracted ray. If only the Kepler of 1611 and the Dioptrice, in which he discovered the law of total reflection, had reread his book of 1604, he would have seen that the data of Vitello had been erroneous and thus he might have found the correct form of the law of refraction.

Kepler investigated the function of the eye more profoundly than his predecessors. He discovered that the eye is not at rest but moves around a point inside it, and he investigated the importance of this motion on our vision. He drew attention to the fact that the image on the retina is inverted. All this is in his Supplements to Vitello.

His Dioptrice , although a relatively small book, was of importance to the development of optics. In it, Kepler assumes that for small angles the angle of refraction is proportional to the angle of incidence. From this assumption, he develops the laws of firstorder optics for thin lenses and systems of thin lenses with finite separations. He describes, among other things, the Keplerian telescope, in which two positive lenses are placed so that the back focus of the first is at the front focus of the second, and compares it with the form named after Galileo, consisting of a positive and a negative lens placed so that the focus of the former s at the virtual focus of the latter. He gives the focal length correctly for all forms of lenses. His book contains many practical proposals for lens design and is highly recommended to those who are interested in these problems.

Rene Descartes (1569-1650) must be considered as one of the discoverers of the correct law of refraction, for he published the sine law in his Dioptrique in 1637. After his death, he was violently attacked by Isaac Voss, who accused him of stealing the law from Willebrord Snel* (1591-1626). Descartes had indeed visited Leyden in 1630, where Snell, a professor of physics, had written a book on optics containing the refraction law. Although the book had never appeared, Voss, in his posthumous attack, claimed that Descartes had seen it on his visit after Snell's death, and Voss quoted Snell's proof.

The question of priority has been much discussed in the literature. A recent study has been published by Boegehold, who refers to a paper by P. Kramer. Boegehold found that Descartes, in some letters written in 1627, had suggested making plano-convex lenses with hyperbolic surfaces. Since such a lens would give a sharp point image for an infinite object if and only if one assumes the sine law, it is fair to conclude that Descartes knew of the law three years before his visit to Leyden in 1630. It is strange that Voss should have years after waited for twenty-five years-twelve Descartes's death-to make his accusation. Credit for discovering the law of refraction should therefore be divided between Descartes and Snel, with a fair share going to Johannes Kepler, who failed to find the sine law only because he trusted the tables of Vitello.

Descartes's Dioptrique is beautifully written. It tries to explain the law of refraction by means of the theory of elasticity. A light ray is compared to the path of a ball, first being bounced off the ground, then being slowed down as it goes into a denser medium. Descartes was the first to suggest determining the velocity of light from the position of the moon at an eclipse.

He also made experiments with human and animal eyes. If the eye is removed and put in the opening of a window in a darkened room, one can see the images of distant objects on the retina, and by pressing the eye a little, one can see the image of near-by objects, showing that the eye can accommodate.

In the eighth chapter of his book, Descartes investigates surfaces that image a point sharply-the socalled Cartesian surfaces, of which the conic sections constitute a special case. He proposes machines for grinding lenses and demonstrates the colors of the rainbow with small glass spheres filled with water.

But much of Descartes's reasoning in connection with the refraction law was faulty and was rejected by the great mathematician Pierre de Fermat (1601-1665), who pointed out that water would have to be less dense than air to fit Descartes's explanation. Fermat returned to the formulation of the refraction law given by Hero, deriving it from the assumption that light has different velocities in different media and takes the quickest path between any two points. Like Hero, he illustrates this idea with a drawing of a convex surface, for which the minimum principle is correct, rather than a concave surface, for which it can be wrong. But he was called on the carpet by Abbé Mersenne, as was mentioned earlier, one of the great scientists and letter writers of his day. Fermat's answer is a classic example of double talk: he says that the path would be a minimum if the surface were a plane, and the incoming ray sees so little of the surface that it cannot distinguish whether the surface is plane or curved!

1.1.5 Newton and Huygens

Sir Isaac Newton (1642-1727), in his Lectiones opticae of 1669-1671 (published posthumously), showed how to calculate the spherical aberration of a plano-convex lens for an infinite object point. This book contains the formula for locating the image, still known by his name, in terms of the distances of the object and the image from the focal points. It also contains a thorough investigation of reflection in prisms, including the position of minimum deviation. It even goes into some aberrations, in particular, image errors as functions of color, and it compares the size of the color aberration with the spherical aberration of a given lens and prism.

Although Newton's best known work in optics is his Opticcs , I should like to draw your attention especially to his beautiful paper on color, in which he describes the separation of white light into the colors of the spectrum by means of prisms. This charming paper should be read by every student of optics, especially since the writer not only gives his final conclusions but describes many false attempts at interpreting his experiments and the refutation of these attempts.

Newton was misled into believing that all materials have the same relative dispersion, since he found this to be true for such different media as glass and water. The conclusion that he drew-that telescopes should not contain refracting surfaces-was erroneous. But it led him to study mirror telescopes and, with his teacher Gregory, he contributed much to the development of these instruments.

Christian Huygens (1629-1695) wrote two books on optics: Dioptrical in 1653, published posthumously , and Traite de la lumiere , written in 1678 but not published for twelve years. The first book shows enormous advances with respect to the understanding of optical image formation. The second proposes a new and revolutionary theory of light, deriving refraction and reflection laws from a completely new principle.

This principle, set forth earlier in the Dioptrica, is a starting point from which all the laws of Gaussian optics can be derived for lens systems with finite thicknesses. It is expressed as follows: if we interchange object and eye in an optical system, the object will have the same apparent size and orientation as before. The law is known to us in the form that the product of lateral magnification and angular magnification is constant in an optical system, and equal to the ratio of the refractive indices.

Huygens found, before Newton, the law that, for a single surface or for a single lens, the image distance is proportional to the object distance when the distances are measured from the respective principal foci. He knew of the aplanatic surface. He was acquainted with the important property of the optical center of a lens (the point which divides the distance between the vertices in the ratio of the radii), namely, that a ray through it emerges parallel to its original direction. He had an approximate formula for the spherical aberration of a thick lens that enables the form for minimum spherical aberration to be determined.

In his Traite de la lumiere, which should be required reading for every physicist and is now readily available, Huygens derives optical laws from the fact that the light emerging from an object point P forms a wave surface S at any given time t. The light rays are then the normals to this surface and are not straight unless the velocity is constant throughout the medium.

The surface can also be constructed in the following way. At an arbitrary time , light from the point source P forms a wave surface S. (assumed to be spherical in the figure). Considering each point of S, as a source sending out a new wave, we see that the envelope of all the waves at the time gives the original wave surface S.

With the help of this concept, Huygens derives the straight paths of light in a medium of constant velocity. He derives not only the reflection and refraction laws but also the law of refraction in the atmosphere and in doubly refracting crystals. He clearly defines the caustic as the evolute of the rays and investigates it after reflection and refraction, in addition to investigating the degenerate form of the wave surfaces in the points of the caustic. He proves Malus' law: that rays form a normal system for a single reflection and refraction.

It should be emphasized, however, that this formulation is mathematically identical with ray optics, as Hamilton showed in his paper of 1831. The assumption that light has a finite wavelength and that it can be described as a sine wave going along a ray is foreign to Huygens. However, once this theory was established, the methods of Huygens enabled Fresnel to derive all the laws of diffraction in a most elegant manner.

Chapter 2

Geometrical Optics

The Euclidean Geometry exists thanks to light. Euclid wanted to undestand the light rays, so he discovered the plane geometry.

"Music is the arithmetic of sounds as optics is the geometry of light."
– Claude Debussy

2.1 The postulates of geometrical optics and the dark Room

We already know the history of the geometrical optics, well before starting the study of this theory we must take back the contributions of Al Azhen and talk about the **postulates of geometrical physics** which are

1. The speed of light is finite.
2. The light propagates in a straight line.
3. The light rays obey the Euclid's postulates of plane geometry.
    (a) A straight line segment can be drawn joining any two points.
    (b) A finite straight line (line segment) can be extended continuously in a straight line indefinitely.
    (c) A circle can be drawn with any point as its center and any distance as its radius.
    (d) All right angles are equal to one another, regardless of their orientation.
    (e) If a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, the two lines, if extended indefinitely, will eventually meet on that same side.
4. In the matter the light's speed is less than in vacuum.

Now, before starting to use these postulates we must understand the physical meaning of them. The first postulate, **The speed of light is finite** is fundamental, it is incredibly important.

How could Hero of Alexandria understand that the speed of light must be finite? To be honest, I don't really know his motivation but, we could get an idea about why this should be like that from simple phenomenological reasoning without the use a mathematics or advanced physics. *Lets imagine a world where the speed of light is infinite*

How are going to be the nights in such world? It should be a complete chaos. Every star in the sky is a source of light, a source of infinite rays which each one travels at an infinite speed. There wont be any night, all the day is going to be full of light and even every event in the universe is going to be seen simultaneously, the big bang, the birth and death of a star. *Each past and present event is going to be seen at the same time.*

Other effect that only can be explained in a universe with light with finite speed. Imagine you're standing in the rain, which is falling vertically. If you start running, the rain hits you in the face. This happens because your movement changes the apparent direction of the rain. An astronomer, James Bradley, observed that the stars didn't appear to be exactly in their calculated positions. To see a star directly overhead, he had to tilt his telescope slightly in the direction of Earth's movement around the Sun, the light from the star "falls" toward Earth at a finite speed. Earth is moving.

For the light beam to travel from the telescope lens to the eyepiece (and for us to see it), Earth has moved a little. Therefore, you have to point the telescope at a slight angle toward where Earth will be when the light reaches the eyepiece, not where it is at the moment it enters the lens.

If the speed of light were infinite, this effect wouldn't exist. The rain (light) would fall so fast that your (Earth's) movement would be irrelevant. The fact that we have to "aim ahead" of the target is irrefutable proof that light travels at a finite speed.

There are many other consequences but lets continue with all the other postulates. Now lets talk about **The light propagates in a straight line**. This does have sense because it is what is seen in any normal day, all the source of light can be seen as sources of infinite many rays which propagates at a finite speed. With tools purely geometrical there is no way to give a proof of this, however, from a wave theory for light, Huygens was able to prove that the rays are, in fact, the normal vectors of the wavefront are the rays, and, from a quantum perspective, Richard Feynman proved that the rays are the superposition of the "watches" that represent the probability amplitudes.

Often we see that light is not an straight line, for example, an easy experiment consists on a translucid bowl full of salted water, we're going to see that the ray presents a deflection, we're going to see that this curves can be generated as a composition of infinitely many straight refractions.

Because we've postulated that the light travels in straight lines, so we have to give a theory for the propagation so **The light rays obey the Euclid's postulates of plane geometry**. A point is going to be a source, a place where two rays meet, the place where two media separates, etc.

The extension of a line segment to a continuously straight line is a way to make or represent ray tracing. The circle postulate talks about the sphere that propagates alongside all the rays (which behave as the radius). The right angles are just a geometrical formalism which does not have a dramatic optical interpretation.

The last postulate is really important, from a physical point of view it tell us that all the rays are going to be propagate on a completely plane space, so we could ask ourselves if we want to study the light's behavior under the gravitational effects of compact objects we must change the fifth Euclid's postulate, and the short answer is yup, we have to. Under gravitational effects the differential geometry is a better tool to study the light-type observers on the curved space-time.

So, we have many arguments that supports the idea the light's speed must be finite, and if it is finite there should be a way to measure it, right? The answer is yes, and here we're going to use the postulates to follow the Fizeau's ideas on how to measure the light's speed.

In 1849, Hippolyte Fizeau performed a groundbreaking experiment that provided the first non-astronomical measurement of the speed of light. His ingenious setup transformed the problem of measuring an extremely short time interval into a problem of measuring a rotation speed, which could be done with much greater precision. Imagine a very fast runner who starts from a point

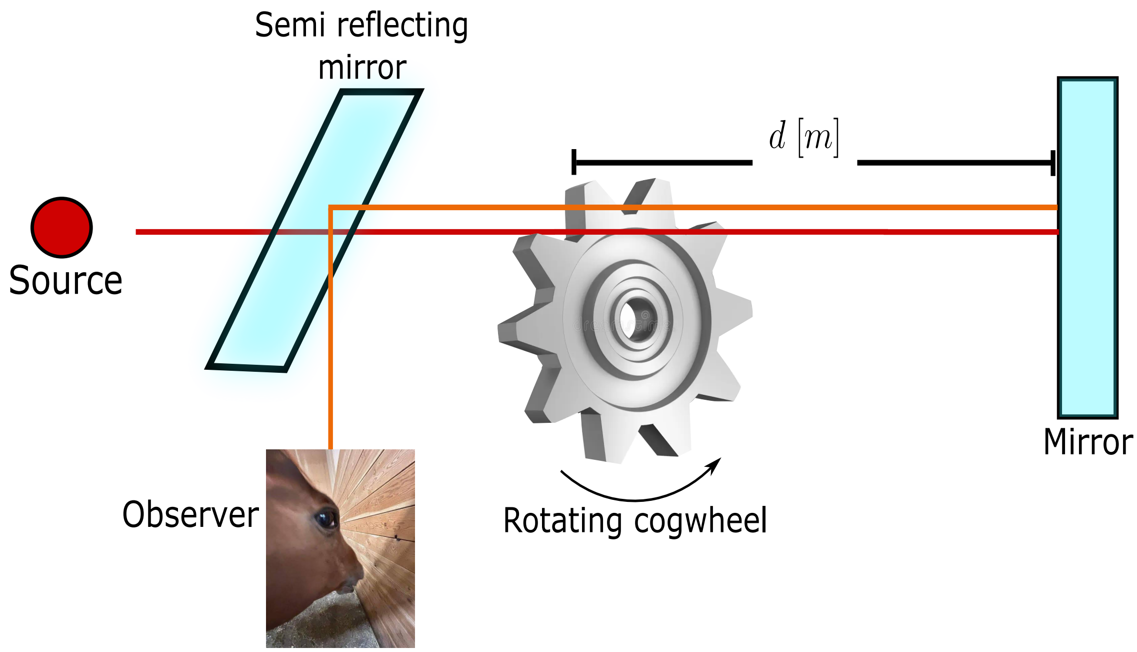


Figure 2.1: Schematic representation of the Fizeau's experiment pretended to measure the light's speed.

A, runs to a point B and immediately returns to A. To measure the runner's speed, you could place a door at A that opens and closes at regular intervals. If you open the door just when the runner leaves, then close it and open it again periodically, you might catch a glimpse of him when he returns if the timing is right. But if the runner is extremely fast, the door might be closed when he gets back. This is exactly the idea Fizeau used with light, replacing the door with a rapidly rotating toothed wheel and the runner with a light pulse.

Fizeau set up his apparatus on two hills in the outskirts of Paris, with the light source and observer on one hill (Montmartre) and a mirror on the other (Mont Valérien), about 8.633 kilometers away. A beam of light from a lamp was directed toward a partially reflecting mirror (a beam splitter) that sent part of the light toward a toothed wheel. The wheel could be spun at a controlled speed and had a large number of teeth and gaps – typically 720 teeth, meaning 720 gaps as well. After passing through a gap, the light traveled to the distant mirror, was reflected, and returned along the same path. On its way back, it would again have to pass through a gap in the wheel to reach the observer's eye via the beam splitter.

If the wheel were stationary, the returning light would pass through the same gap and be seen. But as the wheel began to rotate, the situation changed. When the wheel turned slowly, the gap through which the light had originally passed would have moved only slightly during the light's round trip, so most of the returning light still made it through. As the speed increased, the gap shifted more, and the light began to be partially blocked by the edge of a tooth. At a certain critical speed, the gap moved exactly so that by the time the light returned, the adjacent tooth had taken its place, completely eclipsing the light. The observer would see the light disappear for the first time. By knowing the rotation speed at which this first extinction occurred, the number of teeth, and the distance to the mirror, Fizeau could deduce the time it took for light to travel to the mirror and back.

Let us analyze the situation mathematically. In the figure 2.1 let d be the distance from the wheel to the distant mirror. The total path for the light is $2d$. The wheel has N teeth and consequently N gaps, so the angular width of one tooth (or one gap) is π/N radians (since a full circle has 2π radians and there are $2N$ segments – teeth and gaps – each of equal angular size). For the first extinction, the time required for the light to make the round trip must equal the time needed for the wheel to rotate by the angle corresponding to one tooth (i.e., to replace a gap with a tooth). If the wheel rotates with angular velocity ω (radians per second), the time to turn through an angle π/N is

$$t_{\text{rot}} = \frac{\pi/N}{\omega}. \tag{2.1}$$

The light travel time is

$$t_{\text{light}} = \frac{2d}{c}, \tag{2.2}$$

and equation these gives us

$$\frac{2d}{c} = \frac{\pi}{N\omega}. \tag{2.3}$$

But $\omega = 2\pi f$ where f is the rotation frequency in revolutions per second. Substituting,

$$\frac{2d}{c} = \frac{\pi}{N \cdot 2\pi f} = \frac{1}{2Nf}. \tag{2.4}$$

Solving for c yields

$$c = 4dNf. \tag{2.5}$$

Fizeau used $d = 8.633 \times 10^3$ m, $N = 720$, and observed the first extinction at $f \approx 12.6$ revolutions per second. Plugging in these numbers,

$$c = 4 \times 8.633 \times 10^3 \times 720 \times 12.6 \approx 3.13 \times 10^8 \text{ m/s}, \tag{2.6}$$

a value about 5% higher than the modern accepted value of 2.99792458×10^8 m/s, but remarkably accurate for the time. The discrepancy arose mainly from uncertainties in the distance measurement and the precise determination of the extinction condition. Fizeau's experiment was a triumph of experimental physics, demonstrating that light travels at a finite speed and providing a method that would be refined by later researchers, most notably Léon Foucault, who replaced the toothed wheel with a rotating mirror to achieve even greater precision. The conceptual beauty of Fizeau's approach lies in its translation of an infinitesimal time interval (on the order of 1/20000 of a second) into a measurable mechanical rotation, a perfect illustration of the deep connection between optics and mechanics.

Now we know the speed of light is finit, so lets use these postulates to understand one of the biggest contribution of Al Azhen, the dark room, see figure 2.2.

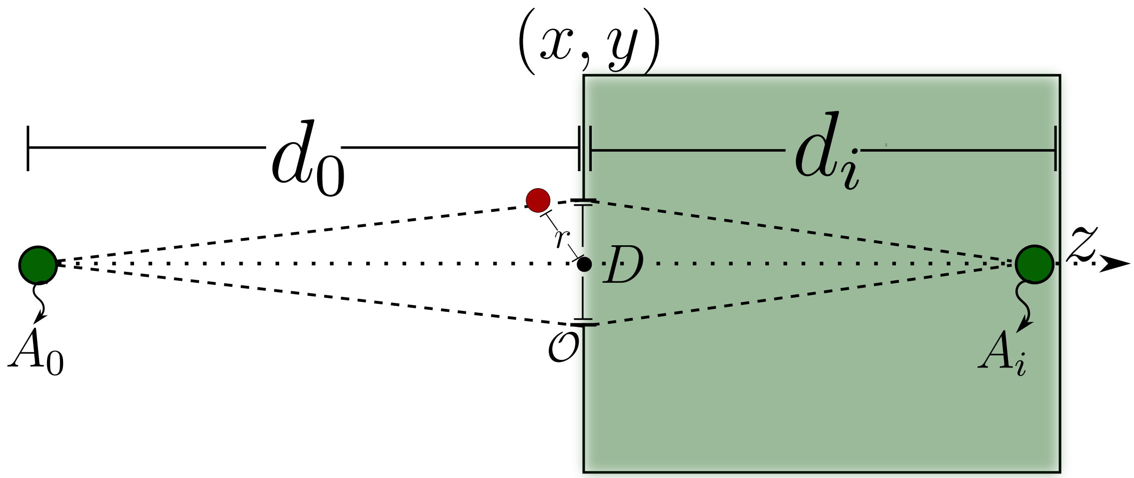


Figure 2.2: Graphical representation of a generic dark room which is a box of an arbitrary material which has a side opened and while it satisfies the Rayleigh's inequality i will form an inverted image.

Before develop mathematically the dark room we're going to set an important convention

- The light propagates from left to right along the z direction.
- A distance measured in the direction of light is positive and a distance measured against the direction of light is negative.

Once defined the convention lets do it. A source in the point A_o is at a distance d_o respect to the opening O of the dark room, (d_o is in fact negative because it is measure against the light direction from the point O), an image if formed at a distance d_i from O thus we must find out a relation between these parameters and the diameter D of the DR. First we're going to calculate the distances $\overline{A_oOA_i}$, $\overline{A_oRA_i}$ where R is the radius of the opening, therefore

$$\overline{A_oOA_i} = d_i - d_o \;\; d_i = \overline{OA_i} \;\; d_0 = \overline{OA_o}, \tag{2.7}$$

$$\overline{A_oRA_i} = \left[d_i^2 + x^2 + y^2\right]^{1/2} - \left[d_o^2 + x^2 + y^2\right]^{1/2}, \tag{2.8}$$

having on count our convention implies that $\sqrt{d_o^2} = -d_o$ so

$$\overline{A_oRA_i} = d_i\left[1 + \frac{r^2}{d_i^2}\right]^{1/2} - d_o\left[1 + \frac{r^2}{d_o^2}\right]^{1/2}. \tag{2.9}$$

Historically it was observed in the dark room that the image that was formed depended on the diameter of the opening and the distance of the object to view, it was observed that the image was formed for far objects, but how far is far? It is relative to the radius of the opening, then the ratios must satisfy that

$$\frac{r^2}{d_o^2} \ll 1\ ,\ \ \frac{r^2}{d_i^2} \ll 1, \tag{2.10}$$

using this and using the Taylor's theorem for a polynomial expansion of the function $\sqrt{1+x}$ then

$$\overline{A_oRA_i} \approx d_i\left[1+\frac{1}{2}\frac{r^2}{d_i^2}+\mathcal{O}(\delta^2)\right] - d_o\left[1+\frac{1}{2}\frac{r^2}{d_o^2}+\mathcal{O}(\delta^2)\right], \tag{2.11}$$

$$\overline{A_oRA_i} \approx (d_i - d_o) + \frac{r^2}{2}\left(\frac{1}{d_i}-\frac{1}{d_o}\right) = \overline{A_oOA_i} + \frac{r^2}{2}\left(\frac{1}{d_i}-\frac{1}{d_o}\right) \tag{2.12}$$

with this we can express this in terms the difference of the light's path and the diameter of the dark room

$$\triangle L = \frac{r^2}{2}\left(\frac{1}{d_i}-\frac{1}{d_o}\right) = \frac{D^2}{8}\left(\frac{1}{d_i}-\frac{1}{d_o}\right). \tag{2.13}$$

In the equation 2.13 lies the hearth of the dark room, it was found by Rayleigh a certain relation between the light's wavelength and the light's path difference between the upper and lower part of the opening of the dark room, this relation (which will be discussed physically in a moment) is:

$$\triangle L < \frac{\lambda}{4}. \tag{2.14}$$

Using this relation we will find the maximum diamater for the dark room

$$\frac{D^2}{8}\left(\frac{1}{d_i}-\frac{1}{d_o}\right) < \frac{\lambda}{4}, \tag{2.15}$$

$$D_{max} = \left(\frac{2\lambda d_i d_o}{d_o - d_i}\right)^{1/2}. \tag{2.16}$$

In the real life there is no monochromatic light, it is forbidden by the laws of quantum mechanics, we can have quasi-monochromatic light but it is only in advanced laboratories so for a polychromatic source in the equation 2.16 we should do the calculations with the maximum wavelength that conforms the beam the origin of this expression have many origins which converges to the same phenomena. If we want to form an image the electromagnetic field coming from the object must arrive at the same time at the image point, if there is some kind of phase difference there will be aberrations as astigmatism or chromatic aberration (even if it is not a lens we can see it from the equation 2.16) from a quantum point of view this is related to an interpretation of the *size* of a photon (Santos et al., 2018).

Now we're going to analyze how does the light interact (from a simple and geometrical point of view) with transparent media. Suppose a source radiates from a point A and the ray light arrives at B, see figure 2.3.

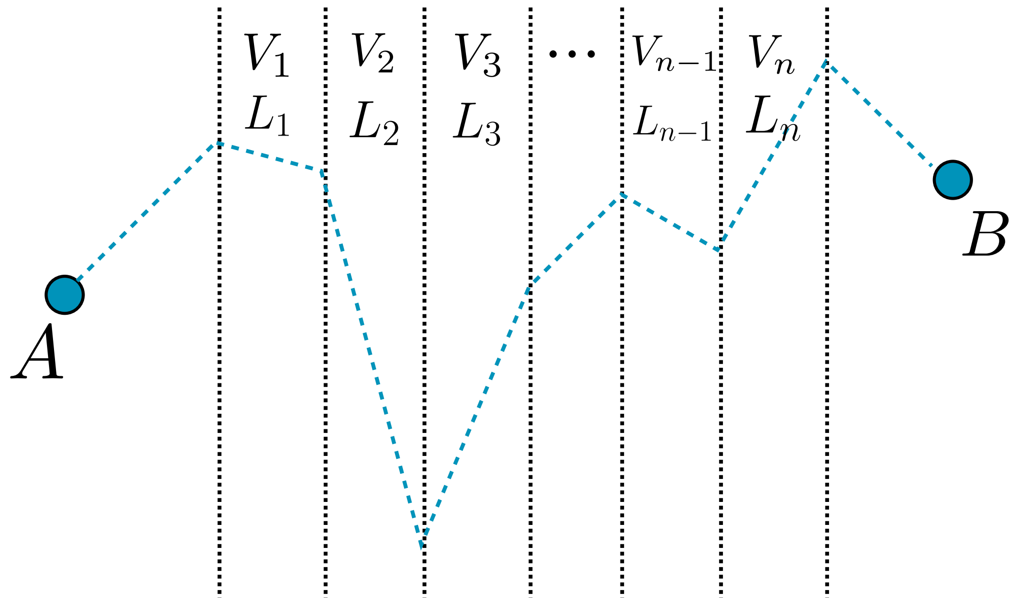


Figure 2.3: A schematic representation of a ray beam from A to B which goes through n different transparent media and in each one it had a different speed and traveled different length.

There are n different transparent media each one hast a different thickness and the light in each one is different, therefore the time that takes the light to go from A to B is

$$t_{AB} = \sum_{i=1}^{n} \frac{L_i}{v_i} = \sum_{i=1}^{n} \frac{L_i C}{v_i} \frac{1}{C} \; [s]. \tag{2.17}$$

Then we define the refraction index in the ith medium as the relation between the speed of light in vacuum and the speed of light in the ith medium. *It is a scalar in linear and homogeneous and isotropic media but mostly it is a tensor.*

$$t_{AB} = \frac{1}{C} \sum_{i=1}^{n} n_i L_i = \frac{L_{AB}}{C} \; [s], \tag{2.18}$$

Definition 1 *(Optical path): The optical path L_{AB} is the distance that must travel in vacuum the light to have the same time to go from a point A to a point B across n different media each one characterized by a different (or equal) refraction index.*

The optical path has a length greater than the geometrical path of the light, that is why it is called optical path. This can be easily generalized to continuous media using a Lebesgue's integral (or Riemann integral if it is well behaved)

$$L_{AB} = \int_A^B n(S)ds\ [m]. \tag{2.19}$$

Before going further we must have to set some definitions.

Definition 2 *(Phase): The phase is the relative angular displacement the ray has.*

Definition 3 *(Wavefront): The geometrical space where the phase of the light is constant is called the wavefront of the light.*

2.2 The Snell-Descartes law and the Fermat's principle

As we wad discussed in the historical chapter it was Heron of Alexandria who realizes that in light reflection it travels the least distance and Al Azhen proposes a maximum time, it was Fermat who formalizes it as a principle the **Fermat's principle**

Principle 1 *(Fermat's principle): The light travels a trajectory such that the optical path is stationary*

$$\delta L = 0. \tag{2.20}$$

The Fermat's principle can be "deduced¨ from the Hamilton principle or even more from a variational principle, lets give a look to it, lets look for an stationary curve for the time the light travels

$$t = \int_{t_0}^{t_f} dt = \int_A^B \frac{1}{C} dL = \int_{\tau_0}^{\tau_f} L d\tau, \tag{2.21}$$

from here we see that, in a mathematical way, if we want to find the relation between the trajectory of the light and the time it takes to go from A to B we will have to solve the Euler-Lagrange equation. Now lets use the Fermat's principle to see how the light interacts with a diopter (a refracting surface) in a vector way, known as the *vectorial Snell-Descartes law.*

Definition 4 *(Diopter): A Diopter is a surface $\Sigma \subset \mathbb{R}^2$ such that it separates two media with different refraction index.*

In the figure 2.4 we see two media separated by a transparent surface (diopter) with refractive index n_o,n_1, respectively, a source A emits a beam which reaches the diopter in I and it is refracted to B, from A there is an unitary vector $\hat{u}_o$ in the direction $\overline{AI}$ and the observer O analyzes a variation of A as A' and the incident point on the diopter is I', then lets analyze it vectorially

$$L = \left|\left|\vec{I} - \vec{A}\right|\right|, \tag{2.22}$$

$$\triangle\vec{I} = \vec{I'} - \vec{I}, \tag{2.23}$$

$$\triangle\vec{A} = \vec{A'} - \vec{A}, \tag{2.24}$$

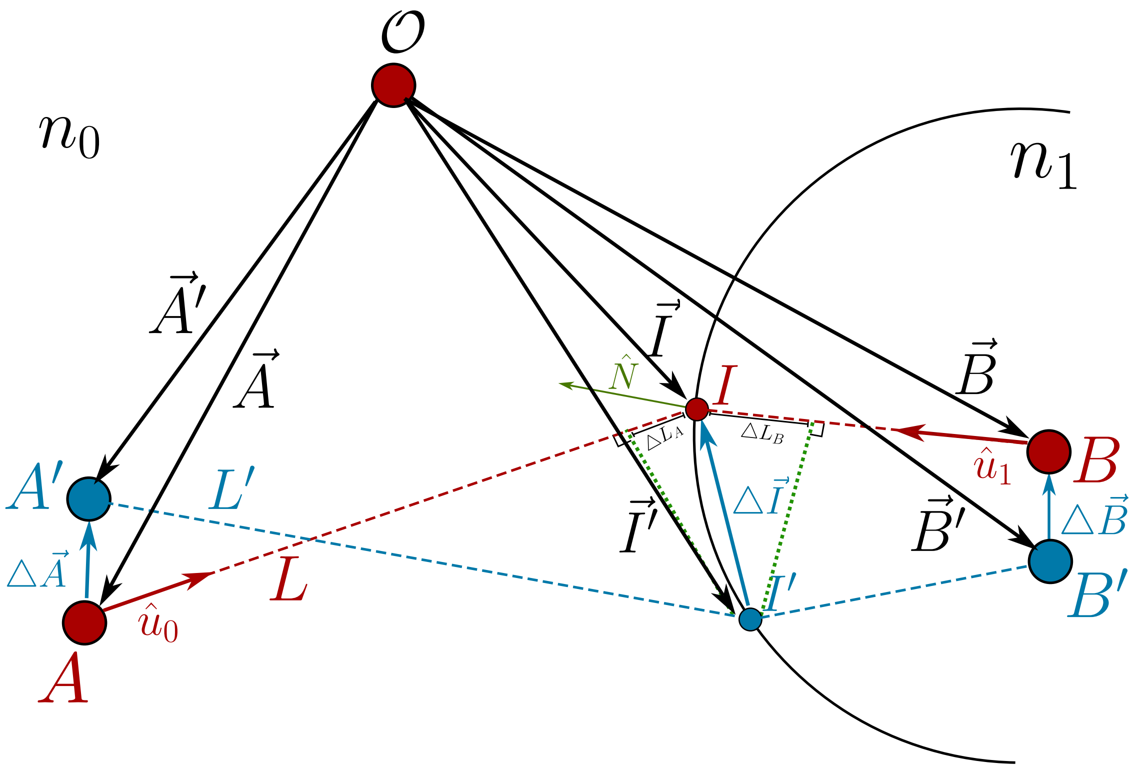


Figure 2.4: two media separated by a diopter which an observer O measures the path the light will follow from A to B and proposes a displacement of the source A to A' to see how must be the light path.

with this we can calculate the path difference $L - L'$ vectorially as a proyection of the unitary vector in the direction of L given by $\hat{u}_o$ along $\triangle\vec{I} - \triangle\vec{A}$

$$\triangle L_A = \frac{\left(\vec{I} - \vec{A}\right) \cdot \left(\triangle\vec{I} - \triangle\vec{A}\right)}{\left|\left|\vec{I} - \vec{A}\right|\right|} = \hat{u}_o \cdot \left(\triangle\vec{I} - \triangle\vec{A}\right), \tag{2.25}$$

now for the optical path

$$\mathcal{L}_A = n_o \triangle L = n_o \hat{u}_o \left(\triangle\vec{I} - \triangle\vec{A}\right), \tag{2.26}$$

doing the same for the right side of the diopter but for the point B we then have got the total optical path difference as

$$\triangle\mathcal{L} = n_o \hat{u}_o \cdot \left(\triangle\vec{I} - \triangle\vec{A}\right) + n_1 \hat{u}_1 \cdot \left(\triangle\vec{B} - \triangle\vec{I}\right) \tag{2.27}$$

because the points A and B are fixed then $\triangle\vec{A} = 0$ and $\triangle\vec{B} = 0$ thus

$$\triangle\mathcal{L} = (n_o\hat{u}_o - n_1\hat{u}_1) \cdot \triangle\vec{I}. \tag{2.28}$$

If the Fermat's principle is satisfied the equation 2.28 implies straightfully that $\triangle\vec{I}$ is a tangent vector to the diopter this implies then

$$n_o\hat{u}_o - n_1\hat{u}_1 = k\hat{N}. \tag{2.29}$$

The equation 2.29 is known as the vectorial Snell-Descartes law, from this equation we can arrive to the law of reflection and refraction to do it we only have to do the cross product of the equation 2.29

$$\hat{N} \times (n_0\hat{u}_0 - n_1\hat{u}_1) = k\hat{N} \times \hat{N} = 0, \tag{2.30}$$

$$n_0 sin(\theta_0) = n_1 sin(\theta_1). \tag{2.31}$$

2.2.1 The critical angle, and the refraction by a prism

We had deduced the vectorial Snell-Descartes law, which is a strong law and we are going to apply it to some physical situations, but before doing that we are going to talk briefly about where is the physical regime where this law can work on.

We are working under the postulates of the geometrical optics, but lets analyze a bit the second one, *the light propagates in a straight line* we haven't say anything about the light's shape, if we say it propagates in a straight line we can image light rays as circular cylinders but what about the radius of this ray? Well, in fact there are more shapes we can imagine of the wavefront and the beam's profile, but we are going to talk about it in the chapter of electromagnetic optics, for now keep with the cylinder image.

If we make this ray to inside over a perfectly polished mirror as shown in 2.5 then we can see an prove easily experimentally the Snell-Descartes law will be fulfilled but what happen if we have not got a polished surface but an extremely irregular surface? we will have reflections in all the points over the surface as shown in the figure 2.6.

The pehonomena shown in the figure 2.6 is, often, called *diffuse refraction* but it is a kind of macroscopic effect of the Snell-Descartes law but there are other behaviors of the refraction index which give a particular behavior to the SD law, which are related to the electromagnetic properties of the matter, in fact, the refraction indez n is a complex quantity, which its real part is related to the refractive properties of the material and the imaginary part is related to absorption an losses in the energy but we will work on this later.

The are some other special materials where the equations 2.29 has an amazing behavior. The meta materials which has a negative refraction index, but how can we interpret this? It is easy to see in the equation 2.29 that a variable changes of $n \to -n$ implies a physical effect of $\theta \to -\theta$ this

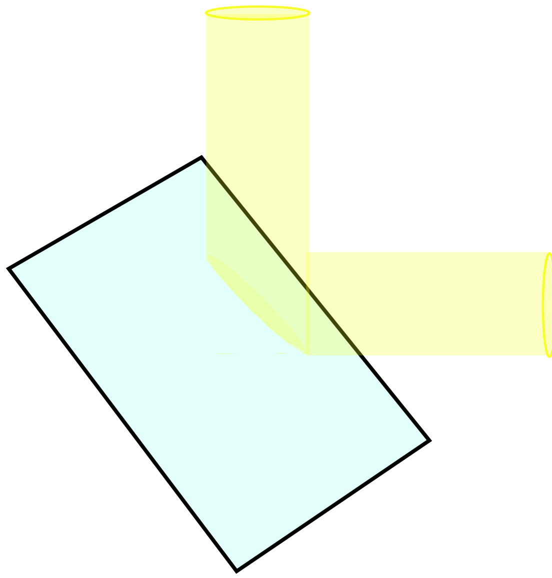


Figure 2.5: Representation of a cylindrical ray light being reflected by a perfectly polished mirror.

means, the rays will be reflected or refracted to the other side!

Well then, lets come back to our well defined Behavior, where the SD law 2.29 will be fulfilled. Can we find a condition between two material such that there is no refraction? It is worth to ask about it because in nature the reflection and refraction happen at the same time both simultaneously because a translucent surface is not perfectly transparent and a refractive surface is not perfectly reflective this is related to the Fresnel's coefficients but this will be hard work of the chapter of electromagnetic optics, for now have on mind reflection and refraction happen at the same time. To try to kill the refraction we have to study two cases between two media.

1. $n_1 > n_2$
2. $n_1 < n_1$

For the first case then we have from 2.29

$$n_1 sin\theta_1 = n_2 sin\theta_2, \tag{2.32}$$

$$\frac{n_1}{n_2} sin\theta_1 = sin\theta_2. \tag{2.33}$$

If $\theta_1 = 0$ then $\theta_2 = 0$ thus if the incident ray is perfectly parallel to the normal vector to surface there wont be any refracted ray, only a completely reflected in the same plane as the incident but with opposite direction, this is said to get a phase shift of π. If the ray's incidence angle is perpendicular to the normal vector to surface $\theta_1 = \pi/2$ then we will have

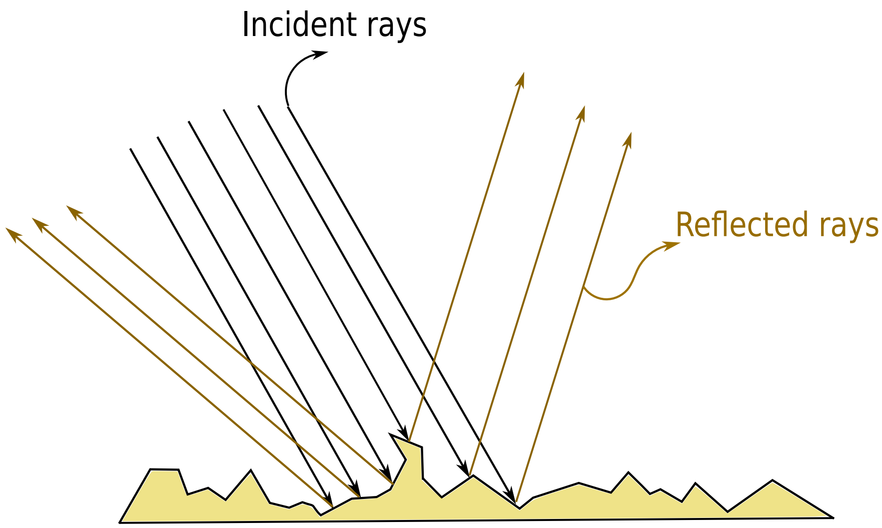


Figure 2.6: Representation of a cylindrical bunch of rays being reflected by an inhomogeneus surface.

$$\frac{n_1}{n_2} = sin\theta_2, \tag{2.34}$$

but $n_1 > n_2$ then $n_1/n_2 > 1$ which has no sense with the right hand side of the equation 2.34 because $0 \leq sin\theta_2 \leq 1$ therefore for $\theta_1 = \pi/2$ there is no refracted ray this is illustrated in the figure 2.7.

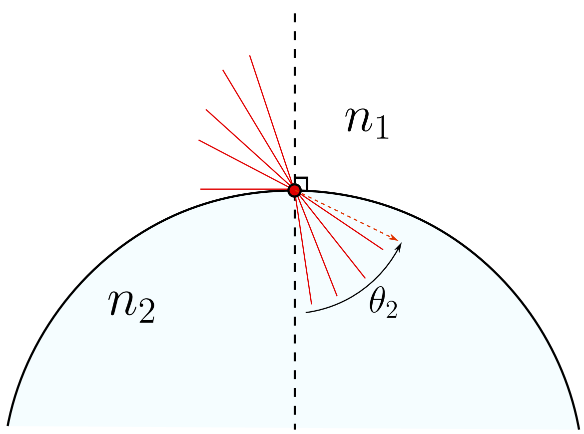


Figure 2.7: Representation of all the incident and refracted rays from a media with refraction index n_1 to another with refraction index n_2 such that $n_1 > n_2$. The angle θ_2 here shown wont never reach a value equal to $asin\left(\frac{n_1}{n_2}\right)$.

Now, we have to consider and study the case where $n_1 < n_2$ the obvious case is when $\theta_1 = 0$ therefore $\theta_2 = 0$ and there wont be any refracted ray, but the interesting thing comes when $\theta_1 = \pi/2$

then again we have got the same equation 2.34 but here is the main difference, now we've got $n_1 < n_2$ or $n_1/n_2 < 1$ therefore there exists this angle and it is called the *total reflection angle* or *critical angle* θ_c this is illustrated in the figure 2.8.

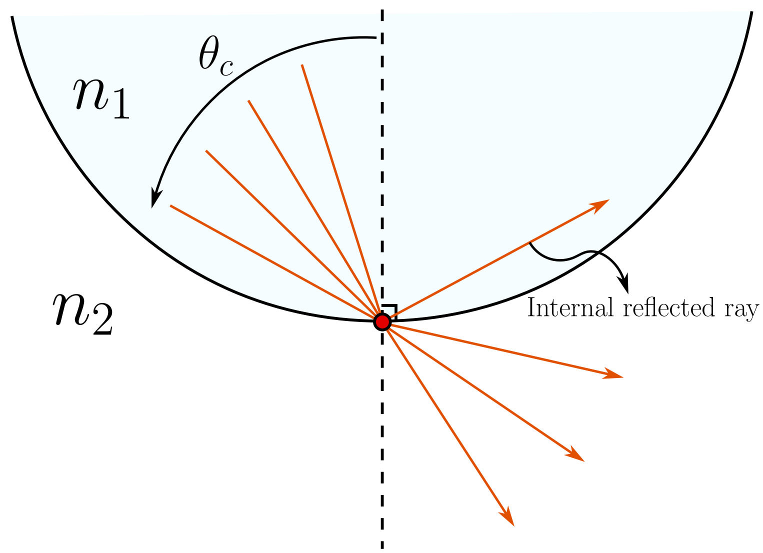


Figure 2.8: Representation of all the incident and refracted rays from a media with refraction index n_1 to another with refraction index n_2 such that $n_1 < n_2$. The angle θ_c here separates the rays that will be refracted , (angles less than θ_c), and the rays that will be totally reflected internally.

2.3 The Descartes's ovoid and diopters

From here on we are going to start to study the theory of image formation where the concepts of **object** and **image** are really important because there are real and virtual objects and image, this depends on the relation between these points ant the diopter, in fact, something is called virtual if we put a photo detector in that point and we wont measure any field if an object is to the left of a diopter is called real and if it is to the right is called virtual, in the same order of ideas if the image is to the left of the diopter it is virtual and if it is to the right is real.

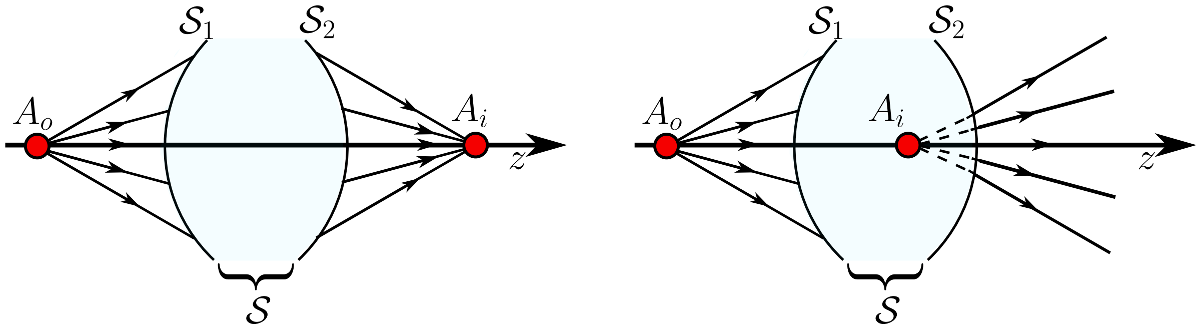
Before starting we have to give some fundamental definitions

Definition 5 *(Optical system) an optical system is a set of translucid media (diopters) and reflective media (catopters) and we say that the* ***system is centered*** *if all the diopters are revolution surfaces and shared the same symmetry axis.*

The systems conformed only by refractive surfaces all called ***dioptric systems*** *and the system conformed only by reflective surfaces are called* ***catoptric systems*** *and the systems formed by these two kind of surfaces are called* ***catadioptrics systems****.*

Definition 6 *(Real and virtual points) Given a optical system $\mathcal{S}$ we say that an object point A_o is real if it is to the left and outside the optical system and we way that the object point is virtual if it is to the right and inside the optical system.*

In the same way, we say that an image point A_i is real if it is to the right and outside the optical system, and we way that the image point is virtual if it is inside the optical system.



(a) Optical system $\mathcal{S}$ formed by two diopters $\mathcal{S}_1, \mathcal{S}_2$ where the point A_o is a real object and A_i is a real image point.

(b) Optical system $\mathcal{S}$ formed by two diopters $\mathcal{S}_1, \mathcal{S}_2$ where the point A_o is a real object and A_i is a virtual image point.

Figure 2.9: Representation of two optical systems where appear real object points, real and virtual image points.

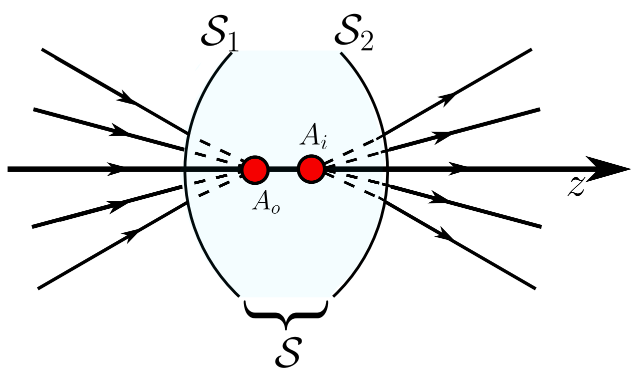


Figure 2.10: Optical system $\mathcal{S}$ formed by two diopters $\mathcal{S}_1, \mathcal{S}_2$ where the point A_o is a virtual object and A_i is a virtual image point.

To understand better these two definitions lets follow the figures 2.9 there we can see two examples of optical systems. In the first one we can see an object point located to the left and outside of the system $\mathcal{S}$ so it is a real object point, and the point A_i is located to the right and outside the optical system, therefore it is a real image point. In the second optical system we can see that the same real object point and that the point A_i is located inside the optical system, therefore it is a virtual image point for A_o.

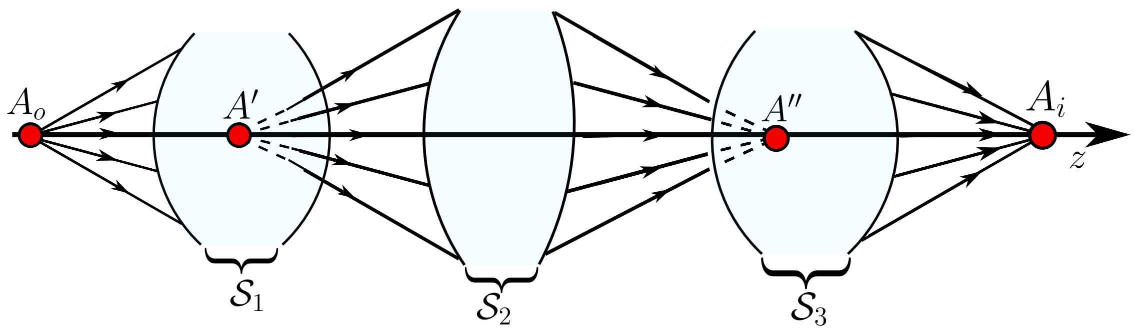


Figure 2.11: Optical system $\mathcal{S}$ formed by three optical systems $\mathcal{S}_1, \mathcal{S}_2, \mathcal{S}_3$ where the point A_o is a real object point, A' is a virtual image point for $\mathcal{S}_1$, but a real object point for $\mathcal{S}_2, \mathcal{S}_3$, A'' is a virtual image for the real object A' and A_i is the real image of the virtual object A''.

In the figure 2.10 we can see two diopter $\mathcal{S}_1, \mathcal{S}_2$ which forms the optical system $\mathcal{S}$ where the point A_o is located inside the optical system, then it behaves as a virtual object point and A_i is to the right of A_o and also inside the optical systen, therefore it is a virtual image point.

In the optical system 2.11 we can see three optical systems $\mathcal{S}_1, \mathcal{S}_2, \mathcal{S}_3$ and a configuration of many points. The first one, A_o is completely to the left and outside any optical system, so it is a real object point, and the point A' is inside the optical system $\mathcal{S}_1$ so it is the **the virtual image of the real object** A_o but, as we can see, A' is to the left and outside the optical system $\mathcal{S}_2$ so it is a real object point for the rest of the total optical system. The point A'' is inside the optical system $\mathcal{S}_3$ so it is the virtual image of the real object A', and in the very last there is the point A_i which is completely to the right and outside any optical system, so it is the real image point of the virtual object point A''.

Definition 7 *(Stigmatism) From a **geometrical point of view** we say that an optical system $\mathcal{S}$ is **stigmatic** if for any given point source, all the rays that pass through $\mathcal{S}$ converge to a point source. In other words, a optical system $\mathcal{S}$ is stigmatic if for any object point there is one unique image point.*

We can also, define the stigmatism from a wavefront perspective. An rigorously stigmatic system is one that transforms an spherical wavefront $\mathcal{A}$ to a new spherical wavefront $\mathcal{A}'$ whose curvature centers, A_o and A_i are the rigorously stigmatic points (see figure 2.12). Thus, the spherical wavefront $\mathcal{A}'$ is the image of the spherical wavefront $\mathcal{A}$.

Understanding that an image-forming optical system is one that maps each point source of an object's electromagnetic field to the point sources of the image's electromagnetic field, corresponding to a homothety between these fields, the interest arises in studying what these optical systems are and what their characteristics are. In general, these systems can be composed of multiple elements, but initially we will study the simplest configuration for two reasons:

1. because it most easily allows us to understand the principles of these systems.

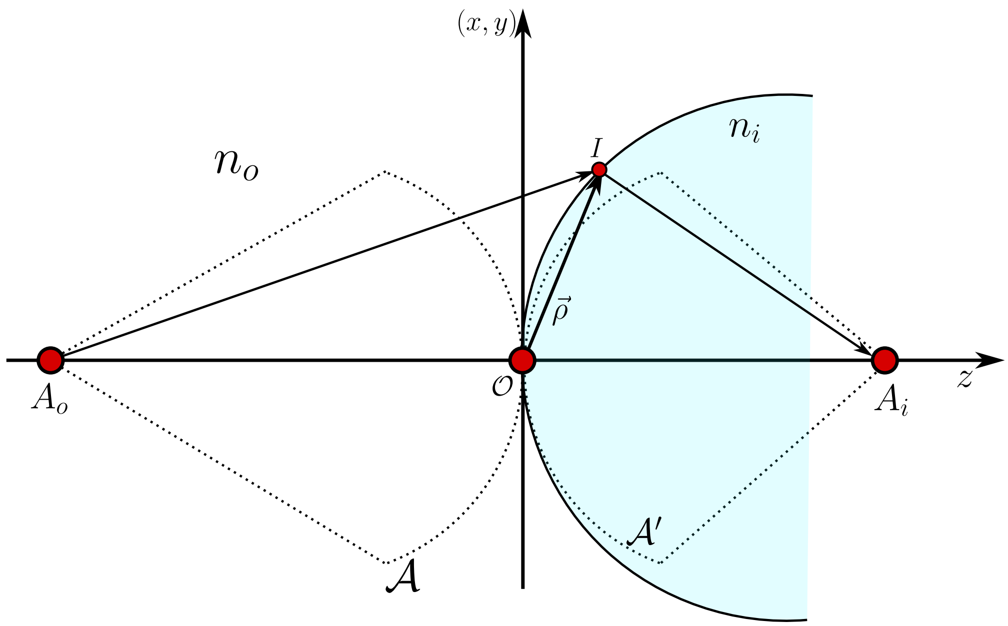


Figure 2.12: The stigmatic system transform the incident wavefront $\mathcal{A}$ to another new spherical wavefront $\mathcal{A}'$.

2. because later, more complex systems can be understood as compositions of these.

These elementary systems are made up of a single optical surface, either refractive or reflective, which is capable of perfectly focusing all the rays emitted by a point source into a perfect point image. In geometric optics, this means that all the rays travel the same optical path length. Initially, this property associated with a single pair of points (stigmatism) will be studied, and then the conditions under which this property becomes invariant to the translation of these points, which is known as aplanetism, will be studied.

Now we are going to develope the *rigurous stigmatism* thes are system capable of making perfect images, this means from a point source there is a perfect point image. Let a diopter which separates two media with refraction index n_o and n_i to the left of the diopter there is an object A_o at a point $[0, 0, z_o]$ the field from A_o reaches a point I, measured from the diopter's vertex v, from the point I the beam is refracted and forms an image A_i at a point $[0, 0, z_i]$ then we are going to find the relations between all these parameters to make all the rays from A_o arrive at A_i making a perfect image. From the Fermat's principle we have got

$$n_0 l_0 + n_i l_i = cte \tag{2.35}$$

the, as we can see in the figure 2.13 we then have

$$-n_0[x^2 + y^2 + (z - z_o)^2]^{1/2} + n_i[x^2 + y^2 + (z - z_i)^2]^{1/2} = cte \tag{2.36}$$

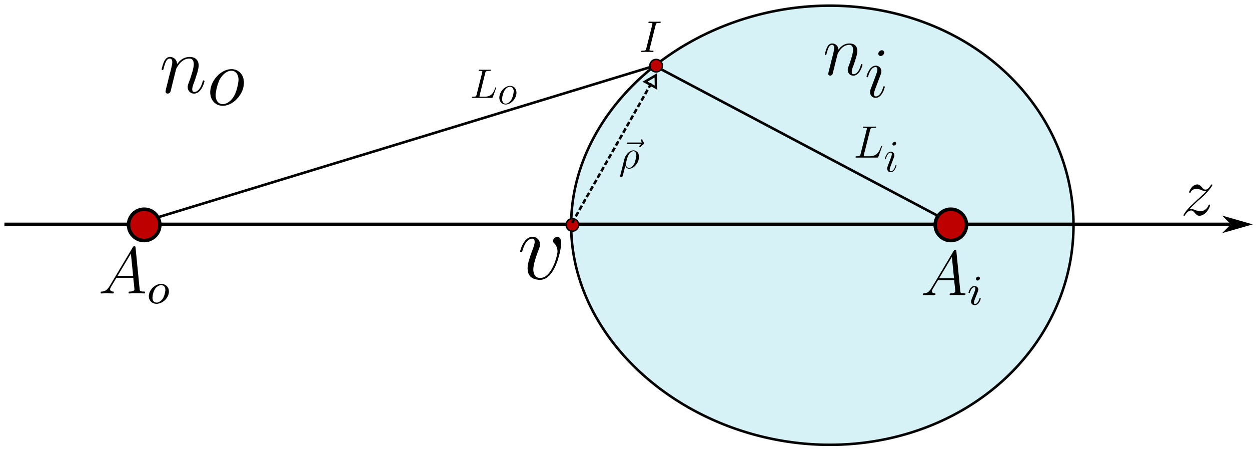


Figure 2.13: Scheme representation of the development of a rigorously stigmatic system, from a point source A_o at a position $[0, 0, z_o]$ the diopter with a vertice v forms a perfect image A_i at $[0, 0, z_i]$.

and, remembering our sign convention $\sqrt{z_o^2} = -zo$, $z_i^2 = z_i$ and to find the constant we can see that if we want to make all the rays from A_o arrive to A_i then the rays that travel parallel to z satisfy also the Fermat's principle, thus

$$n_i z_i - n_o z_o = cte, \tag{2.37}$$

$$n_i[x^2 + y^2 + (z - z_i)^2]^{1/2} - n_0[x^2 + y^2 + (z - z_o)^2]^{1/2} = n_i z_i - n_o z_o, \tag{2.38}$$

now, we are going to do the following change of variables given by

$$\rho^2 = x^2 + y^2 + z^2, \tag{2.39}$$

which represents the square of the vector from the vertex to the interface shown in 2.13, so, having this on count, the equation becomes

$$n_i \left[\rho^2 + z_i^2 - 2zz_i\right]^{1/2} - n_0 \left[\rho^2 + z_0^2 - 2zz_0\right]^{1/2} = n_i z_i - n_0 z_0. \tag{2.40}$$

Now, the main goal is to transform the equation 2.40 into a more explicit and useful algebraic equation, so, the first thing we're going to do is to square both side of the equation 2.40 and then we'll have

$$n_o^2 \left[\rho^2 + z_o^2 - 2zz_o\right] + n_i^2 \left[\rho^2 + z_i - 2zz_i\right] - 2n_o n_1 \left[\rho^2 + z_o^2 - 2zz_o\right]^{1/2} \left[\rho^2 + z_i^2 - 2zz_i\right]^{1/2} = (n_i z_i - n_o z_o)^2$$

Now, we're going to re-order all the equation for convenience as follows

$$n_o^2\left[\rho^2+z_o^2-2zz_o\right]+n_i^2\left[\rho^2+z_i-2zz_i\right]-(n_iz_i-n_oz_o)^2=2n_on_1\left[\rho^2+z_o^2-2zz_o\right]^{1/2}\left[\rho^2+z_i^2-2zz_i\right]^{1/2},$$

on the left hand side we have got the following

$$\begin{aligned}&n_o^2\left[\rho^2+z_o^2-2zz_o\right]+n_i^2\left[\rho^2+z_i-2zz_i\right]-(n_iz_i-n_oz_o)^2=\\&n_o^2\left[\rho^2+z_o^2-2zz_o\right]+n_i^2\left[\rho^2+z_i-2zz_i\right]-\left(n_i^2z_i^2+n_o^2z_o^2-2n_on_iz_oz_i\right)=\\&n_o^2\left[\rho^2-2zz_o\right]+n_i^2\left[\rho^2-2zz_i\right]+2n_on_iz_oz_i,\end{aligned}$$

where we just only simplified the common terms. Having this on count, the equation reduces to

$$n_o^2\left[\rho^2-2zz_o\right]+n_i^2\left[\rho^2-2zz_i\right]+2n_on_iz_oz_i=2n_on_1\left[\rho^2+z_o^2-2zz_o\right]^{1/2}\left[\rho^2+z_i^2-2zz_i\right]^{1/2}. \quad (2.41)$$

Now, we must square again, the whole equation as follows

$$\begin{aligned}&\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]^2+(2n_on_iz_oz_i)^2+4n_on_iz_oz_i\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]=\\&4n_o^2n_i^2\left(\rho^2+z_o^2-2zz_o\right)\left(\rho^2+z_i^2-2zz_i\right),\end{aligned}$$

now we are going to expand the product on the right hand side in the following way

$$\begin{aligned}&\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]^2+(2n_on_iz_oz_i)^2+4n_on_iz_oz_i\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]=\\&4n_o^2n_i^2\left(\rho^2-2zz_o\right)\left(\rho^2-2zz_i\right)+4n_o^2n_i^2\left[z_o^2\left(\rho^2-2zz_i\right)+z_i^2\left(\rho^2-2zz_o\right)+z_o^2z_i^2\right].\end{aligned}$$

Now, we must do a simplification, to do this, we must expand the first term of the left hand side like this

$$\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]^2=\left[n_o^2\left(\rho^2-2zz_o\right)\right]^2+\left[n_i^2\left(\rho^2-2zz_i\right)\right]^2+2n_o^2n_i^2\left(\rho^2-2zz_o\right)\left(\rho^2-2zz_i\right),$$

so the first term of the left hand side have on common in its expansion the same term than the expansion of the term on the right hand side, so, we could simplify then and look at the binomial is the same but with a minus sign instead of a plus sign, thus

$$\begin{aligned}&\left[n_o^2\left(\rho^2-2zz_o\right)-n_i^2\left(\rho^2-2zz_i\right)\right]^2+(2n_on_iz_oz_i)^2+4n_on_iz_oz_i\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]=\\&4n_o^2n_i^2\left[z_o^2\left(\rho^2-2zz_i\right)+z_i^2\left(\rho^2-2zz_o\right)+z_o^2z_i^2\right].\end{aligned}$$

Now, we are going to forget, momentarily the first term of the left hand side and focus on all the rest of the equations because there are going to be many important simplifications. Then, having this on count we'll have the following:

$$(2n_on_iz_oz_i)^2+4n_on_iz_oz_i\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]-4n_o^2n_i^2\left[z_o^2\left(\rho^2-2zz_i\right)+z_i^2\left(\rho^2-2zz_o\right)+z_o^2z_i^2\right]=0$$

$$4n_on_i\left[z_oz_in_o^2\left(\rho^2-2zz_o\right)+z_oz_in_i^2\left(\rho^2-2zz_i\right)-n_on_iz_o^2\left(\rho^2-2zz_i\right)-n_on_iz_i^2\left(\rho^2-2zz_o\right)\right.$$
$$\left.-n_on_izo^2z_i^2+n_on_iz_o^2z_i^2\right],$$

now, we have to factorize the common factors

$$\begin{aligned}
&4n_on_i\left[\left(\rho^2-2zz_o\right)\left(z_oz_in_o^2-n_on_iz_i^2\right)+\left(\rho^2-2zz_i\right)\left(z_oz_in_i^2-n_on_iz_o^2\right)\right]=\\
&4n_on_i\left[z_in_o\left(\rho 2-2zz_o\right)\left(z_on_o-n_iz_i\right)+z_on_i\left(\rho^2-2zz_i\right)\left(z_in_i-z_on_o\right)\right]=\\
&4n_on_i\left[n_iz_i-n_oz_o\right]\left[n_iz_o\left(\rho^2-2zz_i\right)-n_oz_i\left(\rho^2-2zz_o\right)\right]=\\
&4n_on_i\left[n_iz_i-n_oz_o\right]\left[\rho^2\left(n_iz_o-n_oz_i\right)+2zz_iz_o\left(n_o-n_i\right)\right].
\end{aligned}$$

So, joining this result with the first term we will have the following

$$\left[n_o^2\left(\rho^2-2zz_o\right)-n_i^2\left(\rho^2-2zz_i\right)\right]^2+4n_on_i\left[n_iz_i-n_oz_o\right]\left[\rho^2\left(n_iz_o-n_oz_i\right)+2zz_iz_o\left(n_o-n_i\right)\right]=0,$$

so, the last things we have to do are is re organize in the following way

$$\left[\left(n_i^2-n_o^2\right)\rho^2-2z\left(n_o^2z_o-n_i^2z_i\right)\right]^2-4n_in_o\left[n_iz_i-n_oz_o\right]\left[\left(n_oz_i-n_iz_o\right)\rho^2+2zz_oz_i\left(n_i-n_o\right)\right]=0. \tag{2.42}$$

The equation 2.42 is the algebraic expression for the Descartes Ovoid the rigorously stigmatic surface we wanted.

2.4 The spherical diopter

Now, we are going to see that the Descartes Ovoid 2.42 contains the information of all the other diopters known. So, the first one we are going to deduce is the spherical diopter this is obtained in a vicinity of the vertex shown in the figure 2.13. Therefore, to deduce this particular case, we are going to take the equation 2.42 and rewrite in the following convenient way

$$\left[\left(n_o^2-n_i^2\right)\rho^2+2zz_oz_i\left(\frac{n_i^2}{z_o}-\frac{n_o^2}{z_i}\right)\right]^2-4n_in_o\left[n_iz_i-n_oz_o\right]\left[\left(n_oz_i-z_iz_o\right)\rho^2+2zz_oz_i\left(n_i-n_o\right)\right]=0. \tag{2.43}$$

So, the main idea is to construct the sphere that is located in the vertex v. To do this, we must see that the squared term in 2.43 looks very similar to the right factor term in the second term, so, we want to equate them and then simplify, which leads us to the following equations

$$n_o^2-n_i^2=c\left(n_oz_i-n_iz_o\right), \tag{2.44}$$
$$\frac{n_i^2}{z_o}-\frac{n_o^2}{z_i}=c\left(n_i-n_o\right) \tag{2.45}$$

where c is a constant we must calculate. The second thing we are going to do, is to re-order the equation 2.44 in the following way

$$n_o^2 - cz_i n_o = n_i^2 - cz_o n_i, \tag{2.46}$$

and if we re-order the equation 2.45 we'll have

$$z_i n_i^2 - z_o n_o^2 = cz_o z_i (n_i - n_o),$$
$$z_i n_i^2 - cz_o z_i n_i = z_o n_o^2 - cz_o z_i n_o,$$
$$z_o \left(n_o^2 - cz_i n_o\right) = z_i \left(n_i^2 - cz_o n_i\right). \tag{2.47}$$

So if we use 2.46 on 2.47 we have got the following two equations

$$(z_o - z_i)\left(n_i^2 - cz_o n_i\right) = 0, \tag{2.48}$$
$$(z_o - z_i)\left(n_o^2 - cz_i n_o\right) = 0, \tag{2.49}$$

using the equations 2.48 and 2.49 we will have

$$z_o = \frac{n_i}{c}, \qquad z_i = \frac{n_0}{c}, \tag{2.50}$$

and using the equation 2.44 we will have

$$z_0 = z_i \qquad z_{0,i} = \frac{n_o + n_i}{c}. \tag{2.51}$$

Now, before going further analyzing this results, we have to use the equations 2.44, 2.45 on 2.43

$$\left[c\left(n_o z_i - n_i z_o\right)\rho^2 + 2zz_o z_i c(n_i - n_o)\right]^2 - 4n_i n_o\left[n_i z_i - n_o z_o\right]\left[\left(n_o z_i - z_i z_o\right)\rho^2 + 2zz_o z_i\left(n_o - n_o\right)\right] = 0,$$
$$\left[\left(n_o z_i - n_i z_o\right)c\rho^2 + 2zz_o z_i c(n_i - n_o)\right]\left[\left(n_o z_i - n_i z_o\right)\rho^2 + 2zz_o z_i(n_i - n_o) - 4n_i n_o\left(n_i z_i - n_o z_o\right)\right] = 0,$$

and by the fundamental theorem of Algebra, we have got the equation for the spherical diopter

$$\left(n_o z_i - n_i z_o\right)\rho^2 + 2zz_o z_i(n_i - n_o) = 0. \tag{2.52}$$

2.4.1 The Young-Weirstrass points

From the case 2.50 we have that

$$n_i z_i - n_o z_o = 0,$$

and using the equation 2.52 we will have

$$\rho^2\left(n_o^2 - n_i^2\right) + 2czz_o n_i(n_i - n_o) = 0, \tag{2.53}$$

now, dividing by $n_o^2 - n_i^2$ and then we must complete squares as follows

$$x^2+y^2+z^2+\frac{2czz_oz_i}{n_i+n_o}=x^2+y^2+z^2+\frac{2czz_oz_i}{n_i+n_o}+\left(\frac{cz_oz_i}{n_i+n_o}\right)^2=\left(\frac{cz_oz_i}{n_i+n_o}\right)^2,$$

$$x^2+y^2+\left(z-c\frac{z_oz_i}{n_o+n_i}\right)^2=c^2\frac{z_o^2z_i^2}{(n_i+n_o)^2}, \tag{2.54}$$

which corresponds to the equation of an sphere whose radius is given by

$$R=c\frac{z_oz_i}{n_o+n_i}\longrightarrow x^2+y^2+(z-R)^2=R^2, \tag{2.55}$$

and using 2.50 $z_o=n_i/c,\ z_i=n_o/c$ we will have

$$R=c\frac{z_in_i}{c(n_o+n_i)}\longrightarrow z_i=R\left(1+\frac{n_o}{n_i}\right), \tag{2.56}$$

$$R=c\frac{z_on_o}{c(n_o+n_i)}\longrightarrow z_i=R\left(1+\frac{n_i}{n_o}\right) \tag{2.57}$$

which are known as **Young-Weirstrass** points or the aplanetic points. One of these points is inside and the other outside the sphere as is shown in 2.14.

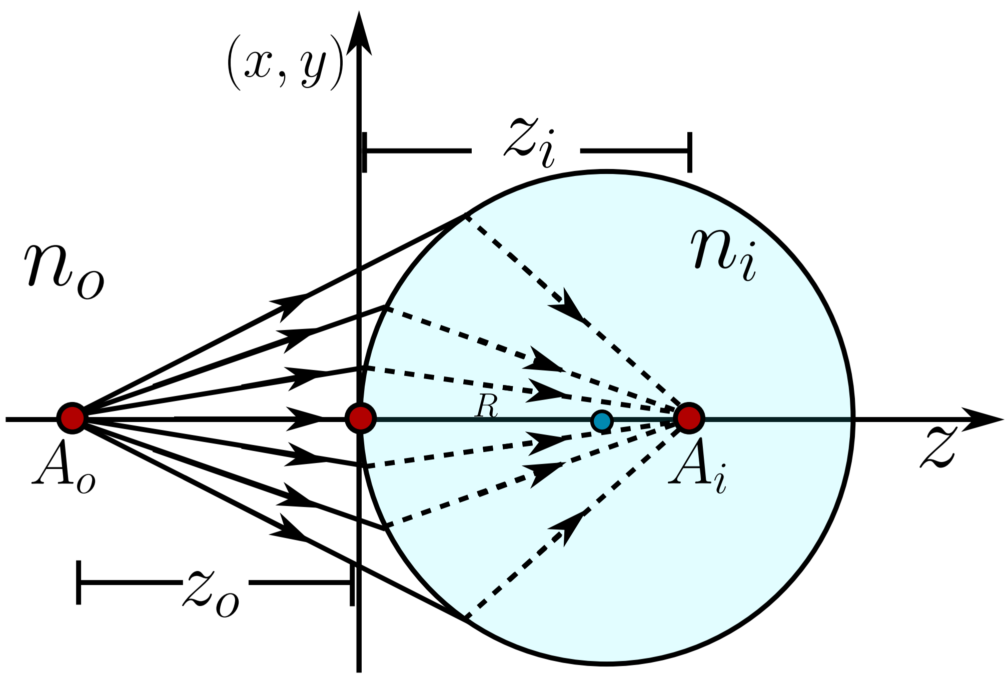


Figure 2.14: The spherical Diopter for the first case, where the object point is real and it is outside the sphere but the image point is virtual and inside the diopter.

2.4.2 The second spherical diopter

So, for the second case we have 2.51 and the equation 2.52 then let z_c given by

$$z_c = z_o = z_i = \frac{n_i + n_o}{c},$$

then we will have

$$\rho^2 (n_o z_c - n_i z_c) + 2z z_c^2 (n_i - n_o) = 0, \tag{2.58}$$
$$\rho^2 - 2z z_c = 0, \tag{2.59}$$

again we must complete squares in the following way

$$x^2 + y^2 + z^2 - 2z z_c + z_c^2 = z_c^2, \tag{2.60}$$
$$x^2 + y^2 + (z - z_c)^2 = z_c^2. \tag{2.61}$$

The equation 2.61 is an spherical diopter where the object point is on the center and the image point is also on the center.

2.5 Objects and images to the infinity

Some other cases arise when either the object or the image is at infinity, which is a convenient condition when the object is very far away, for example, in telescopes.

But even in microscopy, systems are designed to form the image at infinity, since under these conditions, an eyepiece facilitates visual accommodation for observation.

For an object in the infinity we divide by z_o^2 and take the limit $z_o \to -\infty$ in 2.42 in the following way

$$\lim_{z_o \to -\infty} \left[\frac{\rho^2 \left(n_o^2 - n_i^2\right)}{z_o} + 2z \frac{n_o^2 z_o + n_i^2 z_i}{z_o} \right]^2 - \lim_{z_o \to -\infty} \frac{4 n_o n_i \left(n_o z_o + n_i z_i\right)}{z_o} \left[\frac{\rho^2 \left(n_o z_i + n_i z_o\right)}{z_o} - \frac{2z z_o z_i (n_i - n_o)}{z_o} \right] = 0,$$
$$(2z n_o^2)^2 - 4 n_o 2 n_i^2 \left[\rho^2 n_i - 2z z_i (n_i - n_o)\right] = 0,$$
$$4z^2 n_o^4 - 4 n_o^2 n_i^2 \rho^2 + 8 z z_i n_o^2 n_i^2 - 8 z z_i n_o^3 n_i = 0, \qquad \text{Dividing all the equation by } 4n_o^2,$$
$$z^2 \left(n_o^2 - n_i^2\right) + 2z n_i z_i (n_i - n_o) - n_i (x^2 + y^2) = 0,$$
$$[n_o - n_i] \left[z^2 (n_o + n_i) - 2z n_i z_i\right] - n_i (x^2 + y^2) = 0.$$

Now, to complete the algebraic expression we must multiply all the equation by $(n_o + n_i)$ and complete the squares as follows

$$
\begin{aligned}
&(n_o - n_i)\left[(n_o+n_i)(n_o+n_i)z^2 - 2zn_iz_i(n_o+n_i)\right] - n_i^2(n_o+n_i)(x^2+y^2) = 0,\\
&(n_o - n_i)\left[(n_o+n_i)^2z^2 - 2zn_iz_i(n_o+n_i) + n_i^2z_i^2 - n_i^2(n_o+n_i)(x^2+y^2)\right] = (n_o-n_i)n_i^2z_i^2\ ,\\
&(n_o - n_i)\left[(n_o+n_i)z - n_iz_i\right]^2 - n_i^2(n_o+n_i)(x^2+y^2) = (n_o-n_i)n_i^2z_i^2
\end{aligned}
$$

and dividing all by $(n_o - n_i)n_i^2z_i^2$ we will have

$$\left(\frac{n_i+n_o}{n_iz_i}\right)^2\left[z - \frac{n_iz_i}{n_i+n_o}\right]^2 + \frac{n_i+n_o}{z_i^2(n_i-n_o)}(x^2+y^2) = 1. \tag{2.62}$$

The result 2.62 represents a conical section of revolution around the z-axis from this equation we are going to obtain the Ellipsoid and the hyperboloid.

2.5.1 Revolution Ellipsoid

if $n_i > n_o$ we are going to define the following scalars

$$a = \left(\frac{n_i-n_o}{n_i+n_o}\right)^{1/2}|z_i|, \qquad b = \left(\frac{n_i}{n_i+n_o}\right)|z_i|, \tag{2.63}$$

and notice that $b > a$ so we will have **prolate ellipsoid** (see figure 2.15 given by

$$\frac{1}{b^2}(z-b)^2 + \frac{1}{a^2}(x^2+y^2) = 1. \tag{2.64}$$

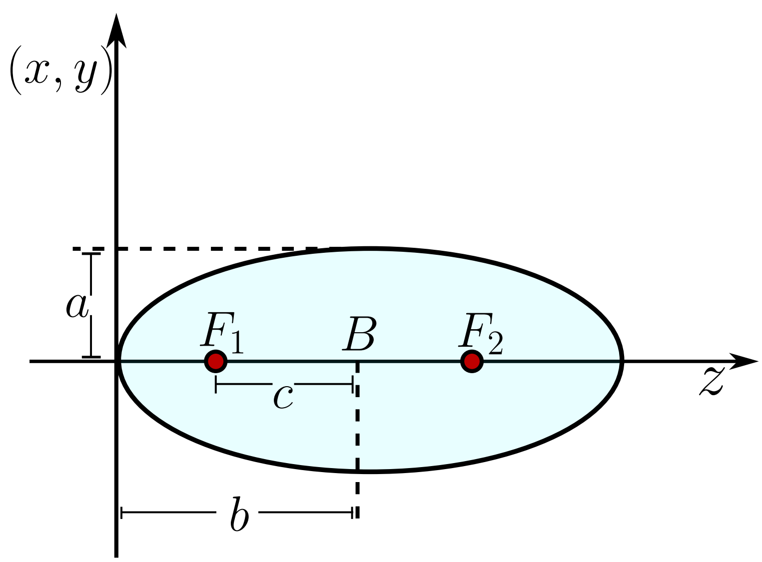


Figure 2.15: Prolate Ellipsoid centered and $z = B$.

We have got two cases, when $z_i > 0$ and $z_i < 0$. For both cases what is next is to study the eccentricity of the ellipse and the foci of the ellipsoids.

When $z_i > 0$ lets define the segment $c = \overline{F_1 B}$ such that the eccentricty is given by

$$\epsilon = \frac{c}{b} = \frac{\sqrt{b^2 - a^2}}{b} = \sqrt{1 - \frac{a^2}{b^2}} = \sqrt{\frac{n_o^2}{n_i^2}} = \frac{n_o}{n_i} < 1, \tag{2.65}$$

therefore the foci F_1 is centered to a distance

$$f_1 = b - c = b - \epsilon b = (1 - \epsilon) b = \frac{n_i - n_o}{n_i + n_o} z_i, \tag{2.66}$$

and the second foci is at

$$f_2 = b + c = b + \epsilon b = (1 + \epsilon) b = z_i. \tag{2.67}$$

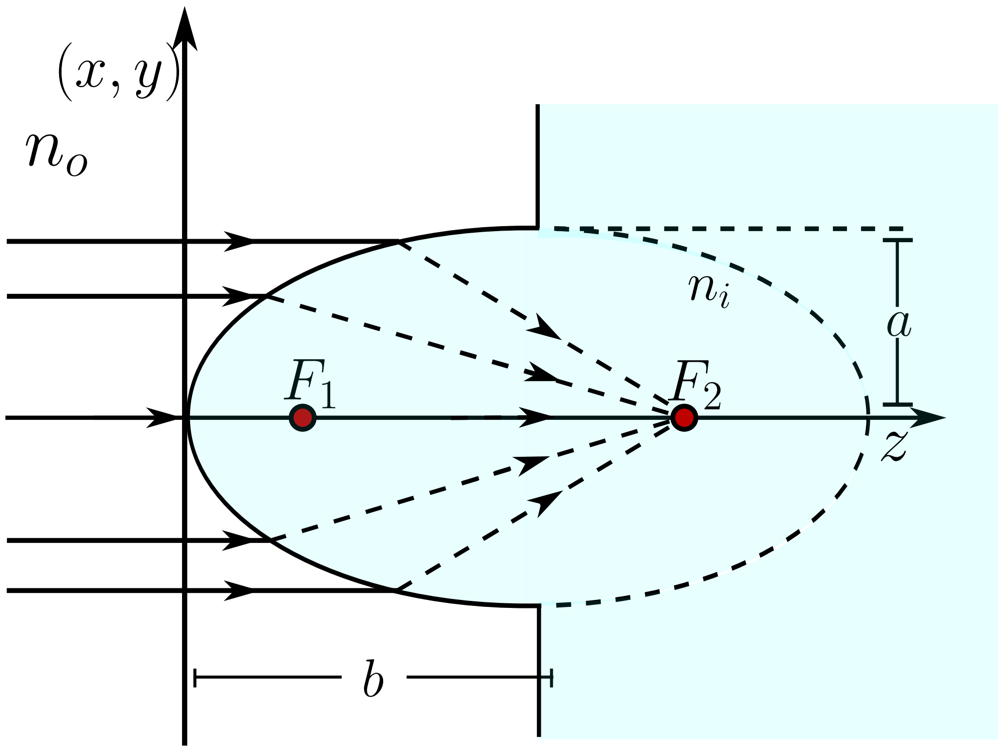


Figure 2.16: Ellipsoid diopter when $n_i > n_o$.

This means, that for this first case that all the rays that comes parallel to the z-axis from infinity converges to the second Foci of the ellipsoid.

If $z_i < 0$ by the convention we have that $\sqrt{z_i^2} = -z_i$ and we define

$$b = -\frac{n_i}{n_i + n_o} z_i, \tag{2.68}$$

then the conical section equation 2.62 becomes

$$\frac{1}{b^2}(z+b)^2 + \frac{1}{a^2}(x^2+y^2) = 1. \tag{2.69}$$

Again, we define $c = \overline{F_1 B}$ so the eccentricty is given by

$$\epsilon = \frac{c}{b}\frac{\sqrt{b^2-a^2}}{b} = \sqrt{1-\frac{a^2}{b^2}} = \frac{n_o}{n_i} < 1,$$

and the foci is now going to be to a distance

$$f_1 = -b - c = -b - \epsilon b = -b(1+\epsilon) = z_i, \tag{2.70}$$

and the second one is at

$$f_2 = -b + c = -b + \epsilon b = (\epsilon - 1)b = \frac{n_i - n_o}{n_i + n_o} z_i. \tag{2.71}$$

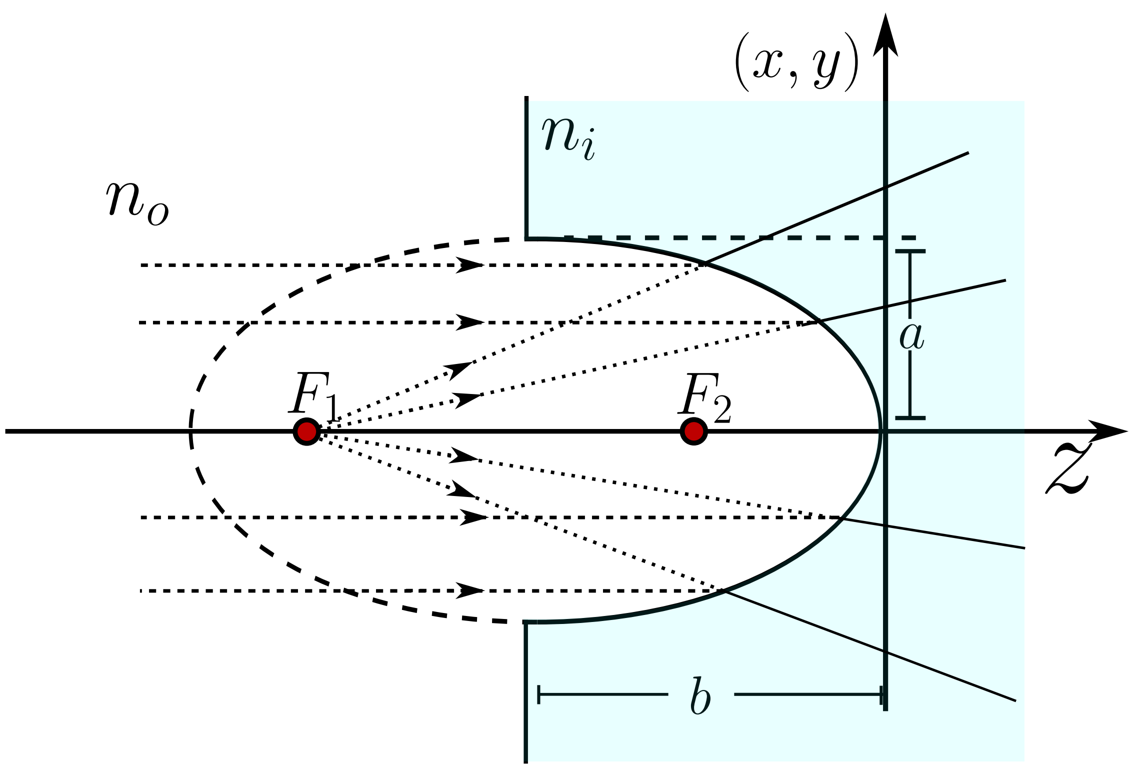


Figure 2.17: Ellipsoid diopter when $n_i > n_o$ and $z_i < 0$.

Having all these results on count, we can see in figure 2.17 that all the rays that becomes parallel to the z-axis and from infinity diverges as they would come from a point source from F_1.

2.5.2 Revolution Hyperboloid

In the case $n_i < n_o$ we have got that the equation 2.62 becomes an hyperboloid and, again, there are two possibilities, $z_i < 0, z_i > 0$, so the conical equation becomes

$$\left(\frac{n_i + no}{n_i z_i}\right)^2 \left[z - \frac{n_i z_i}{n_i + n_o}\right] - \frac{n_i + n_o}{z_i^2 (n_o - n_i)}(x^2 + y^2) = 1. \tag{2.72}$$

If $z_i > 0$ we define again the following scalars

$$b^2 = \frac{n_i^2}{(n_i + n_o)^2} z_i^2,$$
$$a^2 = \frac{n_o - n_i}{(n_i + n_o)^2} z_i^2,$$

so the equation of the hyperboloid (see figure 2.18) becomes

$$\frac{1}{b^2}(z - b)^2 - \frac{1}{a^2}(x^2 + y^2) = 1. \tag{2.73}$$

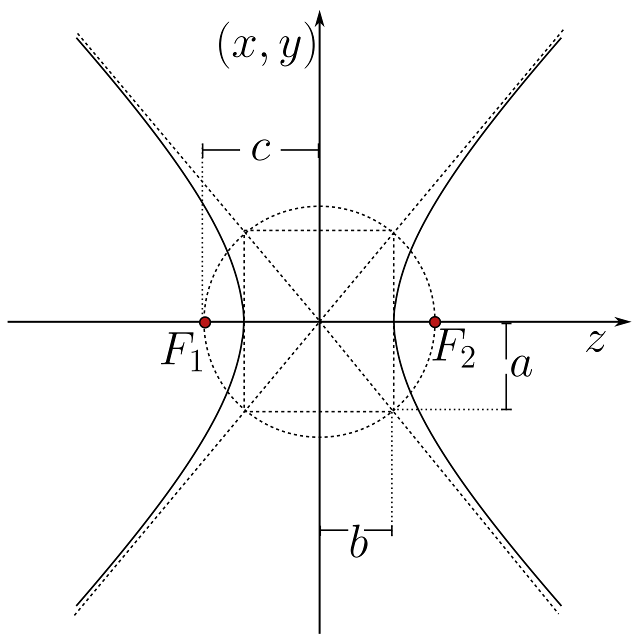


Figure 2.18: Geometrical representation of the two leaf hyperboloid with the parameters a, b and foci F_1 and F_2.

Again, to give a characterization of the conical surface we are going to define the segment $c = \overline{F_1 B}$ such that the eccentricity is given by

$$\epsilon = \frac{c}{b} = \frac{\sqrt{a^2 + b^2}}{b} = \sqrt{1 + \frac{a^2}{b^2}} = \frac{n_o}{n_i} > 1.$$

Thus the two Foci F_1 and F_2 are located at the distances

$$f_1 = b - c = b - \epsilon b = (1-\epsilon)b = \frac{n_i - n_o}{n_o + n_i} z_o, \tag{2.74}$$

$$f_2 = b + c = b + \epsilon b = (1+\epsilon)b = z_i. \tag{2.75}$$

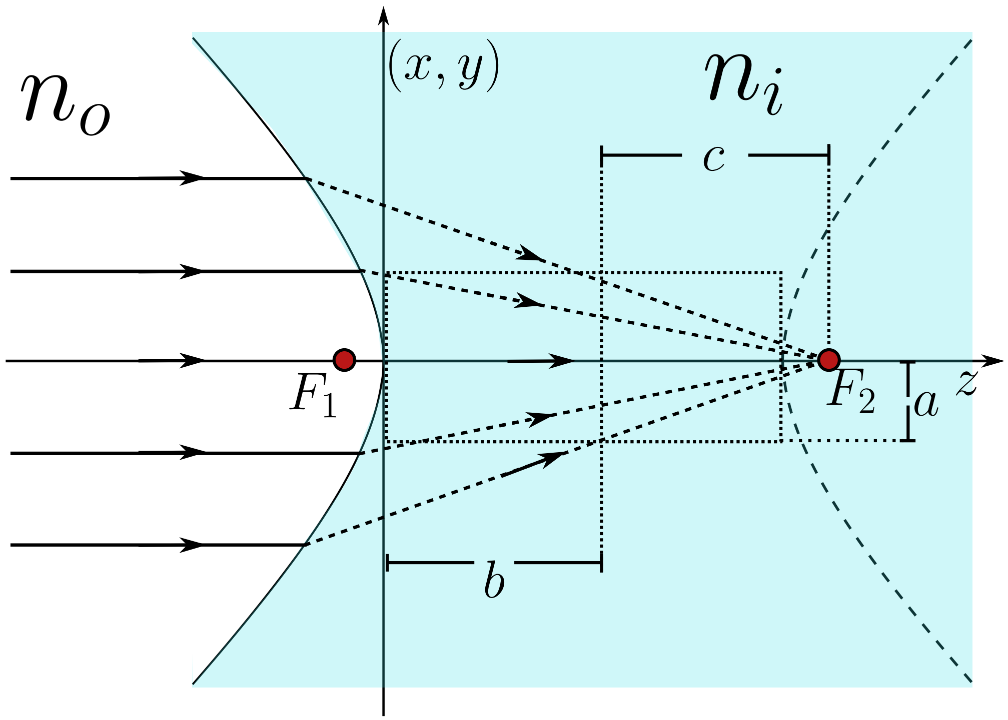


Figure 2.19: Geometrical representation of the first case of a two leaf hyperboloid.

In the figure 2.19 we can see the geometrical behavior of this first case, all the rays comes from the infinity and converges to a point which is located in the second foci F_2.

Now, for the second case, if $z_i < 0$ we define

$$b = -\frac{n_i}{n_i + n_o} z_o,$$

in such way that $b > 0$ with this we write

$$\frac{1}{b^2}(z+b)^2 - \frac{1}{a^2}(x^2+y^2) = 1.$$

Again, we define the segment $c = \overline{F_1 B}$ such that the eccentricity is given by

$$\epsilon = \frac{c}{b} = \frac{\sqrt{a^2+b^2}}{b} = \sqrt{1 + \frac{a^2}{b^2}} = \frac{n_o}{n_i} > 1.$$

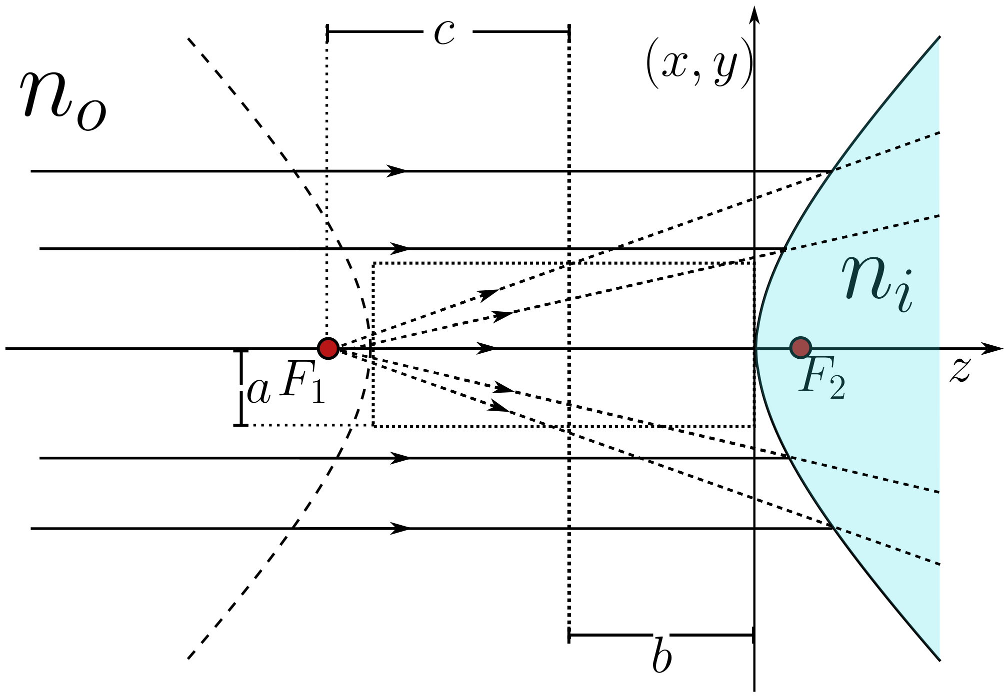


Figure 2.20: Geometrical representation of the first case of a two leaf hyperboloid.

As we can see in the figure 2.20 all the rays come from the infinity and interact with the hyperboloid diopter an the rays refracted in such geometrical way that they behave as they come from a object point source located in the foci of the first leaf at F_1.

2.6 Rigurously stigmatic surfaces by reflection

The reflective surfaces are the main components in the morden giant telescopes, for example, the James Webb telecopes is an array of many reflective surfaces which are electronically displaced such that the image formation gets the least aberration possible. Also, the reflective surfaces are better than the refractive ones in certain cases, because the optical systems can be smaller and don't have chromatic aberrations.

Tradionally, the reflective surface are developed indepentdenly from the refractive surface. Under this formulations, the rigurously stigmatic surfaces by reflection can be obtained doing $n_i = -n_o$, where the Descartes ovals become a conic surface. To this case, the equation 2.42 becomes

$$\begin{aligned}
&\left\{\left[(-n_o)^2 - n_o^2\right]\rho^2 - 2z\left[\left(n_o^2 z_o - (-n_o)^2 z_i\right]\right\}^2 \\
&- 4(-n_o)n_o\left[(-n_o)z_i - n_o z_o\right]\left\{\left[n_o z_i - (-n_o)z_o\right]\rho^2 + 2zz_oz_i\left[(-n_o) - n_o\right]\right\} = 0,
\end{aligned}$$

simplyfing we've got

$$\left\{-2zn_o^2\left[z_o-z_i\right]\right\}^2-4n_o^3\left[z_i+z_o\right]\left\{n_o\left[z_i+z_o\right]\rho^2-4zz_oz_in_o\right\}=0,$$
$$4z^2n_o^4\left(z_o-z_i\right)^2-4n_o^4\left(z_o+z_i\right)^2\rho^2+16n_o^4zz_oz_i\left(z_o+z_i\right)=0,$$

dividing all by $-16n_o^4z_oz_i$ an having on count that $\rho^2=x^2+y^2+z^2$ we've got

$$\frac{(z_o+z_i)^2}{4z_oz_i}\left(x^2+y^2\right)+\frac{(z_o-z_i)^2}{4z_oz_i}z^2-\frac{(z_o+z_i)^2}{4z_oz_i}z^2-z(z_o+z_i)=0,$$
$$\frac{(z_o+z_i)^2}{4z_oz_i}\left(x^2+y^2\right)+z^2-z(z_o+z_i)=0,$$

and completing the squares we've got

$$\left(z-\frac{z_o+z_i}{2}\right)^2+\frac{(z_o+z_i)^2}{4z_oz_i}\left(x^2+y^2\right)=\left(\frac{z_o+z_i}{2}\right)^2, \tag{2.76}$$

which is precisely the equation of a conic of revolution around the z axis, exactly as it happened with the refractive surfaces, only that now the two indices have collapsed into a single reflective condition. The whole family of stigmatic mirrors is hidden inside 2.76, and the sign of the product z_oz_i is what decides which conic we are looking at. When $z_oz_i>0$ the coefficient multiplying x^2+y^2 is positive and the surface closes onto itself as an ellipsoid, whereas when $z_oz_i<0$ that coefficient turns negative and the surface opens as a hyperboloid. There is also a beautiful limit waiting in 2.76, because if we push the object to infinity, $z_o\to-\infty$, the conic degenerates into a paraboloid, which is the case that rules every telescope. We are going to walk through each of these situations, since each one corresponds to a real optical instrument.

2.6.1 The ellipsoidal mirror

Lets start with $z_oz_i>0$, so that the object and the image sit on the same side of the vertex and the mirror is an ellipsoid. Dividing 2.76 by $\left(\frac{z_o+z_i}{2}\right)^2$ the equation takes the canonical shape

$$\frac{\left(z-\frac{z_o+z_i}{2}\right)^2}{\left(\frac{z_o+z_i}{2}\right)^2}+\frac{x^2+y^2}{z_oz_i}=1, \tag{2.77}$$

and if we baptise the two semi-axes as

$$b^2=\frac{(z_o+z_i)^2}{4},\qquad a^2=z_oz_i, \tag{2.78}$$

we recognise a prolate ellipsoid centered at $z=b=\frac{z_o+z_i}{2}$, that is

$$\frac{1}{b^2}\left(z-b\right)^2+\frac{1}{a^2}(x^2+y^2)=1. \tag{2.79}$$

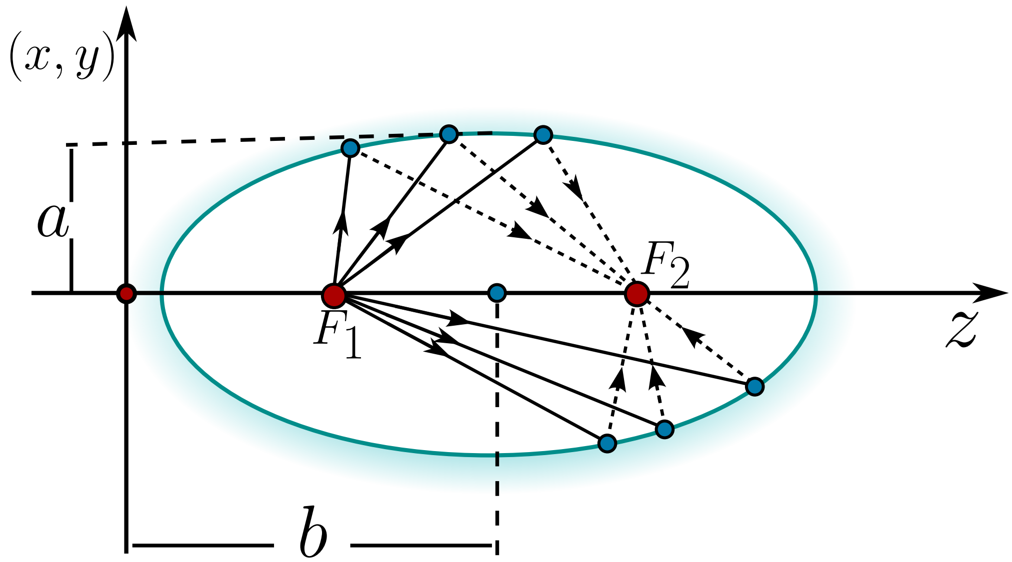


Figure 2.21: Ellipsoidal mirror. The object F_1 and the image F_2 coincide with the two foci of the ellipse, so every ray leaving one focus is reflected exactly onto the other.

That the ellipsoid comes out prolate, and not oblate, is not an accident, because

$$b^2 - a^2 = \frac{(z_o + z_i)^2}{4} - z_o z_i = \frac{(z_o - z_i)^2}{4} \geq 0,$$

so the semi-axis along the optical axis is always the largest one and the mirror is stretched along z. The distance from the center to each focus is then $c = \sqrt{b^2 - a^2} = \frac{|z_o - z_i|}{2}$, and the eccentricity becomes

$$\epsilon = \frac{c}{b} = \frac{\sqrt{b^2 - a^2}}{b} = \sqrt{1 - \frac{a^2}{b^2}} = \frac{|z_o - z_i|}{z_o + z_i} < 1, \tag{2.80}$$

which confirms once more that we are dealing with an ellipse. The really interesting thing is where the foci land, because measured from the vertex they sit at $z = b \pm c$, that is

$$b - c = \frac{z_o + z_i}{2} - \frac{|z_o - z_i|}{2}, \qquad b + c = \frac{z_o + z_i}{2} + \frac{|z_o - z_i|}{2}, \tag{2.81}$$

and no matter which of the two conjugate points is the larger one, these two numbers are nothing but z_o and z_i themselves. **The two foci of the ellipsoid are exactly the object point and the image point.**

This is the whole magic of the ellipsoidal mirror and the reason it is rigorously stigmatic. An ellipse has the geometric property that any ray leaving one focus is reflected straight into the other focus, because the sum of the two focal distances is constant along the whole curve, and a constant optical path is exactly the Fermat's principle at work. Placing a point source at one focus therefore produces a perfect point image at the other one, with no aberration at all. This is the principle behind the elliptical reflectors used to concentrate the light coming from one focus onto a detector sitting at the other, and it is the same property that makes the whispering galleries carry a voice from one focus of the room to the other.

When both z_o and z_i are negative the algebra is identical once we define $b = -\frac{z_o+z_i}{2} > 0$, so that the ellipsoid is now centered at $z = -b$ and its equation reads $\frac{1}{b^2}(z+b)^2 + \frac{1}{a^2}(x^2+y^2) = 1$. The eccentricity is the same and the foci again fall on z_o and z_i, the only difference being that both conjugate points now live on the opposite side of the vertex.

2.6.2 The hyperboloidal mirror

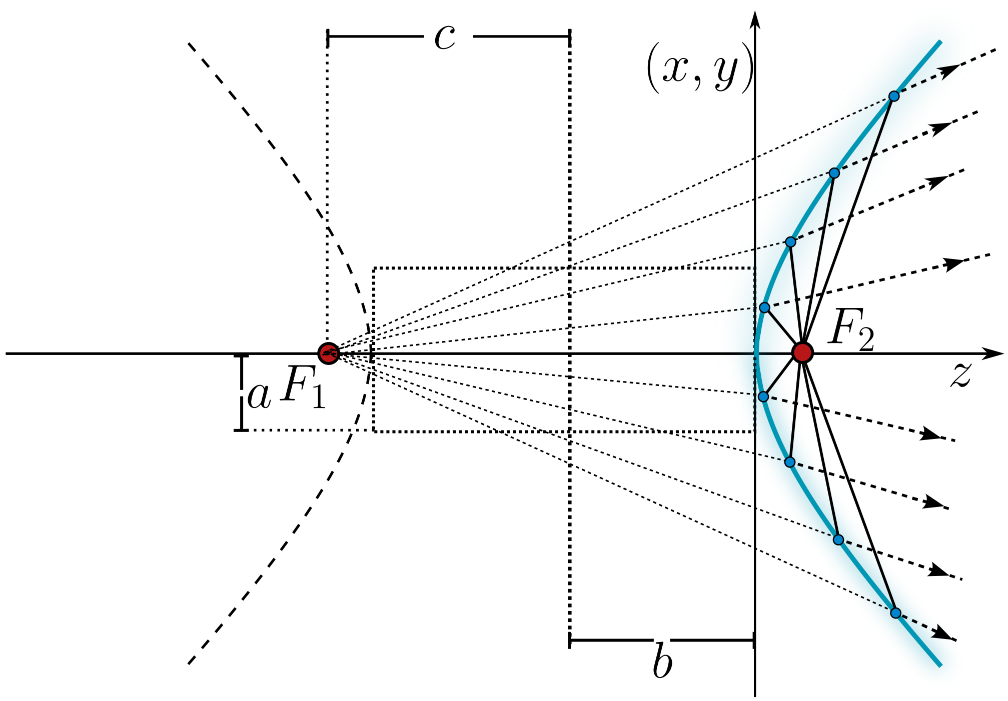


Figure 2.22: Hyperboloid mirror. The rays from the object F_1 diverge as they were emited by a virtual image locaed in F_2.

Now we let the product change sign, $z_o z_i < 0$, which happens when the object and the image sit on opposite sides of the mirror and one of them is virtual. The coefficient of $x^2 + y^2$ in 2.76 becomes negative, so writing $a^2 = -z_o z_i > 0$ while keeping $b^2 = \frac{(z_o+z_i)^2}{4}$ the canonical form is now

$$\frac{1}{b^2}(z-b)^2 - \frac{1}{a^2}(x^2+y^2) = 1, \tag{2.82}$$

a hyperboloid of two sheets around the optical axis. The focal distance is measured with a plus sign this time, $c = \sqrt{a^2+b^2}$, and a short computation gives again $c = \frac{|z_o-z_i|}{2}$, so the eccentricity is

$$\epsilon = \frac{c}{|b|} = \frac{\sqrt{a^2+b^2}}{|b|} = \sqrt{1+\frac{a^2}{b^2}} = \frac{|z_o-z_i|}{|z_o+z_i|} > 1, \tag{2.83}$$

which is greater than one exactly because $z_o z_i < 0$ forces $|z_o - z_i| > |z_o + z_i|$. And just as before, the foci $z = b \pm c$ collapse onto z_o and z_i, so once more the two conjugate points are the two foci of the conic.

The hyperboloidal mirror is therefore stigmatic between its two foci as well, but with a twist, because since the foci lie on opposite sheets one of the conjugate points is always virtual. This is precisely the behavior exploited by the convex secondary mirror of a Cassegrain telescope, where the light that is already converging toward the focus of the primary mirror is intercepted by a hyperboloidal secondary and redirected, without adding any aberration, to a more comfortable focus behind the primary. The stigmatic pair of the hyperboloid is what lets these compact folded telescopes keep a perfect image on the axis.

2.6.3 Special cases: the spherical and the plane mirror

Two very familiar mirrors are hiding inside 2.76 as degenerate cases. The first one appears when the object and the image coincide, $z_o = z_i = R$. The coefficient of $x^2 + y^2$ becomes $\frac{(2R)^2}{4R^2} = 1$ and the conic reduces to $z^2 - 2Rz + x^2 + y^2 = 0$, and completing the square by adding R^2 to both sides we get

$$(z - R)^2 + x^2 + y^2 = R^2, \tag{2.84}$$

which is simply a sphere of radius R centered at $z = R$. This tells us that a spherical mirror is rigorously stigmatic for one very special pair of points, both of them sitting together at the center of the sphere, since a source placed at the center sends every ray along a radius, the ray strikes the surface perpendicularly and comes straight back on itself, forming a perfect image right on top of the source. For any other position the sphere is only approximately stigmatic, and this is the origin of the spherical aberration that spoils cheap mirrors and that the conics above were designed to cure.

The second degenerate case is $z_o = -z_i$, for which the sum $z_o + z_i$ vanishes. Both the linear term and the whole right hand side of 2.76 die at once, and the equation collapses to $z^2 = 0$, that is, the plane $z = 0$. We recover the ordinary plane mirror, which is stigmatic for every point and always produces an image symmetric to the object at $z_i = -z_o$, which is nothing more than the everyday statement that a plane mirror puts your reflection as far behind the glass as you stand in front of it.

2.6.4 The parabolic mirror and objects at infinity

The most important mirror of all comes from letting the object recede to infinity, $z_o \to -\infty$, which is the situation of a telescope pointed at a star. Going back to 2.76 and expanding the square on the left, the terms $\frac{(z_o+z_i)^2}{4}$ appear on both sides and cancel,

$$z^2 - z(z_o + z_i) + \frac{(z_o + z_i)^2}{4} + \frac{(z_o + z_i)^2}{4 z_o z_i}(x^2 + y^2) = \frac{(z_o + z_i)^2}{4},$$

leaving the expanded conic

$$z^2 - z(z_o + z_i) + \frac{(z_o + z_i)^2}{4 z_o z_i}(x^2 + y^2) = 0.$$

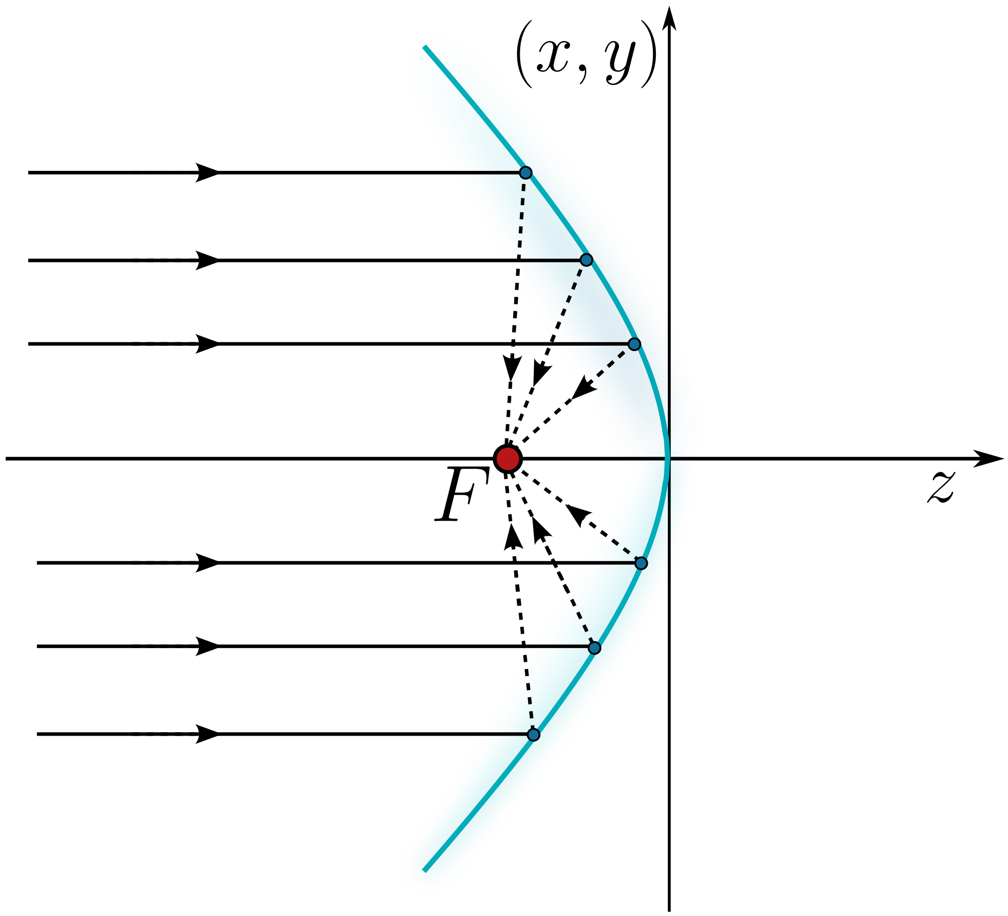


Figure 2.23: Parabolic mirror obtained as the limit $z_o \to -\infty$. The rays coming parallel to the axis from an object at infinity are all focused onto the single point $z = z_i$.

Dividing everything by z_o and taking the limit $z_o \to -\infty$, the term z^2/z_o vanishes, the factor $\frac{z_o+z_i}{z_o} \to 1$ and $\frac{(z_o+z_i)^2}{4z_o^2 z_i} \to \frac{1}{4z_i}$, so we are left with the remarkably clean surface

$$x^2 + y^2 = 4z_i z, \tag{2.85}$$

a paraboloid of revolution whose opening is fixed by the sign of z_i.

The paraboloid is the crown jewel of the reflective optics. A parabola has the defining property that every ray arriving parallel to its axis is reflected exactly through the focus, so a bundle of parallel rays coming from a star, which is an object at infinity, is gathered into a single perfect point at $z = z_i$, the focus of the mirror. There is no chromatic aberration, because the reflection does not depend on the wavelength, and there is no spherical aberration, because the surface is a true paraboloid instead of a sphere. This is why every serious reflecting telescope, from the first design of Newton to the segmented primary of the James Webb, uses a parabolic mirror as its light collecting surface, and it is also why the same shape reappears in the satellite dishes and the searchlights, where the ray tracing simply runs the other way and a source placed at the focus is turned into a parallel beam.

Chapter 3

Electromagnetic Optics and Polarisation

Light is the slender thread that ties electricity and magnetism into a single weave.

"We can scarcely avoid the inference that light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena."

– James Clerk Maxwell

3.1 A historical context of electromagnetic optics and polarisation

The idea that light is an electromagnetic phenomenon - and that its transverse vibration carries a hidden orientation we call polarisation - is one of the great unifications of nineteenth-century physics. Before light could be understood as a travelling disturbance of electric and magnetic fields, it had to shed two older skins: Newton's corpuscles and the purely mechanical aether of the early wave theorists. The story of polarisation, in particular, is the story of how a seemingly minor curiosity - light reflected from a window behaving differently from ordinary light - forced physicists to accept that the optical vibration is transverse, and ultimately vectorial.

The first quantitative encounter with polarisation came in 1808, when Étienne-Louis Malus, looking through a calcite crystal at sunlight reflected from the windows of the Luxembourg Palace in Paris, noticed that the intensity of the transmitted beam changed as he rotated the crystal. From these observations he extracted the law that bears his name, $I = I_0 \cos^2 \theta$, relating the transmitted intensity to the angle between the analyser and the plane of polarisation Malus (1809). Malus had discovered that reflection could endow light with the same directional property that double refraction revealed, without invoking any crystal at all. A few years later, in 1815, David Brewster established the law that the polarisation upon reflection is complete at a particular angle, the Brewster angle, for which the reflected and refracted rays are perpendicular Brewster (1815).

These facts were deeply puzzling within the longitudinal wave picture inherited from sound. The resolution came from Augustin-Jean Fresnel and François Arago, who between 1816 and 1819 showed experimentally that two beams polarised at right angles do not interfere. Fresnel drew the inevitable, audacious conclusion: the optical vibration must be *transverse* to the direction of propagation. From this single hypothesis he derived the laws governing the amplitudes of reflected and refracted light - the Fresnel equations - which remain in use today. Transversality made polarisation a natural attribute of light: a wave oscillating in a plane perpendicular to its motion can do so along infinitely many directions, and the choice of that direction *is* its state of polarisation.

A decisive step toward a quantitative description of partially polarised light was taken by George Gabriel Stokes in 1852. Recognising that realistic beams are never perfectly polarised, Stokes introduced four measurable quantities - today called the Stokes parameters - that completely characterise the state of polarisation, including the degree of polarisation, of any beam, coherent or not Stokes (1852). Decades later, Henri Poincaré gave these parameters a beautiful geometric home: every state of polarisation corresponds to a point on the surface of a sphere, the Poincaré sphere, on which the poles represent circular polarisation and the equator linear polarisation. This geometric picture is the one that reappears, in the scalar reduction of diffraction theory, when one decomposes the field onto two opposite states of the sphere.

The grand synthesis arrived with James Clerk Maxwell. In his 1865 memoir *A Dynamical Theory of the Electromagnetic Field* Maxwell (1865), Maxwell showed that the equations governing electricity and magnetism admit wave solutions propagating at a speed numerically equal to the measured speed of light, and concluded that "light is an electromagnetic disturbance." Polarisation was thereby identified with the direction of the electric field vector $\vec{E}$, and the whole edifice of physical optics became a chapter of electromagnetism. The experimental confirmation came in 1887, when Heinrich Hertz generated and detected electromagnetic waves in the laboratory, verifying that they reflect, refract and polarise exactly as light does Hertz (1893).

In the twentieth century the description of polarisation was recast in the language of linear algebra. R. Clark Jones, in a celebrated series of papers beginning in 1941, introduced the calculus of two-component complex vectors and 2×2 matrices that now bears his name, ideally suited to fully polarised, coherent light Jones (1941). For partially polarised or depolarising systems, the complementary 4×4 Mueller calculus acting on the Stokes vector provides the appropriate tool. The modern treatment of all these matters - transversality, the Fresnel equations, the StokesPoincaré formalism and the Jones and Mueller calculi - is presented with classic completeness in Born and Wolf's *Principles of Optics* Born and Wolf (1999), the reference against which later treatments are still measured.

From Malus's rotating crystal to the polarisation-encoded qubits of quantum information, the orientation of the optical field has proven to be far more than an ornament: it is a genuine degree of freedom of light, vectorial in nature, and the bridge between the scalar optics of rays and the full electromagnetic description developed in the chapters that follow.

3.2 The vector behavior of the electromagnetic radiation

Until this part of the lectures, we've studied the light as a ray, however, a better model for the light becomes with the 4 Maxwell's equations, which are:

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \qquad \textit{Gauss Law for the electric field} \tag{3.1}$$

$$\nabla \cdot \vec{B} = 0, \qquad \textit{Gauss Law for the magnetic field} \tag{3.2}$$

$$\nabla \times \vec{E} = -\frac{\partial}{\partial t}\vec{B}, \qquad \textit{Faraday-Lenz law} \tag{3.3}$$

$$\nabla \times \vec{B} = \mu_0\epsilon_0\frac{\partial}{\partial t}\vec{E} + \mu_0\vec{J}, \qquad \text{Ampére-Maxwell Law..} \tag{3.4}$$

The equations from 3.1 to 3.4 have got such elegant form thanks to the vector notation given by the physicist Oliver Heaviside and thanks to some of the most useful theorems of the mathematical analysis, the divergence (Gauss) and curl (Stokes) theorems, which allow a smoth transition from the integral form of the Maxwell equations to the differential ones.

The Gauss Law for the electric field 3.1 is really easy to read when we know the geometric interpretatio4n of the divergence operator $\nabla\cdot$ which quantifies how much the electric field $\vec{E}$ goes into (also gets out) from a point, but the right hand side of the equation 3.1 tells us that the presence of any *charge density* ρ produces sources and sinks of electric field (positive charge densities produce sources and negative charge densities produce sinks of electric field).

The Gauss Law for the magnetic field 3.2 tells us that **alwyas the sources of magnetic fields are compensated by sinks of magnetic fields**, also it is interpreted as **there are no magnetic fields monopoles**.

The equation 3.3 is known as the Faraday-Lenz law, the Faraday law tells us that every circulation of electric field produces a variation of magnetic field and every time variation of magnetic field produces circulations of electric field, however there is something importart, because if we take the equation 3.3 as $\nabla \times \vec{E} = \partial_t \vec{B}$ without the minus sign there wont be energy conservation, so the minus sign is introduced by Lenz and this equation can describe the beahavior of eddy currents.

The fourth equation 3.4 describes the circulations of magnetic fields, due to the Gauss Law 3.2 there wont be any polar magnetic field, it is only possible to have axial-like magnetic fields so, this circulations can only be created by two ways, from the presence of density currents $\vec{J}$ or by time variation of the electric fields, Maxwell added this term such that the continuity equation

$$\frac{\partial}{\partial t}\rho + \boldsymbol{\nabla} \cdot \vec{J} = 0, \tag{3.5}$$

is satisfied.

Now,because we're interested in the study of the electromagnetic radiation *light* we must look for a wave equation, to do this, we're going to use the following mathematical identity:

$$\boldsymbol{\nabla} \times \left(\boldsymbol{\nabla} \times \vec{B}\right) = \boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \vec{A}\right) - \nabla^2 \vec{A}. \tag{3.6}$$

From the Faraday-Lenz law 3.3 we've got applying the curl to both sides

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \times \vec{E}\right) = -\boldsymbol{\nabla} \times \left(\frac{\partial}{\partial t}\vec{B}\right),$$

now, befores we're going further we're going to suppose that the electromagnetic fields are smooth vector functions, or at least of class $\mathcal{C}^2 := \left\{f : D^2 f \in \mathcal{C}\right\}$ (this means all the second derivaties are continuos), therefore because these are continuous we can use the identity 3.6 and interchange the curl with the time derivative on $\vec{B}$, thus

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \vec{E}\right) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t}\left(\boldsymbol{\nabla} \times \vec{B}\right),$$

to solve the left hand side we are going to use the Gauss law of the electric field 3.1 and for the right hand side we're going to use the Ampére-Maxwell law 3.4 such that we've got

$$\boldsymbol{\nabla}\left(\frac{\rho}{\epsilon_0}\right) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t}\left(\mu_0 \epsilon_0 \frac{\partial}{\partial t}\vec{E} + \mu_0 \vec{J}\right),$$

$$\nabla^2 \vec{E} - \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\vec{E} = \frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho - \mu_0 \frac{\partial}{\partial t}\vec{J}, \tag{3.7}$$

where $C^2 = \frac{1}{\epsilon_0 \mu_0}$.

The 3.7 is the formal wave equation for the electric field. In the left hand side we find the traditional well known wave equation but on the right hand side we can see some physical information about how can those waves be generated. We can see that any gradient of electric charg densities produces electric waves and also time variations of the current density, and the minus sign isn't there just to be beauty, the minus sign comes from the Lenz-law, therefore that minus sign guarantees the energy conservation on the electric field waves. Now, we've shown that there are electric waves so we're going to proof that there are also magnetic waves.

From the Ampére-Maxwell law 3.4 we've got applying the curl on both sides

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla}\times\vec{B}\right)=\boldsymbol{\nabla}\times\left(\mu_0\epsilon_0\frac{\partial}{\partial t}\vec{E}+\mu_0\vec{J}\right),$$

now, under our assumptions and also $\rho,\vec{J}$ are at leat $\mathcal{C}^2$, then we can apply the identity 3.6 and interchange the time derivatide with the curl on $\vec{E}$, thus

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\vec{B}\right)-\nabla^2\vec{B}=\mu_0\epsilon_0\frac{\partial}{\partial t}\left(\boldsymbol{\nabla}\times E\right)+\mu_0\boldsymbol{\nabla}\times\vec{J}.$$

On the left side we can use the Gauss law for the magnetic field 3.2, and on the right hand side we can use the Faraday-Lenz law thus

$$-\nabla^2\vec{B}=\mu_0\epsilon_0\frac{\partial}{\partial t}\left(-\frac{\partial}{\partial t}\vec{B}\right)+\mu_0\boldsymbol{\nabla}\times\vec{J},$$

$$\nabla^2\vec{B}-\frac{1}{C^2}\frac{\partial^2}{\partial t^2}\vec{B}=-\mu_0\boldsymbol{\nabla}\times\vec{J}. \tag{3.8}$$

The equation 3.8 is the wave equation for the magnetic field, on the left hand side we can recognize the traditional wave terms and on the right hand side we can see the sources of magnetic waves, because there are no magnetic monopoles there is not term for gradients of such charges, only the circulations of density currents produces this magnetic waves.

Befores going any further we must stop and think about something important, if the light is both electric and magnetic radiation, why the equations 3.7 and 3.8 are only coupled by the density current, or even more, in the vacuum, both equation are independent, so if we think on photons we can question ourselves, ***Where are the photons? Are the photons on the electric or magnetic field*** We are so used to learn they're on both, but how?

3.3 Maxwell equations are the Schrödinger equation for photons

The sentence *photons are in the electromagnetic radiation* is easy to beliave but hard to proof from a classical point of view. We could be tempted by the devil and add both equation 3.7 and 3.8 and get something like

$$\nabla^2\left(\vec{E}+\vec{B}\right)-\frac{1}{C^2}\frac{\partial^2}{\partial t^2}\left(\vec{E}+\vec{B}\right)=\frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho-\mu_0\left(\frac{\partial}{\partial t}+\boldsymbol{\nabla}\times\right)\vec{J}, \tag{3.9}$$

which is correct, completely correct **in a world where does not exist physics and analysis**.

From an analysis point of view, the electric field $\vec{E}$ and the magnetic field $\vec{B}$ have got completely different natures, the electric field is a polar vector function, and the magnetic field is an axial vector function, so both fields live on differente vector spaces and the addition $\vec{E}+\vec{B}$ does not have any sense.

To solve this problem, there is an interesing vector field called the ***Riemann-Silberstein field*** which is a correct mathematical (and physical as we we'll see) superposition of the electric and magnetic fields given by

$$\vec{F} = \vec{E} + iC\vec{B}. \tag{3.10}$$

That imaginary unit i may be seen innocent, however adding this complex topology to $\vec{E}, \vec{B}$ solves the superposition problem. It is called for Riemann because he was the first one who studied (alongside Gauss) the electromagnetism using complex numbers, and for Riemann because he used the vector field 3.10 to rewrite the Einstein's field equations in a quaternions formulation. This may be seen as an innocent superpostion but we're going to see that this will allow us, from a classical point of view, to arrive to the correct *photon wave function* and showing that the Maxwell equations acting on 3.10 (which behaves as the photon field) are also quantum equations.

Lets rewrite the Maxwell equations 3.1-3.4 in terms of the photon field $\vec{F}$. First, just multiply the Gauss law for the magnetic field 3.2 by iC and add to the electric Gauss law 3.1 and we'll have

$$\boldsymbol{\nabla} \cdot \left(\vec{E} + iC\vec{B}\right) = \frac{\rho}{\epsilon_0}. \tag{3.11}$$

And, also multiply the Ampére-Maxwell law 3.4 by iC and add it to the Faraday-Lenz law 3.3 we will have

$$\boldsymbol{\nabla} \times \left(\vec{E} + iC\vec{B}\right) = \frac{i}{C}\frac{\partial}{\partial t}\left(\vec{E} + iC\vec{B}\right) + iC\mu_0\vec{J}. \tag{3.12}$$

Joning both equations and using the definitions of $\vec{F}$ we'll wave that the four Maxwell's equations becomes two

$$\boldsymbol{\nabla} \cdot \vec{F} = \frac{\rho}{\epsilon_0}, \qquad \text{Gauss law for the photon field} \tag{3.13}$$

$$\boldsymbol{\nabla} \times \vec{F} = \frac{i}{C}\frac{\partial}{\partial t}\vec{F} + iC\mu_0\vec{J}, \qquad \text{Faraday-Lenz-Ampére-Maxwell laws for the photon field.} \tag{3.14}$$

The physical interpretaion of the equations 3.13 and 3.14 becomes clear. The Gauss law 3.13 tells us that the presence of charge densities produces sources and sinks of photon fields (electromagnetic fields), and the second equation 3.14 says that time variatons of the photon field produces circulations of photon fields and also, the mere presence of density currents produces also circulations of this photon fields. Now it is becoming clearer the relation between where are the photons and how can it be seen in the elctromagnetic field at classical levels.

Now, we're going to proof that the set of equations 3.13 and 3.14 contains two wave equation. One traditional wave equation for $\vec{F}$ and an Schrödigner wave equation for $\vec{F}$, showing this beauty behavior both classical and quantum of the Maxwells equation. Lets apply the curl of the second law 3.14 to both sides as follows

$$\boldsymbol{\nabla} \times \left(\boldsymbol{\nabla} \times \vec{F}\right) = \frac{i}{C}\boldsymbol{\nabla} \times \left(\frac{\partial}{\partial t}\vec{F}\right) + iC\mu_0\boldsymbol{\nabla} \times \vec{J},$$

applying the identity 3.6 on the left han side and permutting th time derivative and the curl in the right hand side as

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \vec{F}\right) - \nabla^2\vec{F} = \frac{i}{C}\frac{\partial}{\partial t}\left(\boldsymbol{\nabla} \times \vec{F}\right) + iC\mu_0\boldsymbol{\nabla} \times \vec{J},$$

using the Gauss law 3.13 and the FRAM law 3.14 we'll have

$$\frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho - \nabla^2\vec{F} = \frac{i}{C}\frac{\partial}{\partial t}\left(\frac{i}{C}\frac{\partial}{\partial t}\vec{F} + iC\mu_0\vec{J}\right) + iC\mu_0\boldsymbol{\nabla}\times\vec{J},$$

and ordergin the terms we are going to have the *classical wave equation for the photon field*

$$\nabla^2\vec{F} - \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\vec{F} = \frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho - \mu_0\left(\frac{\partial}{\partial t} + \boldsymbol{\nabla}\times\right)\vec{J}. \tag{3.15}$$

We can see that looks very similar to the sinister equation 3.9 but that was a really forced one insted of our beauty 3.15 which was calculated from the Maxwell's equations, and we see the origin of this electromagnetic or photon fields, these waves are originated as from charge density gradientes, and time variations and circulations of the density current, now the behavior of the photons, from a classical perspective can be seen in the presence of charges and currents.

Now, to find out the quantum wave equation from the classical equation 3.14 we are going to use, the following algebraic identity:

$$\vec{A}\times\vec{B} = i\left(\vec{A}\cdot\vec{S}\right)\vec{B}, \qquad \vec{S} = S_x\hat{i} + S_y\hat{j} + S_z\hat{k} \tag{3.16}$$

$$S_x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{bmatrix},\ S_y = \begin{bmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{bmatrix},\ S_z = \begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The matrices S_x, S_y, S_z are called teh photon spin 1 matrices, and $\vec{A}, \vec{B}$ are arbitrary vectors. Uusing the identity 3.16 on 3.14 we'll have

$$\begin{aligned}
\boldsymbol{\nabla}\times\vec{F} &= -i\left(\boldsymbol{\nabla}\cdot\vec{S}\right)\vec{F}, \quad \text{using the product rule}\\
\boldsymbol{\nabla}\times\vec{F} &= -i\left(\vec{S}\cdot\boldsymbol{\nabla}\right)\vec{F},\\
\frac{i}{C}\frac{\partial}{\partial t}\vec{F} &= -i\left(\vec{S}\cdot\boldsymbol{\nabla}\right)\vec{F} - iC\mu_0\vec{J},
\end{aligned}$$

multipliying both sides by $C\hbar$ we'll have

$$i\hbar\frac{\partial}{\partial t}\vec{F} = C\left(\vec{S}\cdot\frac{\hbar}{i}\boldsymbol{\nabla}\right)\vec{F} - i\hbar C^2\mu_0\vec{J}, \tag{3.17}$$

which is the Schrödinger-like equation for the photon field $\vec{F}$ with the perturbation of the density currents $\vec{J}$ if we're on the vacuum, we'll have the well known as the ***Birula's photon wave function*** Białynicki-Birula (1994)

$$i\hbar\frac{\partial}{\partial t}\vec{F} = C\left(\vec{S}\cdot\frac{\hbar}{i}\boldsymbol{\nabla}\right)\vec{F}. \tag{3.18}$$

Therefore the Maxwells equation by themselves are classical and quantum, contains the wave functions that describes the photon behavior and, as we we'll see further in this lectures, we can carry on this result to general relativity using a tensor that I named **Riemann-Silberstein tenstor** which even condensates the four equations to just one.

3.4 Properties of the electromagnetic waves

From now on we're going to focus on the electric wave equation 3.7 and not the complete electromagnetic wave equation 3.15 for three reasons:

1. For now, we're going to focus on the radiation propagation on the vacuum, so the electric field intensity $||\vec{E}||$ is a lot bigger than the magnetic field intensity $||\vec{B}||$ with a relation approximadelety like $||\vec{E}|| \approx C||\vec{B}||$.

2. The interaction between light and matter is really hard to study when we consider the electric polarization and even more when we consider all the magnetic properties and effects. This topics are going to be study in the lectures on quantum optics.

3. This is still a lecture on fundamentals, so we're going to start from the basic notions and getting richer models while we advance.

3.4.1 Principal solutions of the electric wave equation

Due to the last reasons, we're going to to focus on the study of the electric wave equation

$$\nabla^2 \vec{E}(\vec{r}, t) = \frac{1}{C^2} \frac{\partial^2 \vec{E}(\vec{r}, t)}{\partial t^2}. \tag{3.19}$$

We're going to solver this equation in a physical way (latter on this lectures we're going to solve it in a more sophisticaed way). We know from a physical point of view that $\vec{E}$ should be a propagation function with a constant velcity $\vec{v}$ (because there is no matter so the momentum is conserved), therefore we could suppose the electric wave have got the following expression

$$\vec{E}(\vec{r}, t) = \vec{u}(\vec{k} \cdot \vec{r} \pm \omega t, t), \tag{3.20}$$

where we have got that $\omega^2/k^2 = C^2$. The two most popular solutions in the basics literature are the plane wave equation

$$\vec{E}(\vec{r}, t) = E(\vec{r}, t) e^{i\left(\vec{k}\cdot r - \omega t + \phi\right)} \hat{e}(\vec{r}, t), \tag{3.21}$$

where $E(\vec{r}, t)$ is the amplitude, $\vec{k}$ is the waver number, a kind of spatial frequency, ω is the angular frequency, ϕ is the proper fase, an $\hat{e}$ is the unit vector that describes the vector behavior of the electric wave, also we will identify $\hat{e}$ as the *polarisation*.

The other one, very popular is the spherical wave

$$\vec{E}(\vec{r}, t) = \frac{E(t)}{||\vec{r}||} e^{i(kr - \omega t + \phi)} \hat{e}. \tag{3.22}$$

Both expressions are usually written down as if they had fallen from the sky, but each of them is a genuine solution of the wave equation 3.19 and can be obtained directly from it. Let us do it carefully, first for the plane wave and then for the spherical one.

In the vacuum the wave equation 3.19 acts on every Cartesian component of $\vec{E}$ in exactly the same way and without mixing them, so for the moment we can forget the vector character, write $\vec{E}(\vec{r},t)=\psi(\vec{r},t)\hat{e}$ with $\hat{e}$ a fixed unit vector, and solve the scalar equation

$$\nabla^2\psi=\frac{1}{C^2}\frac{\partial^2\psi}{\partial t^2}. \tag{3.23}$$

A natural way to attack it is to separate the space and time dependence, proposing $\psi(\vec{r},t)=R(\vec{r})T(t)$. Substituting into 3.23 and dividing by RT we get

$$\frac{\nabla^2 R}{R}=\frac{1}{C^2}\frac{\ddot{T}}{T}.$$

The left hand side depends only on the position and the right hand side only on the time, so the only way for them to be equal for every $\vec{r}$ and every t is that both are the same constant. We call that constant $-k^2$, where the minus sign is chosen so that the time part stays bounded and oscillatory, as a wave must be. The time equation then becomes

$$\ddot{T}+C^2k^2T=0,$$

and defining $\omega=Ck$, which is precisely the relation $\omega^2/k^2=C^2$ we anticipated, its solution is $T(t)=e^{\mp i\omega t}$. The spatial part becomes the Helmholtz equation

$$\nabla^2R+k^2R=0. \tag{3.24}$$

Now we ask for the simplest possible geometry of the wavefronts. If we want the surfaces of constant phase to be planes, the field must depend on the position only through a single linear combination of the coordinates, that is, through $\vec{k}\cdot\vec{r}$ for some fixed vector $\vec{k}$. This suggests

$$R(\vec{r})=e^{i\vec{k}\cdot\vec{r}},$$

and since $\nabla^2 e^{i\vec{k}\cdot\vec{r}}=-\left|\left|\vec{k}\right|\right|^2 e^{i\vec{k}\cdot\vec{r}}$, the Helmholtz equation 3.24 is satisfied as long as $\left|\left|\vec{k}\right|\right|=k$. The vector $\vec{k}$ that was born as a mere separation constant turns out to point along the direction of propagation and to have a length fixed by the frequency. Putting the space and time parts back together, restoring the amplitude E_0 and a constant phase ϕ, and recovering the polarisation vector, we arrive at

$$\vec{E}(\vec{r},t)=E_0\,e^{i\left(\vec{k}\cdot\vec{r}-\omega t+\phi\right)}\hat{e},$$

which is exactly the plane wave 3.21. If instead of the constant E_0 and the fixed $\hat{e}$ we allow a slowly varying amplitude $E(\vec{r},t)$ and polarisation $\hat{e}(\vec{r},t)$, we recover the modulated waves that we will use later, whose wavefronts are still approximately planes over regions small compared with the modulation.

There is still one condition we have not used, and it is the one that makes the wave truly electromagnetic. In the vacuum the Gauss law 3.1 reads $\boldsymbol{\nabla}\cdot\vec{E}=0$, and evaluating it on our solution

$$\boldsymbol{\nabla}\cdot\left(E_0\,e^{i\left(\vec{k}\cdot\vec{r}-\omega t+\phi\right)}\hat{e}\right)=i\,E_0\left(\vec{k}\cdot\hat{e}\right)e^{i\left(\vec{k}\cdot\vec{r}-\omega t+\phi\right)}=0,$$

which can only hold if $\vec{k} \cdot \hat{e} = 0$. The polarisation is forced to be perpendicular to the direction of propagation, and we recover, now as a theorem instead of a hypothesis, the transversality that Fresnel had to postulate a century and a half earlier.

The plane wave describes a field whose wavefronts are infinite parallel planes, an idealisation that is excellent far from the source but says nothing about how the radiation spreads out from a small emitter. For that situation the natural symmetry is not translational but spherical, so we now look for a field that depends on the position only through the distance $r = ||\vec{r}||$ to the source. Working again with the scalar amplitude $\psi(r, t)$, we need the Laplacian of a function that depends on r alone, which in spherical coordinates is

$$\nabla^2\psi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right).$$

This expression hides a very useful identity. If we expand it and compare it with the second derivative of the product $r\psi$, we find

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) = \frac{1}{r}\frac{\partial^2}{\partial r^2}(r\psi), \tag{3.25}$$

which turns the awkward radial Laplacian into an ordinary second derivative acting on $r\psi$. Using 3.25, the wave equation 3.23 for a spherically symmetric field becomes

$$\frac{1}{r}\frac{\partial^2}{\partial r^2}(r\psi) = \frac{1}{C^2}\frac{\partial^2\psi}{\partial t^2}.$$

Multiplying both sides by r and noticing that r does not depend on the time, so it can go inside the time derivative on the right, we get

$$\frac{\partial^2}{\partial r^2}(r\psi) = \frac{1}{C^2}\frac{\partial^2}{\partial t^2}(r\psi).$$

If we now baptise the combination $u(r, t) = r\,\psi(r, t)$, this is nothing but the one dimensional wave equation

$$\frac{\partial^2 u}{\partial r^2} = \frac{1}{C^2}\frac{\partial^2 u}{\partial t^2}, \tag{3.26}$$

whose general solution, by d'Alembert, is the sum of a disturbance travelling outwards and one travelling inwards,

$$u(r, t) = f(r - Ct) + g(r + Ct).$$

The term $g(r + Ct)$ represents a wave collapsing towards the source, so for radiation escaping from an emitter we keep only the outgoing part $u(r, t) = f(r - Ct)$. Undoing the change $u = r\psi$ we finally obtain

$$\psi(r, t) = \frac{f(r - Ct)}{r},$$

and the factor $1/r$, which we had to guess before, now appears on its own as a direct consequence of the geometry. Choosing a monochromatic profile $f(r-Ct) = E_0\, e^{ik(r-Ct)} = E_0\, e^{i(kr-\omega t)}$, again with $\omega = Ck$, and restoring the phase and the polarisation we reach the spherical wave 3.22,

$$\vec{E}(\vec{r},t) = \frac{E_0}{r}\, e^{i(kr-\omega t+\phi)}\hat{e}.$$

It is worth pausing on the $1/r$. The amplitude of the field falls off as the inverse of the distance, so its square, which measures the intensity, falls off as $1/r^2$. Since the area of a spherical wavefront grows exactly as r^2, the total power crossing every sphere centred on the source is the same, and energy is conserved as the wave expands. This is why the field goes as $1/r$ and not as $1/r^2$: the inverse square law belongs to the intensity, not to the field itself.

The plane wave and the spherical wave look like very different objects, and yet they are intimately related, because far enough from the source the spherical wave is indistinguishable from a plane wave. To see it, place ourselves at a large distance r_0 from the emitter along a fixed direction $\hat{n} = \vec{r}_0/r_0$, and look at the field only inside a small region around that point, writing $\vec{r} = \vec{r}_0 + \vec{\delta}$ with $\left|\left|\vec{\delta}\right|\right| \ll r_0$. The distance to the source is then

$$r = \left|\left|\vec{r}_0 + \vec{\delta}\right|\right| = r_0\sqrt{1 + \frac{2\,\hat{n}\cdot\vec{\delta}}{r_0} + \frac{\left|\left|\vec{\delta}\right|\right|^2}{r_0^2}} \approx r_0 + \hat{n}\cdot\vec{\delta},$$

where we kept only the first order in $\vec{\delta}/r_0$. The phase of the spherical wave becomes, up to the constant kr_0,

$$kr \approx kr_0 + k\,\hat{n}\cdot\vec{\delta} = kr_0 + \vec{k}\cdot\vec{\delta}, \qquad \vec{k} = k\,\hat{n},$$

so it depends on the position through the single linear combination $\vec{k}\cdot\vec{\delta}$, which is exactly the signature of a plane wave travelling along $\hat{n}$. The amplitude, on the other hand, is $E_0/r \approx E_0/r_0$, which over the small region hardly changes and can be treated as the constant E_0/r_0. Putting both together, in the neighbourhood of the point the field reads

$$\vec{E}(\vec{r},t) \approx \frac{E_0}{r_0}\, e^{i\left(kr_0+\vec{k}\cdot\vec{\delta}-\omega t+\phi\right)}\hat{e},$$

which is precisely a plane wave 3.21 of amplitude E_0/r_0 and wavevector $\vec{k} = k\hat{n}$. In physical terms, a small patch of a sphere of enormous radius is almost flat, and the observer far from the source sees the curved wavefronts as if they were planes. This is the regime in which the plane wave, despite carrying infinite energy across an infinite front, becomes a legitimate and extremely convenient approximation, and it is the reason why so much of optics can be done with plane waves even though every real source radiates spherically.

3.4.2 Beams with a particular phase and structured light

The plane wave and the spherical wave are the two solutions that every textbook writes down first, and for a good reason, since they are the simplest geometries the wave equation admits. It would be a serious mistake, however, to believe that they are the only solutions, or even the most important ones. Both are idealisations that no laboratory ever produces exactly, because the plane wave has an infinite transverse extent and carries an infinite power, and the pure spherical wave demands a perfectly point like source radiating in every direction at once. The light we actually generate and use, above all the light of a laser, comes in the form of *beams*, that is, fields well confined around a propagation axis that carry a finite power. What distinguishes one beam from another, and what grants each one its remarkable properties, is not so much its amplitude as the way its *phase* is organised across the wavefront. This subsection is a short tour through the most relevant of these structured beams.

Before writing any of them it helps to have a common equation from which they all descend. Almost every beam of interest travels mostly along one axis, which we take to be z, with only a gentle spreading in the transverse plane, and this is what we call the *paraxial regime*. We therefore look for solutions of the Helmholtz equation 3.24 of the form

$$\vec{E}(\vec{r},t) = u(x,y,z)\, e^{i(kz-\omega t)}\hat{e},$$

where the exponential carries the fast oscillation along the axis and u is a slowly varying envelope that sculpts the beam. Substituting this into the Helmholtz equation and cancelling the common factor we obtain

$$\nabla_\perp^2 u + \frac{\partial^2 u}{\partial z^2} + 2ik\frac{\partial u}{\partial z} = 0, \qquad \nabla_\perp^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2},$$

and because the envelope varies slowly over a wavelength we may neglect $\partial_z^2 u$ against $2k\,\partial_z u$, arriving at the *paraxial wave equation*

$$i\frac{\partial u}{\partial z} = -\frac{1}{2k}\nabla_\perp^2 u. \tag{3.27}$$

It is impossible not to notice that 3.27 is formally identical to the Schrödinger equation of a free particle in two dimensions, with the propagation coordinate z playing the role of the time and $1/k$ playing the role of $\hbar/m$. This is not a coincidence but another face of the deep link between Maxwell and Schrödinger we met earlier in this chapter, and it tells us that the whole catalogue of structured beams is, in a precise mathematical sense, the catalogue of the wave packets of a free particle.

The simplest and most important solution of 3.27 is the one whose transverse profile is a Gaussian. Writing $\rho^2 = x^2 + y^2$ for the transverse radius, its electric field is

$$\vec{E} = E_0\,\frac{w_0}{w(z)}\exp\left(-\frac{\rho^2}{w(z)^2}\right)\exp\left(i\left[kz - \omega t + \frac{k\rho^2}{2R(z)} - \zeta(z)\right]\right)\hat{e}, \tag{3.28}$$

where the three functions that govern the beam are the width, the radius of curvature and the Gouy phase,

$$w(z) = w_0\sqrt{1+\left(\frac{z}{z_R}\right)^2}, \qquad R(z) = z\left[1+\left(\frac{z_R}{z}\right)^2\right], \qquad \zeta(z) = \arctan\frac{z}{z_R}, \tag{3.29}$$

and $z_R = \frac{kw_0^2}{2} = \frac{\pi w_0^2}{\lambda}$ is the *Rayleigh range*, the distance over which the beam stays approximately collimated, while w_0 is the *waist*, the minimal radius the beam reaches at $z = 0$. Reading the phase of 3.28 term by term makes the physics appear. The piece kz is the ordinary plane wave advance along the axis. The quadratic piece $k\rho^2/2R(z)$ is exactly the phase that a spherical wave of radius $R(z)$ carries near the axis, so it tells us that the wavefronts are neither planes nor full spheres but gently curved caps, flat at the waist where $R \to \infty$, most strongly curved around the Rayleigh range, and flattening again far away where $R(z) \approx z$ and the beam finally looks spherical. The last piece, the *Gouy phase* $\zeta(z)$, is an extra amount of phase, close to π in total, that the beam accumulates as it crosses the focus, an anomaly with no counterpart in the plane wave and measurable by interferometry. The amplitude factor $w_0/w(z)$ in front of the Gaussian keeps the power constant while the beam widens, since as $w(z)$ grows the peak drops so that the integrated intensity is conserved. The beam spreads with a divergence half angle $\theta = \lambda/\pi w_0$, and this reciprocal relation between waist and divergence is nothing but the diffraction limit and, once more, an uncertainty relation between the transverse position and the transverse momentum of the field. The Gaussian beam is the fundamental transverse mode of a laser resonator, the famous TEM_{00}, and it is the most confined beam that diffraction permits, which is why it is the default description of any well behaved laser and the starting point for the design of almost every optical system that works with coherent light.

The Gaussian is only the lowest member of a whole family. Since 3.27 is linear, it admits complete sets of higher order modes that share the same Gaussian envelope but decorate it with nodal structure. In Cartesian coordinates these are the *Hermite-Gauss* modes, indexed by two integers and built from Hermite polynomials, which display a rectangular grid of bright lobes. In cylindrical coordinates the natural family is that of the *Laguerre-Gauss* modes, indexed by a radial number p and an azimuthal number ℓ, whose field is

$$\begin{aligned}\vec{E}_{p\ell} = E_0\,\frac{w_0}{w(z)}\left(\frac{\rho\sqrt{2}}{w(z)}\right)^{|\ell|} L_p^{|\ell|}\left(\frac{2\rho^2}{w(z)^2}\right)\exp\left(-\frac{\rho^2}{w(z)^2}\right)e^{i\ell\varphi} \\ \times\exp\left(i\left[kz-\omega t+\frac{k\rho^2}{2R(z)}-(2p+|\ell|+1)\,\zeta(z)\right]\right)\hat{e},\end{aligned} \tag{3.30}$$

where $L_p^{|\ell|}$ are the associated Laguerre polynomials and φ is the azimuthal angle. Two features of 3.30 deserve attention. The Gouy phase now grows in proportion to the mode order $2p + |\ell| + 1$, so different modes advance their phase at different rates, a fact that governs how a superposition of them rotates and reshapes as it propagates. Far more striking is the azimuthal factor $e^{i\ell\varphi}$, which winds the phase through ℓ complete turns around the axis and forces the field to vanish exactly on the axis, giving these beams their characteristic doughnut profile with a dark core. The wavefront is then no longer a stack of caps but a set of intertwined helices, an *optical vortex* of topological charge ℓ. This helical phase is not a mere decoration, because a field carrying $e^{i\ell\varphi}$ transports orbital angular momentum, and each of its photons carries exactly $\ell\hbar$ of it, a quantity entirely separate from the spin angular momentum $\pm\hbar$ associated with circular polarisation. For this reason the Laguerre-Gauss beams have become one of the central tools of modern optics. They are used in the *optical tweezers* to trap microscopic particles and set them spinning, in super resolution microscopy where the dark core of a doughnut beam switches off the fluorescence everywhere except at the very centre, and in classical and quantum communications alike, where the unbounded integer ℓ offers a new and in principle limitless alphabet that lets a single photon encode much more than a single bit.

All the beams written so far spread as they travel, because they are paraxial and diffraction is unavoidable for any field of finite width. It is natural to ask whether the exact Helmholtz equation admits a beam that does not spread at all, and it does. If we demand a field whose transverse intensity profile is literally independent of z, separation of variables in cylindrical coordinates leads to the *Bessel beam*

$$\vec{E} = E_0\, J_\ell(k_\perp \rho)\, e^{i\ell\varphi}\, e^{i(k_z z-\omega t)}\hat{e}, \qquad k_\perp^2 + k_z^2 = k^2, \tag{3.31}$$

where J_ℓ is the Bessel function of order ℓ. Its transverse pattern, a bright central spot when $\ell = 0$ or a vortex ring when $\ell \neq 0$, surrounded by concentric rings, is exactly the same at every plane z, so the beam is said to be *non-diffracting.* The picture behind 3.31 is illuminating, since it is the coherent superposition of plane waves whose wavevectors all lie on a cone of fixed half angle $\arctan(k_\perp/k_z)$, and the interference of that cone regenerates the same profile plane after plane. This conical construction also explains the beam's most surprising property, its *self-healing*, because if an obstacle blocks the central spot the outer rays that build the cone continue past it and reconstruct the pattern a short distance downstream. The ideal Bessel beam pays for these gifts with an infinite power, since its rings extend forever and each one carries a comparable share of energy, so in practice it is produced with an axicon or a ring aperture and remains non-diffracting only over a finite range. Its long, thin and self-repairing focus makes it invaluable for light sheet microscopy, for precise optical alignment and for material processing whenever a large depth of focus is required.

There is a last beam whose behaviour borders on the paradoxical. Looking again at the paraxial equation 3.27, now in a single transverse dimension, one finds a solution written in terms of the Airy function,

$$u(x, z = 0) = \mathrm{Ai}\left(\frac{x}{x_0}\right) e^{\,ax/x_0}, \qquad a > 0, \tag{3.32}$$

where x_0 sets the transverse scale and the exponential aperture e^{ax/x_0} is introduced to make the total energy finite. As it propagates, the *Airy beam* shows two properties that seem to contradict everything the plane wave taught us. It is non-diffracting, keeping its intensity profile over long distances, and it is *self-accelerating*, since its main lobe does not travel in a straight line but follows a parabolic trajectory $x \propto z^2$, so the beam bends sideways even though no force and no index gradient act on it. Nothing is being violated here, because the centre of mass of the whole field does travel straight, and it is only the intensity pattern that curves, in the same way that the crest of a water wave may move differently from the water itself. Like the Bessel beam it is also self-healing, and these curved, resilient trajectories have been used to route light around obstacles, to guide electric discharges along curved plasma channels, to sweep microscopic particles along a chosen path and, once again, in advanced microscopy.

Seen together, this family teaches a lesson that the plane and the spherical waves, taken alone, keep hidden, namely that what organises a beam and endows it with a useful behaviour is the shape of its phase across the wavefront. A phase linear in the transverse coordinate, $e^{i\vec{k}_\perp\cdot\vec{\rho}}$, simply tilts the beam. A phase quadratic in ρ, $e^{ik\rho^2/2R}$, focuses or defocuses it and is exactly what a thin lens imprints. A phase linear in the azimuth, $e^{i\ell\varphi}$, gives it a vortex and orbital angular momentum. A conical phase produces the non-diffracting Bessel beam, and a cubic phase produces the self bending Airy beam. Modern devices such as the spatial light modulators and the metasurfaces let us paint almost any of these phase profiles onto a wavefront at will, and this ability to engineer the phase is the whole discipline known today as *structured light.* The plane wave and the spherical wave remain useful as building blocks and as limiting cases, but from here on the reader should keep in mind that they are only two convenient points in a vastly richer landscape of solutions.

3.4.3 From 3D wavefronts to 2D wavefronts

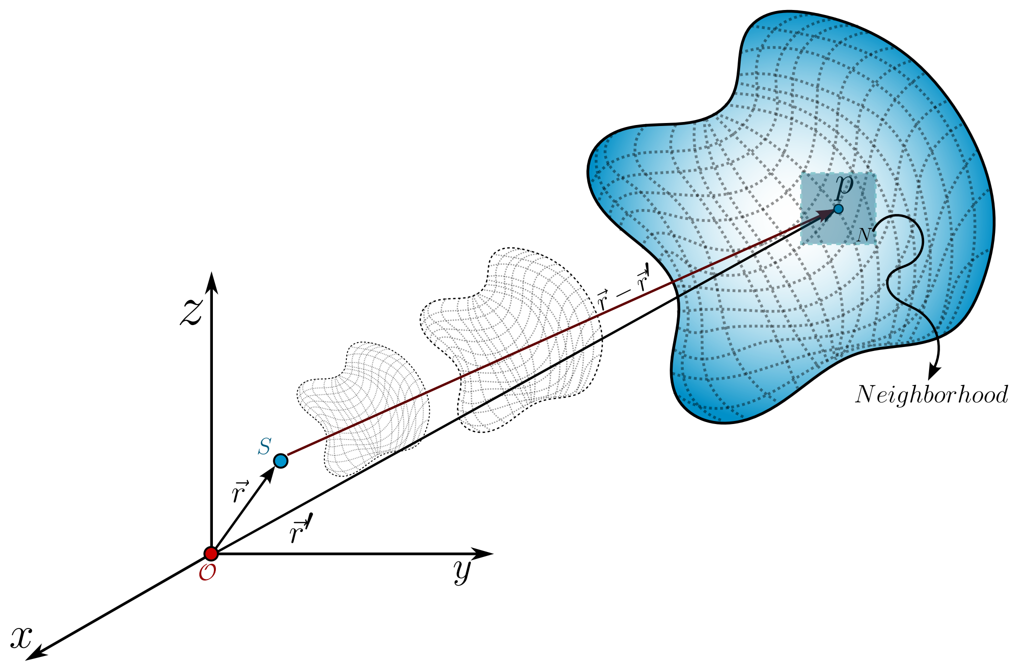


Figure 3.1: A source $\mathcal{S}$ produces an arbitrary kind of wave which propagates in the space-time, then, for any given time and position we can take an arbitrary point p and, due to the smooth manifold nature of this wave, there exists a neighborhood N of p where the wavefront is plane.

As we've seen the electromagnetic radiation has got many presentations, however, there is a very strong formulation where the light is studied in $2D$, in fact, the following chapter treats the electromagnetic radiation as a $2D$ field.

From the mathematical point of view there is a whole theory about arbitrary surfaces or complex geometries, here, we're going to use the properties of *smooth manifolds*[1] This is thanks to the nature of the physical solution of the wave equations, so, given any arbitrary wavefront as shown 3.1 we can choose any point a set and observer on a neighborhood N, and because the wavefront behaves as an smooth manifold then this neighborhood is $\mathbb{R}^2$.

[1]A smooth manifold if a set which can be completely covered by subsets and there are smooth functions from the manifold to $\mathbb{R}^n$.

What we are doing is thinkg on this surface like the earth, because we're son tiny we see the earth as plane but it is not. The same phenomena is here, we step on and observer across the point p on the wavefront and for us there is a neighborhood N where it behaves exactly as $\mathbb{R}^2$ ad that is going to allow us to start the study of the polarisation.

3.5 Light's polarisation

In the electromagnetic theory the word *polarisation* is used as a tendency of the electric dipoles to orientate in a given microscopic volume. However, this word also is used to describe the **electric field orientation** in its propagation. Those two definitions then seem completely indepent, however, as we are going to show in the microscopic theory of radiation these two are completely related.

The polarisation of light is one of the most usefull properties of light, there are many applications on real life which uses this property, for example, the cancer cells stole the glucose from the healthy ones, so there is a higher sugar concentration in cancer cells than the healthy ones, so if we irradiate a sample with linearly polarized light and study the transmited radiation, we're going to see that the light which went across the cancer cells changed their polarization, rotating a given agle (which depends on the concentration of glucose) so we can detected the presence of cancer cells using the polarisation.

3.5.1 Monochromatic and deterministic formulation: Jones spinors and the polarization ellipse

As we've said, we're going to focus on the electric field behavior in a $2D$ formulation, so the electric field is going to be a vector field $\vec{E} \in \mathcal{C}^2\left(\mathbb{C}^2\right)$ which means it is *a complex vector function which their second derivatives are continuous.* Its general form is given by

$$\vec{E}(\vec{r},t;\omega) = \begin{bmatrix} \mathcal{E}_1 e^{i\left(\vec{k}\cdot\vec{r}-\omega t+\phi_1\right)} \\ \mathcal{E}_2 e^{i\left(\vec{k}\cdot\vec{r}-\omega t+\phi_2\right)} \end{bmatrix}, \tag{3.33}$$

where we're considering only one frequency ω so this describes the electric field of a monocrhomatic field and as we can see, this vector field is a plane wave.

We can factor the propagation phase $\exp\left[i\left(\vec{k}\cdot\vec{r}-\omega t\right)\right]$ in both components ad we will have the following vector field

$$\vec{e} = \begin{bmatrix} \mathcal{E}_1 e^{i\phi_1} \\ \mathcal{E}_2 e^{i\phi_2} \end{bmatrix}. \tag{3.34}$$

The vector field 3.34 is known as the *Jones vector* but, in fact, their true nature ist vectorial, the field 3.34 is a **spinor**[2] a we're going to see this spinor behavior in this chapter.

[2]We can see a spinor a 1/2 tensor. From a geometric point of view, we say that a vector $\vec{v}$ is, in fact, and spinor if it needs two complete rotation of 360° to arrive its original position

The Jones spinor 3.34 is a vector that describes how the electric field oscillates, in general, under our assumptions[3] is an ellipse, well known as the polarization ellipse. To get the ellipse from 3.34 we're going to define the following electric field componentes, given by

$$\frac{E_1}{\mathcal{E}_1}e^{-i\phi_2} = \frac{1}{2}e^{i\triangle\phi}, \tag{3.35}$$

$$\frac{E_2}{\mathcal{E}_2}e^{-i\phi_1} = \frac{1}{2}e^{-i\triangle\phi}, \tag{3.36}$$

$$\triangle\phi = \phi_1 - \phi_2.$$

Now, we're going to calculate the magnitude of the difference between 3.35 and 3.36, and because these are complex number, it is given by

$$\left(\frac{E_1}{\mathcal{E}_1}e^{-i\phi_2} - \frac{E_2}{\mathcal{E}_2}e^{-i\phi_1}\right)\left(\frac{E_1}{\mathcal{E}_1}e^{-i\phi_2} - \frac{E_2}{\mathcal{E}_2}e^{-i\phi_1}\right)^* = \frac{1}{4}\left(e^{i\triangle\phi} - e^{-i\triangle\phi}\right)\left(e^{i\triangle\phi} - e^{-i\triangle\phi}\right)^*,$$

where $*$ represents the complex conjugate. In the right hand side we can see tha the difference between those two exponentials are the sine of the phase difference $\triangle$ thus

$$\frac{1}{4}\left(e^{i\triangle\phi} - e^{-i\triangle\phi}\right)\left(e^{i\triangle\phi} - e^{-i\triangle\phi}\right)^* = \frac{1}{4}\left(2i\sin\triangle\phi\right)\left(-2i\sin\triangle\phi\right) = \sin^2\triangle\phi. \tag{3.37}$$

In the left hand side we've got the following

$$\left(\frac{E_1}{\mathcal{E}_1}e^{-i\phi_2} - \frac{E_2}{\mathcal{E}_2}e^{-i\phi_1}\right)\left(\frac{E_1}{\mathcal{E}_1}e^{-i\phi_2} - \frac{E_2}{\mathcal{E}_2}e^{-i\phi_1}\right)^* =$$
$$\frac{|E_1|^2}{\mathcal{E}_1^2} - \frac{E_1E_2^*}{\mathcal{E}_1\mathcal{E}_2}e^{i\triangle\phi} - \frac{E_2E_1^*}{\mathcal{E}_2\mathcal{E}_1}e^{-i\triangle\phi} + \frac{|E_2|^2}{\mathcal{E}_2^2},$$

the crossed terms in blue corresponds to the same quantity but one is the complex conjugate of the other, so if we factor the minus sign we'll have the real part of that complex number, so, the left hand side is given by

$$\frac{|E_1|^2}{\mathcal{E}_1^2} + \frac{|E_2|^2}{\mathcal{E}_2^2} - 2\frac{Re\left\{E_1E_2^*\right\}}{\mathcal{E}_1\mathcal{E}_2}\cos\triangle\phi. \tag{3.38}$$

Combining the LHS 3.38 and the RHS 3.37 we will have the *polarisation ellipse*

$$\frac{|E_1|^2}{\mathcal{E}_1^2} + \frac{|E_2|^2}{\mathcal{E}_2^2} - 2\frac{Re\left\{E_1E_2^*\right\}}{\mathcal{E}_1\mathcal{E}_2}\cos\triangle\phi = \sin^2\triangle\phi. \tag{3.39}$$

Then we've shown the electric field lives on an geometric space on $\mathcal{C}^2$ which is a real rotated elllipse, before going further we should notice something.

The Jones spinor 3.34 is not an unique thing of the electric field, also describes the magnetic field under these assumptions but also **describes any complex field with 2 degrees of freedom**,

[3]deterministic, $2D$, monocrhomatic, space-time independent, propagation on vacuum

for example the gravitational waves have got a polarisation ellipse, $2D$ phonons have polarisation ellipse, and so on.

From a mathematical point of view the origin of the ellipse is obvious, the Jones spinor 3.34 corresponds to two unit complex circles which rotate asincronically, so the general figure is an ellipse not a circle, but, there is a further a more fundamental question, and it is **why does the nature choose an elllipse as the fundamental description of the orientation of the electric field** The short answer is: $No, thenaturedoesnotchoosethat$, the long answer is going to be shown in the theory of coherence chapter.

How to build the polarisation ellipse: Phasors

Given any Jones spinor 3.34 how can we construct the polarisation ellipse? To answer this we're going to use phasors which is a formal word for a polar form of a complex number.

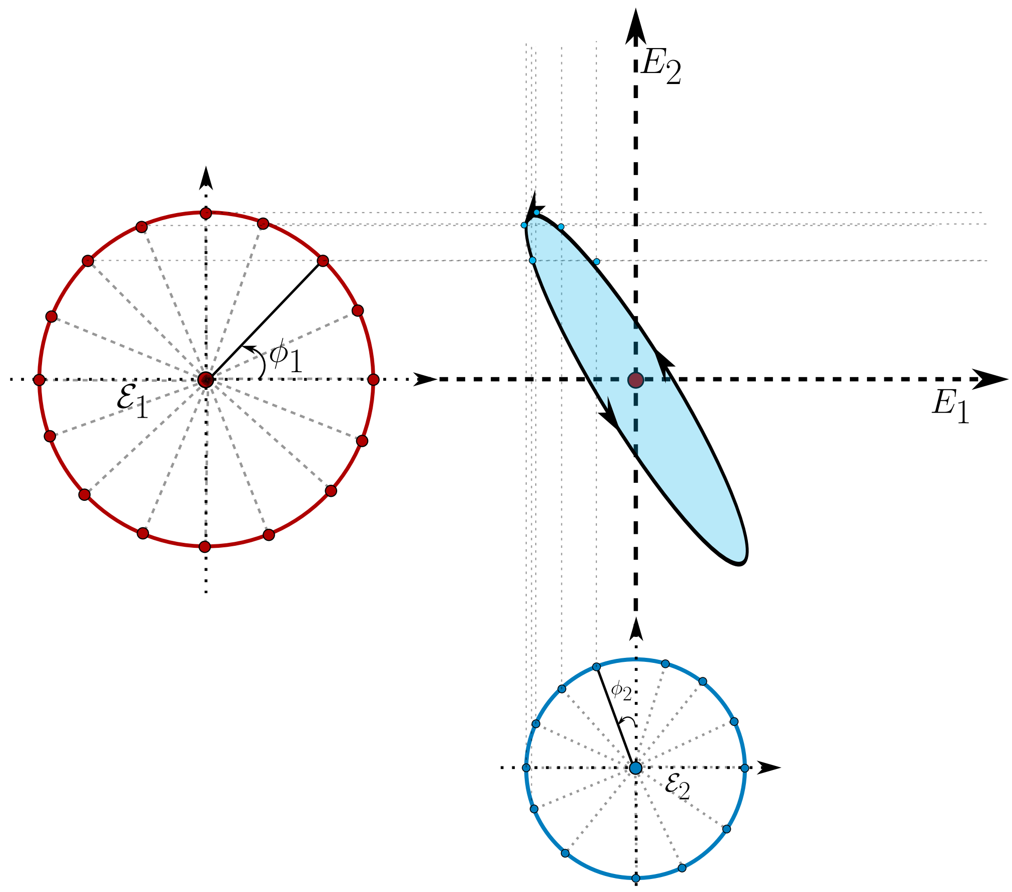


Figure 3.2: The polarisation ellipse can be reconstructred taking all the possible rotations of the two phasors (which corresponds to the compoenentes of the Jones spinor).

Because we've got a $2D$ vector we've got two components, two polar complex numbers $\mathcal{E}_m e^{i\phi_m}$

with $m = 1, 2$ so we can describe these two phasors as two circunferences as is shown in 3.2. The x component or $1 - st$ component is a circunference with radius $\mathcal{E}_1$ and the initial point on this circunference is located at an angle ϕ_1 respect to the $x - axis$ (shown as the red circunference in the figure 3.2). The second phasor, corresponding to the y component of the jones spinor is another circunference but the radius is $\mathcal{E}_2$ and the initial point is located at an angle of ϕ_2 but respect to the $y - axis$.

To construct the polarisation ellipse, we start with the two initial points an draw two straight lines, for the horizontal phasor (the red one) the lines are horizontal, and for the vertical phasor (the blue one) the lines are vertical. In the intersection of the two rects we draw a point, and we desplace both initial points by the same angle, lets call it $\delta\phi$ until we arrive at the initial point. The set of all intersection points draw the polarisation ellipse, which their semi-axes are going to be $\mathcal{E}_1, \mathcal{E}_2$ and the orientation is going to depends on $\triangle\phi$ if $\triangle\phi = \frac{\pi}{2} + n\pi$ with $n \in \mathbb{Z}$ the ellipse isn't going to be orientated also $\triangle\phi$ conditionates if the electric field is going to rotate clockwise or counterclockwise.

How to build the polarisation ellipse: Ellipse angles

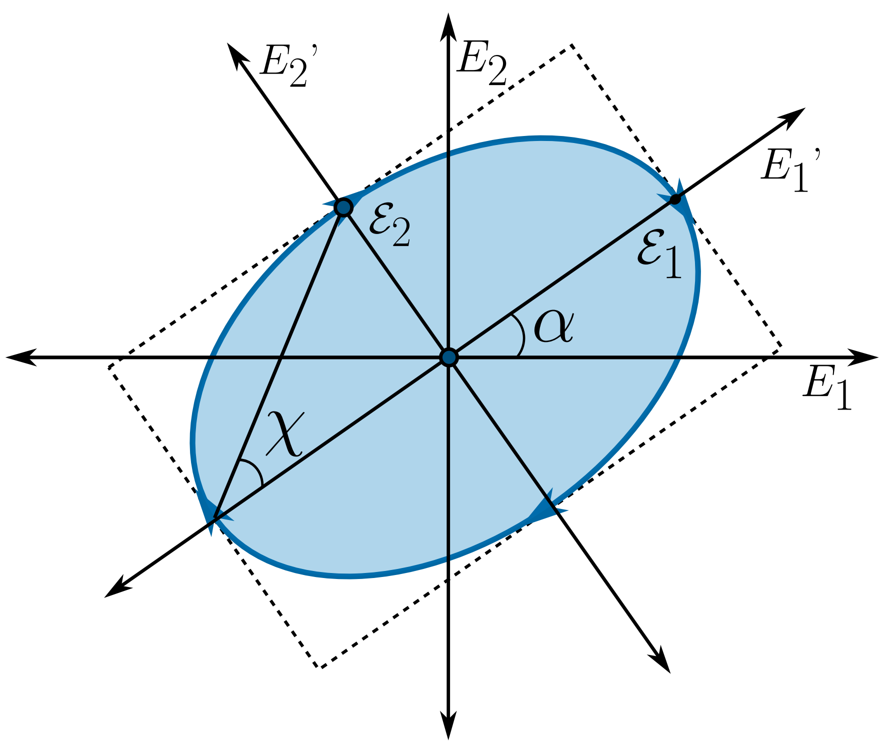


Figure 3.3: The general form of the polarisation ellipse can be described with only two angles; the orientation angle α and the elipticity angle χ.

It is not very comfortable to work with phasor and even we can't see the ellipse inmediately with the Jones spinor, so we would like a more visual or efficient way to interpretate the polarisation. First we must notice that the electric field amplitudes only change the size of the ellipse, but the relative phases ϕ_1, ϕ_2 are the ones who determine the form of the ellipse, so we can work out with a Jones spinor as an unit vector, this means $\vec{e} \to \hat{e}$ such that $\hat{e} \cdot \hat{e} = 1$.

The general polarisation ellipse can be seen in 3.3 which can be fully described using just two angles α and χ.

3.5.2 Time evolution of the polarisations: Stokes paramters and the Poincaré sphere

3.5.3 Transformations of Polarisation

Chapter 4

Interference and Interferometers

When two waves meet, light may add to light and yield darkness; in that paradox lies the whole of physical optics.

"When two undulations, from different origins, coincide either perfectly or very nearly in direction, their joint effect is a combination of the motions belonging to each."
– Thomas Young

4.1 A historical context of interference and interferometers

Interference is the most direct fingerprint of the wave nature of light: when two beams overlap, their amplitudes - not their intensities - add, and the result can be brighter than either beam alone or, astonishingly, complete darkness. From this simple superposition principle grew the interferometer, an instrument that turns the invisible phase of a light wave into a visible pattern of fringes, and which became one of the most precise measuring devices ever built.

Although Robert Hooke and Isaac Newton both studied the coloured fringes produced by thin films - the phenomenon still known as Newton's rings, carefully described in Newton's *Opticks* of 1704 Newton (1704) - neither interpreted them as evidence of waves. Newton's immense authority, and his corpuscular theory, held the field for a century. The decisive break came in 18011804 with Thomas Young, whose double-slit experiment showed that light passing through two narrow apertures produces a pattern of bright and dark bands that can only be explained if the two beams alternately reinforce and cancel one another Young (1804). Young introduced the very word "interference" and, with it, the principle of superposition that underlies all of wave optics.

Young's qualitative insight was forged into a quantitative theory by Augustin-Jean Fresnel, who combined the principle of interference with Huygens's secondary wavelets to compute diffraction and interference patterns in full detail, and whose recognition that light is a transverse wave (developed with Arago) tied interference firmly to polarisation: only components sharing a polarisation can interfere. By the middle of the nineteenth century the wave theory was secure, and interference had become a tool. Hippolyte Fizeau and Léon Foucault used interferometric methods to compare the speed of light in different media, and Fizeau exploited the visibility of fringes to infer stellar properties - an idea far ahead of its time.

The instrument that defines the modern era is the Michelson interferometer. Albert A. Michelson, seeking ever more precise measurements, built in the 1880s a device that splits a beam in two, sends the halves along perpendicular arms, and recombines them so that any difference in optical path appears as a shift of the fringes. With it, Michelson and Edward Morley performed in 1887 their famous experiment to detect the motion of the Earth through the luminiferous aether; the null result Michelson and Morley (1887) became one of the experimental pillars on which Einstein's special relativity would later rest. Michelson also turned his interferometer into a metrological standard, measuring the length of the standard metre in wavelengths of light, work for which he received the Nobel Prize in 1907.

Almost simultaneously, Charles Fabry and Alfred Perot introduced in 1899 an interferometer of a different kind, based on multiple reflections between two highly reflecting parallel surfaces Fabry and Perot (1899). The FabryPerot interferometer produces extremely sharp transmission fringes and offers very high spectral resolving power; it is the direct ancestor of the optical resonators and cavities that define the modes of every laser. Other configurations followed - the MachZehnder interferometer, with its separated paths ideal for probing transparent samples, and the Sagnac interferometer, sensitive to rotation and the heart of modern optical gyroscopes.

The visibility of the fringes in any of these instruments is not merely an experimental nuisance: it is a direct measure of the coherence of the light, and it was precisely the analysis of fringe visibility that motivated the statistical theory of coherence developed in a later chapter. In the twentieth century, interferometry reached almost unimaginable sensitivity: the Laser Interferometer Gravitational-Wave Observatory (LIGO), a pair of kilometre-scale Michelson interferometers, detected in 2015 the passage of a gravitational wave by measuring a change in arm length thousands of times smaller than a proton Abbott et al. (2016). From Young's two slits to the detection of colliding black holes, the interferometer has remained the same idea - read the phase of a wave by letting it interfere with itself - refined to extraordinary perfection.

Chapter 5

Theory of Diffraction

Where geometry promised sharp shadows, light insisted on painting fringes.

"Diffraction is any deviation of light rays from rectilinear paths which cannot be interpreted as reflection or refraction."

– Arnold Sommerfeld

5.1 A historical context of diffraction theory: from shadows to the vector world

Diffraction - that phenomenon in which light bends around obstacles and paints patterns of light and shadow where geometry promised only sharp silhouettes - is one of the richest stories in physics. Its evolution spans from Renaissance observations to the vector equations that today model nanotechnologies, and the tale shows how humankind slowly learned to tame the undulations of light.

The story begins around 1660, when the Italian Jesuit Francesco Maria Grimaldi observed something disconcerting: when sunlight was passed through a small hole into a dark room, the edge of the shadow cast by an object was not sharp; instead, coloured fringes and illuminated zones appeared where the ray geometry of Euclid predicted darkness. In his posthumous book *Physico-Mathesis de Lumine* (1665) Grimaldi (1665), Grimaldi coined the term *diffractio* (from the Latin *dis-* + *fractus*, "to break into pieces"), suggesting that light was not merely reflected or refracted but also "broke" when skirting obstacles. Nevertheless, in an age dominated by Newton's corpuscular theory, these ideas remained in the penumbra.

The first theory capable of qualitatively explaining diffraction arrived in 1690 with Christiaan Huygens. In his *Traité de la Lumière* Huygens (1690), he proposed that every point of a wavefront acts as a source of new secondary wavelets (the **Huygens principle**), and that the envelope of these wavelets generates the propagating front. Although he did not explicitly incorporate interference - which limited the predictive power of his model - his construction already suggested that light could deviate when passing near edges.

The true leap occurred in the nineteenth century. Thomas Young, with his double-slit experiment (1801) Young (1804), demonstrated that light interferes with itself; but it was Augustin-Jean Fresnel who, between 1815 and 1820, fused the Huygens principle with the principle of interference, creating the **HuygensFresnel principle**. In his *Mémoire sur la diffraction de la lumière* Fresnel (1819), Fresnel cast diffraction as an integral superposition of secondary wavelets with relative phases. His expression for the intensity at a point P,

$$I(P) \propto \left| \int_{\text{aperture}} \frac{A e^{i(kr-\omega t)}}{r} K(\theta)\, dS \right|^2, \tag{5.1}$$

where $K(\theta)$ is the *obliquity factor*, accounted quantitatively for the observed diffraction patterns. The climactic moment came in 1818: Siméon Denis Poisson, using Fresnel's theory, predicted that a circular disc should project a bright spot at the very centre of its shadow (the **PoissonArago spot**). When Dominique Arago verified it experimentally, the wave theory triumphed over Newtonian corpuscularism.

Powerful as it was, Fresnel's theory remained heuristic. In 1883 Gustav Kirchhoff gave it mathematical rigour through the wave equation and Green's theorem. In his memoir *Zur Theorie der Lichtstrahlen* Kirchhoff (1883), he derived the **Kirchhoff integral formula**

$$U(P) = \frac{1}{4\pi} \iint_{\Sigma} \left(U \frac{\partial}{\partial n} \left(\frac{e^{iks}}{s} \right) - \frac{e^{iks}}{s} \frac{\partial U}{\partial n} \right) d\Sigma, \tag{5.2}$$

where U is the scalar field (assuming monochromatic light and uniform polarisation), s the distance from the source point to P, and Σ the integration surface. This formulation - although it assumes mutually inconsistent boundary conditions, as pointed out by Henri Poincaré - became the cornerstone of the **scalar theory of diffraction**, valid whenever polarisation effects are negligible.

Kirchhoff's inconsistencies were resolved in 1896 by Arnold Sommerfeld Sommerfeld (1896). By eliminating the need to impose Dirichlet and Neumann conditions simultaneously, he derived rigorous solutions for the diffraction by a plane screen:

$$U(P) = \frac{1}{i\lambda} \iint_{\text{aperture}} U_0 \frac{e^{ikr}}{r} \cos\theta \, dS \quad \text{(RayleighSommerfeld diffraction).} \tag{5.3}$$

This approach, later detailed in his *Vorlesungen über Theoretische Physik* Sommerfeld (1954), allowed diffraction patterns to be computed without unphysical infinities, extending the theory to arbitrary apertures.

The scalar theory, however, ignored the vector nature of light. In 1908 Gustav Mie, studying the diffraction (scattering) of light by spheres - today called **Mie scattering** - solved Maxwell's equations directly for the electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ Mie (1908). Shortly afterwards, Peter Debye, in his 1909 thesis, introduced inhomogeneous vector waves to describe the focusing of beams, laying the foundations of the **vector theory of diffraction**.

The modern synthesis was consolidated by Max Born and Emil Wolf in *Principles of Optics* (1959) Born and Wolf (1999), where it was shown that the scalar approximation fails for systems with high numerical aperture, anisotropic or chiral media, and structures smaller than the wavelength. The vector theory, built directly on Maxwell's curl equations

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \tag{5.4}$$

allows one to model polarisation effects, surface plasmons and metasurfaces.

The invention of the laser in 1960 boosted practical applications. Joseph Goodman, in *Introduction to Fourier Optics* (1968) Goodman (2005), reinterpreted diffraction as a **linear, space-invariant system**, in which the far-field pattern is the Fourier transform of the aperture. This viewpoint enabled the design of holograms, diffractive lenses and computational aberration correction. Today the field enjoys a golden age: sub-wavelength **metasurfaces** that shape wavefronts through vector diffraction Yu et al. (2011), electron and neutron diffraction used to map atomic structures, and diffraction in nonlinear media probed with femtosecond lasers. From Grimaldi's fringes to engineered wavefronts, diffraction theory has continuously challenged and refined our understanding of light. In every diffracted corner we still hear an echo of Fresnel's question: how far can light bend before it breaks our certainties?

5.2 Scalar theory of diffraction

Electromagnetic waves are inherently vectorial: at each point $\vec{r}$ the electric field may exhibit a different state of polarisation. This property is expressed mathematically as

$$\vec{E}(\vec{r},t) = E(\vec{r},t)\,\hat{\vec{e}}(\vec{r},t), \tag{5.5}$$

where $E(\vec{r},t)$ denotes the amplitude of the field and $\hat{\vec{e}}(\vec{r},t)$ represents the polarisation vector, which besides varying in space may evolve in time in a stochastic manner.

The associated wave equation then takes the form

$$\nabla^2 \left[E(\vec{r},t)\hat{\vec{e}}(\vec{r},t)\right] - \mu\varepsilon\frac{\partial^2}{\partial t^2}\left[E(\vec{r},t)\hat{\vec{e}}(\vec{r},t)\right] = 0, \tag{5.6}$$

whose direct solution presents notable complexities. To tackle this problem one resorts to simplifying strategies, such as decomposing the field into scalar components. These scalar solutions may afterwards be combined to reconstruct the full vector behaviour.

The temporal variation of amplitude, phase and polarisation is analysed rigorously through coherence theory (a topic explored in depth in Chapter 6). Nevertheless, under certain approximations the temporal dependence of the amplitude is attributed solely to the intrinsic oscillations of the field, while the polarisation is regarded as static. The electric field then simplifies to

$$\vec{E}(\vec{r},t) = E(\vec{r},t)\,\hat{\vec{e}}(\vec{r}), \tag{5.7}$$

allowing its expansion in an orthogonal polarisation basis,

$$\vec{E}(\vec{r},t) = E_1(\vec{r},t)\hat{\vec{e}}_1 + E_2(\vec{r},t)\hat{\vec{e}}_2. \tag{5.8}$$

Here $\hat{\vec{e}}_1$ and $\hat{\vec{e}}_2$ correspond to opposite polarisation states on the Poincaré sphere. Each scalar component E_i individually satisfies the wave equation

$$\nabla^2 E_i - \mu\varepsilon\frac{\partial^2 E_i}{\partial t^2} = 0 \quad (i = 1, 2), \tag{5.9}$$

thereby decoupling the vector problem into manageable scalar equations.

In situations where one component is negligible (for instance $E_2 \to 0$), the field reduces to

$$\vec{E}(\vec{r},t) = E(\vec{r},t)\,\hat{\vec{e}}, \tag{5.10}$$

defining a purely scalar case. This formalism underlines how the general vector description can emerge from the controlled superposition of scalar solutions.

5.3 The scalar wave equation (Helmholtz equation)

Let $\Psi(\mathbf{r},t)$ be an arbitrary component of the electromagnetic field. This quantity satisfies the classical wave equation

$$\nabla^2\Psi(\mathbf{r},t) - \mu\varepsilon\frac{\partial^2\Psi(\mathbf{r},t)}{\partial t^2} = 0, \tag{5.11}$$

where Ψ represents a polychromatic field. To decompose the field temporally we apply the Fourier transform

$$\Psi(\mathbf{r}, t) = \int_{\mathbb{R}^+} U(\mathbf{r}, \nu) \exp(-2\pi i \nu t)\, d\nu. \tag{5.12}$$

Substituting (5.12) into (5.11), the equation is re-expressed as

$$\int_{\mathbb{R}^+} \left[\nabla^2 U(\mathbf{r}, \nu) + \frac{\omega^2}{v^2} U(\mathbf{r}, \nu)\right] \exp(-2\pi i \nu t)\, d\nu = 0, \tag{5.13}$$

which leads directly to the Helmholtz equation for each spectral component:

$$(\nabla^2 + k^2) U(\mathbf{r}, \nu) = 0. \tag{5.14}$$

Here $k = 2\pi/\lambda$ denotes the wavenumber, and $\lambda = c/(n\nu)$ is the wavelength in a medium of refractive index n. This formulation describes mathematically how each monochromatic component of the field (characterised by its complex amplitude U) obeys an equation independent of the temporal frequency.

5.4 The Kirchhoff scalar boundary-value problem

The scalar theory of diffraction, based on solutions of the wave equation under boundary conditions, describes the propagation of fields in paraxial approximations or with smooth phase modulations. Despite its scalar formulation, the formalism - developed by Kirchhoff by means of Green's functions - is adaptable to vector fields.

The original inconsistency of Kirchhoff, pointed out by Poincaré, lies in the over-determination of the boundary conditions (prescribing both the field and its normal derivative on a plane), which violates the uniqueness theorems of differential equations. Sommerfeld resolved this by redefining the Green's function so as to eliminate the redundancy, guaranteeing mathematical consistency.

We therefore start from the physical entity under study - the electromagnetic field - and as a first step it is appropriate to recall its wave equation

$$\nabla^2 \vec{E}(\vec{r}, t) - \frac{1}{v^2} \frac{\partial^2 \vec{E}(\vec{r}, t)}{\partial t^2} = 0, \tag{5.15}$$

which is of a vector nature. We may restrict our phenomenon to the study of the propagation of its components, grouping the vector character into the polarisation vector $\hat{e}(\vec{r}, t)$. In order to avoid possible spacetime correlations between the evolution of the components and the polarisation of the field, the dependence of the polarisation vector on these variables is dropped, so that

$$\left(\nabla^2 E(\vec{r}, t) - \frac{1}{v^2} \frac{\partial^2 E(\vec{r}, t)}{\partial t^2}\right) \hat{e} \quad \longrightarrow \quad (\nabla^2 + k^2) E(\vec{r}, t) = 0, \tag{5.16}$$

thus obtaining the Helmholtz equation for the scalar electric field.

5.4.1 Green's theorem

Green's theorem, a direct corollary of Gauss's theorem, establishes fundamental relations between volume and surface integrals for scalar fields. Let $f(\vec{r})$ and $g(\vec{r})$ be scalar functions with continuous second derivatives in a domain V. We start from the differential identity

$$\nabla \cdot [f(\vec{r})\nabla g(\vec{r})] = f(\vec{r})\nabla^2 g(\vec{r}) + \nabla f(\vec{r}) \cdot \nabla g(\vec{r}). \tag{5.17}$$

Subtracting the analogous expression with f and g interchanged, we obtain

$$\nabla \cdot [f\nabla g - g\nabla f] = f\nabla^2 g - g\nabla^2 f. \tag{5.18}$$

Integrating this relation over a volume V and applying the Gauss divergence theorem,

$$\int_V \nabla \cdot [f(\vec{r})\nabla g(\vec{r}) - g(\vec{r})\nabla f(\vec{r})]\, dV = \int_V \left(f\nabla^2 g - g\nabla^2 f\right) dV, \tag{5.19}$$

which allows the volume integral on the right-hand side to be converted into a surface integral over ∂V:

$$\oint_{\Sigma(V)} [f(\vec{r})\nabla g(\vec{r}) - g(\vec{r})\nabla f(\vec{r})] \cdot d\vec{\sigma} = \int_V \left(f\nabla^2 g - g\nabla^2 f\right) dV, \tag{5.20}$$

where:

- $d\vec{\sigma} = \hat{n}\, d\sigma$ is the differential surface element oriented along the outward normal $\hat{n}$ to the surface $\Sigma(V)$.
- $\frac{\partial}{\partial n} = \hat{n} \cdot \nabla$ denotes the directional derivative along $\hat{n}$.
- The operator ∇ acts exclusively on the spatial coordinates $\vec{r}$.

This identity allows the solution of inhomogeneous differential equations (e.g. the Helmholtz equation) to be expressed in terms of integrals over the sources and the boundary conditions. The presence of $\hat{n}$ emphasises that the surface contribution depends critically on the geometry of the volume V.

5.4.2 The HelmholtzKirchhoff integral theorem

The HelmholtzKirchhoff integral theorem adapts Green's formalism to the propagation of electromagnetic fields. Starting from Green's identity and adding the term $k^2 fg$ to both sides, one obtains

$$\oint_{\Sigma(V)} [f(\vec{r})\nabla g(\vec{r}) - g(\vec{r})\nabla f(\vec{r})] \cdot d\vec{\sigma} = \int_V f(\vec{r}) \left(\nabla^2 + k^2\right) g(\vec{r})\, dV - \int_V g(\vec{r}) \left(\nabla^2 + k^2\right) f(\vec{r})\, dV, \tag{5.21}$$

where f and g are scalar functions. Choosing $f = U(\vec{r}, \nu)$, a solution of the Helmholtz equation, and g as an arbitrary smooth function, the expression simplifies to

$$\oint_{\Sigma(V)} [U(\vec{r},\nu)\nabla g(\vec{r}) - g(\vec{r})\nabla U(\vec{r},\nu)] \cdot d\vec{\sigma} = \int_V U(\vec{r},\nu) \left(\nabla^2 + k^2\right) g(\vec{r})\, dV. \tag{5.22}$$

The key lies in selecting g as the Green's function $G(\vec{r},\vec{r}')$, defined by

$$\left(\nabla^2 + k^2\right) G(\vec{r},\vec{r}') = 4\pi\delta(\vec{r}-\vec{r}'). \tag{5.23}$$

Remarkably, we may add an arbitrary number of Green's functions while keeping the solution invariant. What we can do is add an infinity of image charges, but *outside* the integration volume bounded by the surface $\Sigma(V)$, as shown in Figure 5.1.

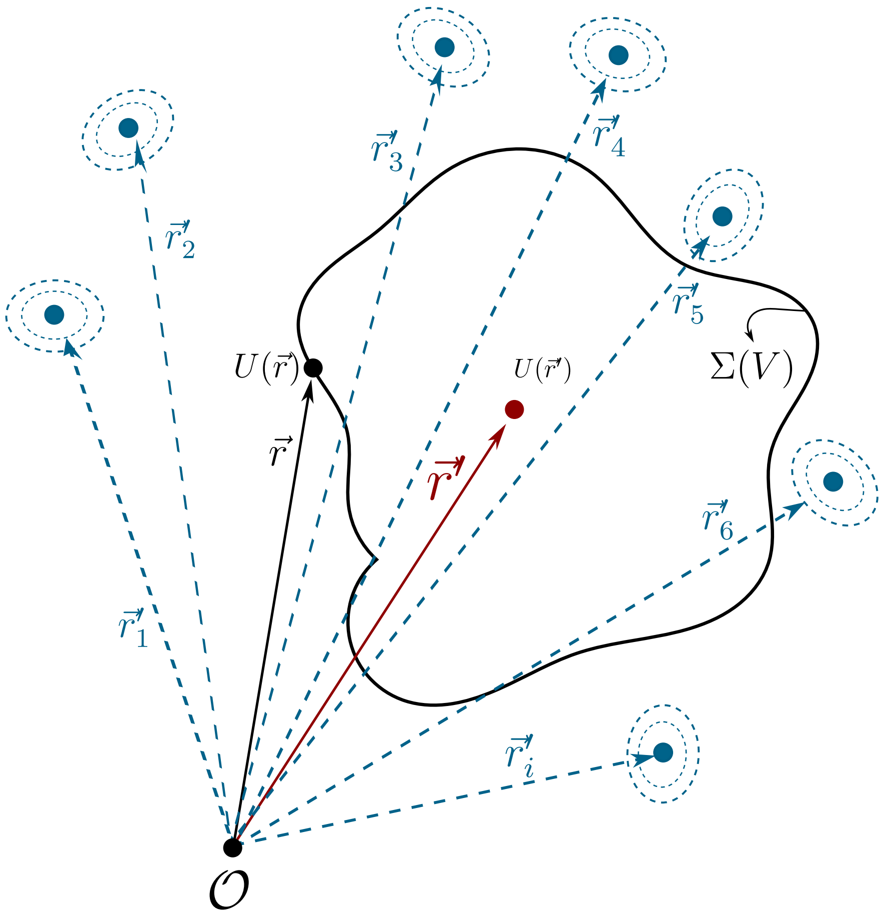


Figure 5.1: On the boundary $\Sigma(V)$ the contribution of the field components $U(\vec{r})$ is shown; inside the region the total field is built from the contribution of all the points on the boundary. One may add infinitely many point Green charges of arbitrary propagation nature; the condition for the invariance of equation (5.23) is that these image charges lie outside $\Sigma(V)$.

In this spirit, equation (5.23) becomes

$$(\nabla^2 + k^2)G(\vec{r}, \vec{r}') = 4\pi\delta(\vec{r} - \vec{r}') + 4\pi \sum_{i=1}^{\infty} \delta(\vec{r} - \vec{r}_i''). \tag{5.24}$$

The explicit solution for a homogeneous medium of equation (5.23) is

$$G(\vec{r}, \vec{r}') = \frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r} - \vec{r}'\|}, \tag{5.25}$$

corresponding to an outgoing spherical wave centred at $\vec{r}'$. Substituting G into Green's identity, one derives

$$U(\vec{r}', \nu) = \frac{1}{4\pi} \oint_{\Sigma(V)} \left[U(\vec{r}, \nu)\nabla G(\vec{r}, \vec{r}') - G(\vec{r}, \vec{r}')\nabla U(\vec{r}, \nu)\right] \cdot d\vec{\sigma}, \tag{5.26}$$

known as the HelmholtzKirchhoff integral formula. This equation allows $U(\vec{r}', \nu)$ to be computed at any point of the volume V by means of an integral over its boundary $\Sigma(V)$.

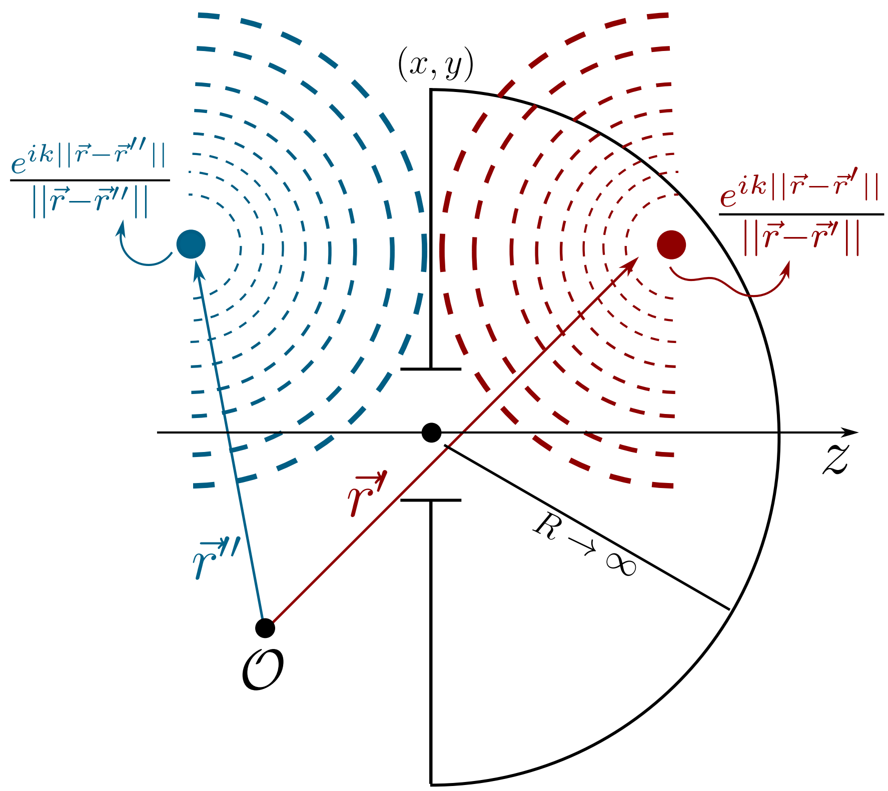


Figure 5.2: Representation of the Green's functions for point sources. Under these boundary conditions the Green's functions behave as point charges, one the mirror image of the other.

Before continuing, it is worth noting that the point-charge nature of the Green's functions $G(\vec{r}, \vec{r}')$ is an assumption taken for granted; there is, however, no restriction on their physical nature. As is well known in electromagnetic theory, radiation sources may have any volume and shape, so our problem (Figure 5.2) can be made more general, as shown in Figure 5.3, and the differential equation for Green's functions with volume would read

$$\left(\nabla^2 + k^2\right) G(\vec{r}, \vec{r}') = 4\pi\rho(\vec{r} - \vec{r}'_+) - 4\pi\rho(\vec{r} - \vec{r}'_-). \tag{5.27}$$

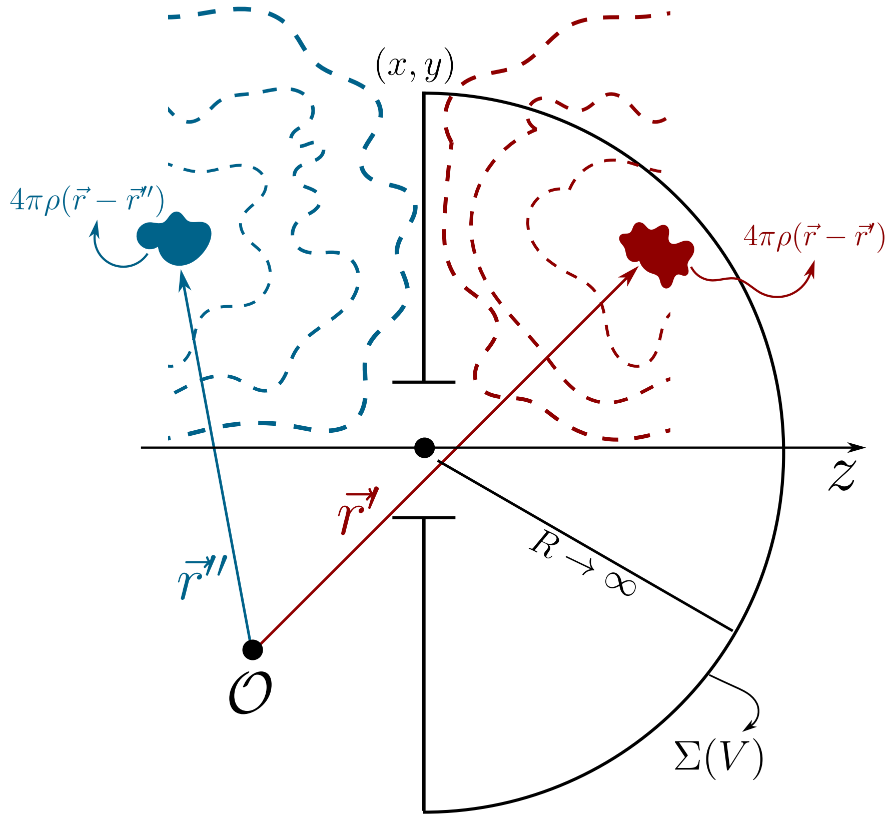


Figure 5.3: Representation of the Green's functions for non-point sources. Under these boundary conditions the Green's functions still behave as mirror images, but now with given volume and geometry; the only constraint on the emission of these sources is that, on the boundary $z = 0$, the total field must vanish, as required by the assumed Dirichlet conditions.

Let us continue with the development for point Green's functions. To simplify the evaluation of the integral, we consider a volume V bounded by an infinite plane Σ_1 ($z = 0$) and a hemisphere Σ_2 of radius R (see Figure 5.2). The contribution of Σ_2 vanishes as $R \to \infty$ (owing to the Sommerfeld radiation condition), reducing the expression to

$$U(\vec{r}', \nu) = \frac{1}{4\pi} \int_{\Sigma_1} \left[U(\vec{r}, \nu) \frac{\partial G}{\partial n} - G(\vec{r}, \vec{r}') \frac{\partial U}{\partial n} \right] d\sigma, \tag{5.28}$$

where $\frac{\partial}{\partial n} = \hat{n} \cdot \nabla$ denotes the normal derivative. However, this formulation requires specifying simultaneously U and $\partial U/\partial n$ on Σ_1, which over-determines the problem. This difficulty motivates the introduction of physically realistic boundary conditions, such as those of Kirchhoff, in order to resolve the ambiguity.

This problem is aggravated by a theorem of complex analysis, attributed to Riemann, which states that if a function and its normal derivative vanish on an open region of a plane, then the function must be identically zero throughout the domain. This restriction motivates the strategic need to eliminate one of the terms in the HelmholtzKirchhoff surface integral.

The key is to exploit the degrees of freedom inherent in the choice of the auxiliary Green's function G. Kirchhoff initially proposed the intuitive boundary conditions

- $U = 0$ outside the aperture on Σ (blocked field),
- $\partial U/\partial n = 0$ on opaque regions (vanishing normal derivative).

Although these conditions reproduced the experimental results of diffraction, they introduced mathematical inconsistencies by violating the CauchyRiemann uniqueness theorem. Poincaré showed that such assumptions generate contradictions in the mixed derivatives of the field. The resolution came with Sommerfeld, who redefined G by means of a double-layer (mirror) Green's function, ensuring that

- $G = 0$ on Σ (Dirichlet condition),
- $\partial G/\partial n$ remains non-singular,

thus eliminating the redundancy in the boundary conditions and preserving the self-consistency of the formalism.

5.4.3 The RayleighSommerfeld formulation

Returning to equation (5.26) and choosing the Dirichlet condition for the function G, we have

$$U(\vec{r}', \nu) = \frac{1}{4\pi} \int_{\Sigma_1} [U(\vec{r}, \nu) \nabla G \cdot \hat{n}] \, d\sigma. \tag{5.29}$$

Given the integration surface under consideration, we may express (5.25) as

$$G(\vec{r}, \vec{r}') = \frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|} - \frac{e^{ik\|\vec{r}-\vec{r}''\|}}{\|\vec{r}-\vec{r}''\|}, \tag{5.30}$$

where

$$\|\vec{r}-\vec{r}''\| = \|\vec{r}-\vec{r}'\|, \qquad x'' = x', \quad y'' = y', \quad z'' = -z', \tag{5.31}$$

since $\vec{r}''$ is the mirror image of $\vec{r}'$ across the plane Σ_1. It follows that

$$\nabla G \cdot \hat{z} = 2 \left[ik - \frac{1}{\|\vec{r}-\vec{r}'\|} \right] \frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|} \cos(\vec{r}-\vec{r}', \hat{z}), \tag{5.32}$$

and with this, (5.29) becomes

$$U(\vec{r}',\nu)=\frac{1}{2\pi}\int_{\Sigma}U(\vec{r},\nu)\left[ik-\frac{1}{\|\vec{r}-\vec{r}'\|}\right]\frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|}\cos(\vec{r}-\vec{r}',\hat{z})\,d\sigma. \tag{5.33}$$

We may now make the approximation appropriate to the RayleighSommerfeld regime, in which $\|\vec{r}-\vec{r}'\|\gg\lambda$, obtaining

$$U(\vec{r}',\nu)=\frac{ik}{2\pi}\int_{\Sigma}U(\vec{r},\nu)\frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|}\cos(\vec{r}-\vec{r}',\hat{z})\,d\sigma, \tag{5.34}$$

a mathematical expression that embodies the HuygensFresnel principle: as can be appreciated, there is one spherical wave per point of the integration surface.

5.5 The Fresnel approximation

Starting from the field

$$U(\vec{r}')=\frac{ik}{2\pi}\int_{\Sigma}U(\vec{r})\frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|}\cos(\vec{r}-\vec{r}',\hat{z})\,\mathrm{d}\sigma, \tag{5.35}$$

it is possible to carry out the so-called mid-field or Fresnel approximation. To this end we take

$$\begin{aligned}\|\vec{r}-\vec{r}'\|&=\left[(x-x')^2+(y-y')^2+(z-z')^2\right]^{1/2}\\&=z\left(1+\frac{(x-x')^2+(y-y')^2}{z^2}\right)^{1/2}\\&\approx z+\frac{(x-x')^2+(y-y')^2}{2z}.\end{aligned} \tag{5.36}$$

It is clear that, for the factor $\frac{1}{\|\vec{r}-\vec{r}'\|}$, an expansion up to the first term of (5.36) suffices; this is not enough for the factor $e^{ik\|\vec{r}-\vec{r}'\|}$, however, since the latter has oscillatory behaviour over its domain, so both terms are required. We additionally take $\cos(\vec{r}-\vec{r}',\hat{z})=\frac{z}{\|\vec{r}-\vec{r}'\|}\approx 1$.

In this spirit, we obtain an expression for the field of the form

$$U(\vec{r}')=\frac{ie^{ikz}}{\lambda z}\int U(\vec{r})\exp\left(\frac{ik}{2z}((x-x')^2+(y-y')^2)\right)d\sigma, \tag{5.37}$$

which describes the propagation or diffraction of the electromagnetic field in the Fresnel regime. It is worth stressing that equation (5.37) describes a parabolic wave, and although, at least conceptually, such propagation is valid for a mid-field, in practice it has proved more than sufficient for sufficiently short distances.

5.6 The Fraunhofer approximation

Finally, from expression (5.37) a second approximation may be carried out, corresponding to the far field or Fraunhofer regime, in which the observation distances are considerably larger than the area over which one wishes to study the propagated field (whether through a slit or another object). We focus on the factor

$$\exp\left(\frac{ik}{2z}((x-x')^2+(y-y')^2)\right)=\exp\left(\frac{ik}{2z}(x^2+y^2)\right)\exp\left(\frac{ik}{2z}(x'^2+y'^2)\right)\exp\left(-\frac{ik}{z}(xx'+yy')\right), \tag{5.38}$$

where we may make the far-field assumption $\exp\left(\frac{ik}{2z}(x^2+y^2)\right)\approx 1$, so that we obtain

$$U(\vec{r}')=\frac{e^{ikz}}{\lambda z}\exp\left(\frac{ik}{2z}(x'^2+y'^2)\right)\int U(\vec{r})\exp\left(-\frac{ik}{z}(xx'+yy')\right)d\sigma. \tag{5.39}$$

There we can still appreciate a quadratic phase term outside the integral, usually called the curvature factor, which shows precisely the dephasing induced by the difference of optical path between a plane called the observation screen and a spherical surface of radius z (the propagation distance of the field along the optical axis) known as the cardinal sphere. The introduction of this spherical surface is rather useful when one wishes to describe Fraunhofer diffraction on a screen as a Fresnel propagation through two confocal cardinal spheres, as shown in Figure 5.4.

This makes more sense once we analyse the effect of each propagation step. It is clear that, without imposing far-field approximations from the outset, propagating the field at $\mathcal{A}$ towards $\mathcal{F}$ with the Fresnel approximation, we would obtain

$$U_{\mathcal{F}}(\vec{r})=\frac{ie^{ikz}}{\lambda z}\exp\left(\frac{ik}{2z}(x^2+y^2)\right)\int u_{\mathcal{A}}\exp\left(\frac{ik}{2z}(x'^2+y'^2)\right)\exp\left(-\frac{ik}{z}(xx'+yy')\right)d\sigma', \tag{5.40}$$

where we may associate $u_{\mathcal{A}}$, i.e. the field on the sphere $\mathcal{A}$, with the field on $\mathcal{P}$ multiplied by its corresponding quadratic factor $u_{\mathcal{P}}=u_{\mathcal{A}}(\vec{r}')\exp\left(\frac{ik}{2z}(x'^2+y'^2)\right)$, while the term $\exp\left(\frac{ik}{2z}(x^2+y^2)\right)$ may be understood as the factor resulting from passing the field from the sphere $\mathcal{F}$ to the screen $\mathcal{Q}$, so that the field on $\mathcal{F}$ would correspond to

$$u_{\mathcal{Q}}(\vec{r})=\frac{ie^{ikz}}{\lambda z}\int u_{\mathcal{P}}\exp\left(-\frac{ik}{z}(xx'+yy')\right)d\sigma', \tag{5.41}$$

thus recovering a Fraunhofer expression for diffraction on $\mathcal{Q}$. It should be noted that one must check whether the study of the propagated field is valid within a paraxial regime, where the field is confined to the neighbourhood of the optical axis and hence the curvature of the cardinal sphere does not differ much from the plane (the screen). In cases where the curvature factor ceases to be negligible, the paraxial regime loses validity and the problem must be addressed from a meta-axial standpoint.

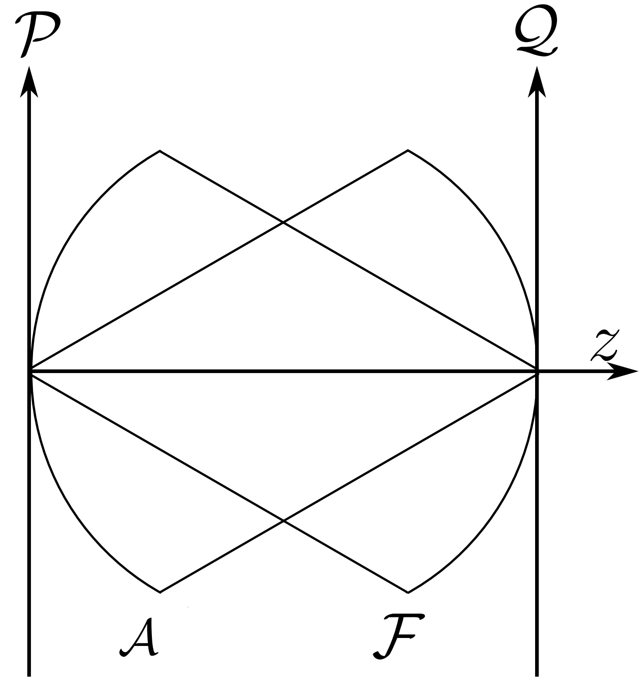


Figure 5.4: Scheme of the confocal cardinal spheres, with radii equal to the propagation distance z of the field.

Studying the propagation of the field through Fresnel diffraction is of great importance because, as will be discussed later, it bears a close relationship with what is known as the fractional Fourier transform, a tool that arises naturally when one chains paraxial propagation steps.

Chapter 6

Theory of Coherence

Coherence is the thread that weaves order out of the chaos of random light.

"Time advances in the direction in which correlations increase."
– attributed to Ludwig Boltzmann

6.1 A historical context of coherence theory

The history of coherence theory is an intricate and fascinating tale rooted in mathematics, statistics and physics, branching in classical and quantum directions with profound implications for our understanding of the world. To tell it, we must go back to the dawn of probability theory, where the idea of measuring relations between events and variables first took shape — a concept that, over the centuries, would evolve into the refined notions of correlation and, eventually, coherence as we understand it today.

It all began in the seventeenth century, when thinkers such as Blaise Pascal and Pierre de Fermat, in their correspondence on games of chance, laid the foundations of probability theory. Their reflections on expected value and combinatorics opened the way to Jacob Bernoulli, who in 1713, in his posthumous *Ars Conjectandi* Bernoulli (1713), formulated the law of large numbers, linking theoretical probability with observed frequency. These advances were consolidated by Pierre-Simon Laplace in 1812 with his *Théorie Analytique des Probabilités* Laplace (1812), where he unified scattered ideas and developed Bayesian inference, essential for understanding how variables relate in complex systems. Yet a mathematical language was still missing to quantify the interdependence between phenomena.

That language began to take form in the nineteenth century. Carl Friedrich Gauss, in his 1809 study of errors in astronomical observations (*Theoria Motus*) Gauss (1809), introduced the method of least squares, a tool that not only optimised measurements but also revealed hidden patterns in correlated data. But it was Francis Galton, in the 1880s, who coined the term "correlation" as we know it. Motivated by his studies of biological inheritance, Galton sought to measure how characteristics such as the height of parents and children varied together Galton (1888). His work inspired Karl Pearson, who in 1896 formalised the Pearson correlation coefficient Pearson (1896), a number between -1 and 1 quantifying the linear relationship between variables. This was a conceptual leap: no longer merely intuiting connections, but assigning them a rigorous numerical value.

In parallel, in the realm of physics, another kind of correlation — coherence — began to reveal itself through the study of light. In 1801 Thomas Young performed his celebrated double-slit experiment, showing that light interferes with itself, creating patterns of bright and dark fringes. This phenomenon, inexplicable under Newton's corpuscular theory, suggested that light was a wave whose phases could be correlated in space and time. Augustin-Jean Fresnel, between 1815 and 1820, developed a mathematical theory of light as a transverse wave, explaining not only interference but also polarisation. Coherence — the ability of waves to maintain a constant phase relationship — was, however, not yet understood in statistical terms.

The crucial step came in the first decades of the twentieth century, when optics met the need to describe realistic light sources, such as the Sun or incandescent lamps, whose emissions are partly random. In 1934 the Dutch physicist Pieter Hendrik van Cittert published a seminal work on the propagation of coherence from an incoherent source van Cittert (1934), followed in 1938 by Frits Zernike Zernike (1938), who formulated the theorem that bears both their names: the **van Cittert–Zernike theorem**. This result established that the light from an extended, incoherent source develops measurable spatial correlations as it propagates to a distant plane. It was the first time coherence had been quantified mathematically as a statistical property of optical fields.

Classical coherence theory reached maturity in the 1950s through Max Born and Emil Wolf, whose treatise *Principles of Optics* (1959) Born and Wolf (1999) became the bible of theoretical optics. In it they defined the **mutual coherence function**

$$\Gamma(\mathbf{r}_1, \mathbf{r}_2, \tau), \tag{6.1}$$

which measures the correlation between the electric field at two points in space ($\mathbf{r}_1$ and $\mathbf{r}_2$) at two instants separated by a delay τ. This function encapsulates both the temporal coherence (how long the light maintains a stable phase, related to its monochromaticity) and the spatial coherence (how correlated the vibrations are at different positions). Leonard Mandel and Emil Wolf, in the following decades, deepened these ideas in their treatise *Optical Coherence and Quantum Optics* Mandel and Wolf (1995), introducing concepts such as the coherence time and the coherence length, key for practical applications.

Coherence was not a mere abstraction: the Michelson interferometer, developed in the 1890s, exploited temporal coherence to measure wavelengths with unprecedented precision. But the technological milestone arrived in 1960 with the invention of the laser, whose light — highly coherent in time and space — is the product of stimulated emission, a phenomenon predicted by Einstein in 1917. Lasers not only confirmed the theories of coherence but opened new frontiers in communications, medicine and industry.

While classical coherence was being consolidated, a parallel revolution was beginning in the quantum world. In the 1960s quantum coherence found its theoretical framework. Roy J. Glauber, in a series of articles published in *Physical Review* (1963) Glauber (1963), reformulated optics from a quantum perspective. He introduced the **coherent states**, quantum analogues of classical waves with well-defined phase, and defined the quantum correlation functions $g^{(n)}$, which measure the probability of detecting several correlated photons. His work, awarded the Nobel Prize in 2005, not only explained phenomena such as photon antibunching but also laid the foundations of modern quantum optics. The mathematical backbone connecting the statistical description of fluctuations with their spectral content — the Wiener–Khinchin theorem Wiener (1930); Khinchin (1934) — remains, as we shall see, one of the central pillars of the whole edifice.

In the twenty-first century, coherence has acquired a new flavour: the idea that it is a quantifiable and exploitable **resource**, akin to energy or information Baumgratz et al. (2014), integrated into the broader scheme of quantum computation. In summary, the history of coherence is a journey from the abstract mathematics of statistical correlations to the engineering of fragile quantum states. In its classical guise it explains why stars twinkle, how lasers work and why holograms capture three-dimensional images; in its quantum facet it challenges our intuition. And although decoherence reminds us that these phenomena are ephemeral, it also underlines their value: in a world where noise and chance seem to reign, coherence — classical or quantum — is the thread that weaves order out of chaos.

6.2 How is statistical similarity quantified?

Suppose we have a completely random phenomenon — such as the electromagnetic radiation emitted by an incandescent bulb that is not in thermodynamic equilibrium, the number of cars in some capital city throughout the day, the motion of an ion in a plasma, or something as simple as throwing a ball into the air under non-ideal conditions.

When we wish to analyse statistically a single object or random variable, the familiar tools appear: the probability density, the mean value, the standard deviation, the kurtosis, and so on. What happens when we want to study at least two random variables and how they relate? When we wish to study the statistics of two or more random variables we must study the statistical similarity between them; but statistical similarity is a rather broad concept, since we may have similarity in the mean values, or in the standard deviations, or in the probability densities. To address this problem we return to linear algebra.

If we have two vectors $\vec{A}, \vec{B} \in \mathbb{R}^n$, the dot product $\vec{A} \cdot \vec{B}$ is interpreted as the magnitude of the projection of $\vec{A}$ onto $\vec{B}$ and vice versa; it can also be read as *how much of $\vec{A}$ lies along $\vec{B}$*. This behaviour can be extrapolated to continuous vector spaces, where one usually employs the notion of inner product: if f_A, f_B are two elements of a Hilbert space L^2 of dimension 1, one can define an inner product as

$$\langle f_A, f_B \rangle = \int_{\mathbb{R}} f_A(x) f_B(x) W(x)\, dx, \tag{6.2}$$

where $W(x)$ is some weight function.[1] The interpretation is still valid: this can be read as how much the function f_A over all the reals coincides with the function f_B over all the reals.

To reach the notion of correlation from linear algebra we use a bit of bra–ket notation, where the ket $|f_A\rangle$ represents a vector of an arbitrary Hilbert space L^2 and the bra $\langle f_A|$ represents an element of the dual of the same Hilbert space,[2] and the inner product between them $\langle f_A, f_A\rangle = \langle f_A|f_A\rangle$ equals the probability density of finding f_A in a state $q \pm \triangle q$, where q represents some generalised coordinate. Suppose that $f_A, f_B \in L^2$ have the form

$$|f_A\rangle = \hat{A}\,|\Psi_A\rangle\,, \quad |f_B\rangle = \hat{B}\,|\Psi_B\rangle\,, \tag{6.3}$$

where $\hat{A}, \hat{B}$ are the operators of the physical quantities of $|\Psi_A\rangle\,, |\Psi_B\rangle$, so that we may interpret $|f_A\rangle\,, |f_B\rangle$ as the wave functions steered by their expected value. Using the inner product (6.2) with negligible weight function, the inner product reads

$$\langle f_A|f_B\rangle = \int_{\mathbb{R}^\ltimes} \hat{A}\hat{B}\,\langle \Psi_A|\Psi_B\rangle\, d^n q = \int_{\mathbb{R}^\ltimes} \hat{A}\hat{B} P(A;B) d^n q, \tag{6.4}$$

[1] $W(x)$ receives this name because, as it says, it gives a weight to each value the integrand may take; there are no uniform contributions, but rather each possible value of the integrand within the domain contributes weighted by the value of W at that point.

[2] The Hilbert space is a particular case of the L^p spaces, the space of p-norm Lebesgue-integrable functionals; for Hilbert $p = 2$, and it has the property that the dual of L^2 is also L^2, which lets quantum mechanics arise naturally in this space.

where we see that the similarity between $|f_A\rangle$ and $|f_B\rangle$ can be viewed as the product of $\hat{A}$ with $\hat{B}$ times the probability density of having states A and B; that is, the similarity, or how *correlated* two variables are, can be seen as the integral of the product of the variables times their joint probability density.

With this analogy in mind we can understand correlation as a first measure of the similarity between one or more variables across space and time. If $x_1(\vec{r}, t_1), x_2(\vec{r}', t_2)$ are two random variables and $p_2(x_2, t'; x_1, t)$ is the joint probability density, then the correlation between x_1, x_2 is defined as

$$\Gamma(\vec{r}, \vec{r}'; t_1, t_2) = E\left\{x_1(\vec{r}, t_1)x_2(\vec{r}', t_2)\right\}, \tag{6.5}$$

that is, as the ensemble average of $x_1 \cdot x_2$.

6.3 The coherence function

The concept of the coherence function is, at the mathematical level, identical to that of the correlation function given in equation (6.5); both represent correlations. However, coherence is a term widely used in the study of electromagnetic radiation; let us see a little of why this is so.

In Figure 6.1 we observe a component of the electric field, represented by $E(\vec{r}, t)$, seen across several observations or realisations. Behaving as a random variable, in each realisation it has a different behaviour in time at a point $\vec{r}$; in general, here we have indexed ${}^{(i)}E(\vec{r}, t)$ with integers, but in general there could be an uncountable set of possible states for the electric field. If we look at Figure 6.1 there are two temporal points, t, t', at which we isolate parts of these random oscillations; we could ask how correlated these oscillations are across the realisations over the interval $t' - t$, or in other words, how *coherent* the oscillations are across the realisations.

We speak of coherence when we want to analyse the correlation of electromagnetic radiation between space–time points. If we have two sources emitting an electric field (for the moment understood as scalar), their coherence function is given by

$$\Gamma(\vec{r}, \vec{r}'; t, t') = E\left\{\mathcal{E}(\vec{r}, t)\mathcal{E}^*(\vec{r}', t')\right\}, \tag{6.6}$$

where we are computing the ensemble average of the components $\mathcal{E}$, which are complex numbers as already discussed in these notes. If we take $\vec{r} = \vec{r}'$ we obtain what is known as the *mutual coherence function*

$$\Gamma(\vec{r}'; t, t') = E\left\{\mathcal{E}(\vec{r}', t)\mathcal{E}^*(\vec{r}', t')\right\}. \tag{6.7}$$

We can go even further by including the concept of *statistical stationarity*, but one must be careful since there are several kinds. Formally, a stochastic process $\{X_t\}$ is considered stationary when its statistical properties are invariant under time translations. This condition manifests in different degrees:

1. **Strict (strong) stationarity**: the most demanding form of stationarity requires that all joint probability distributions be time-invariant:

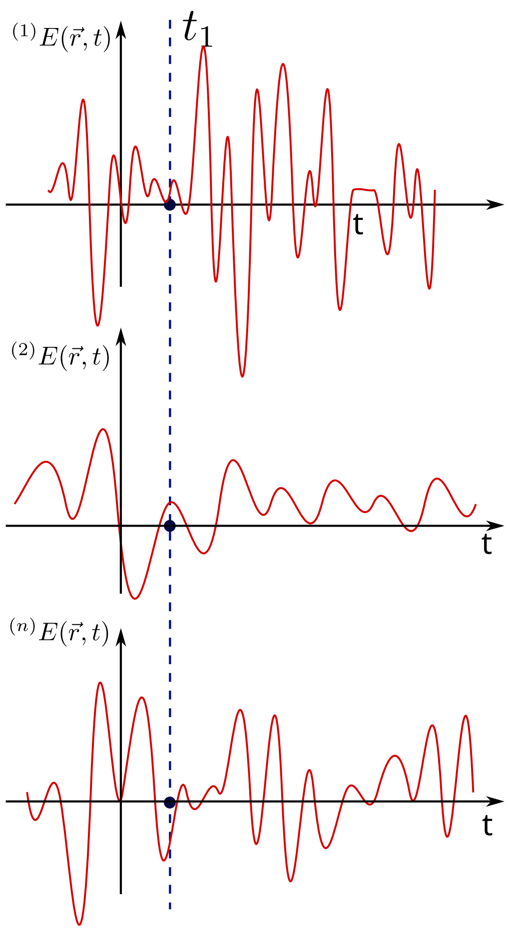


Figure 6.1: Representation of $E(\vec{r},t)$ as a random variable.

$$F_X(x_{t_1+\tau}, \ldots, x_{t_k+\tau}) = F_X(x_{t_1}, \ldots, x_{t_k}) \quad \forall \tau, k, t_i, \tag{6.8}$$

where F_X is the joint distribution function. This implies that:

- all moments (mean, variance, skewness, etc.) are constant;
- the marginal distributions are identical at any instant;
- it is a sufficient but not necessary condition for ARMA models.

2. **Weak (second-order) stationarity**: more practical in applications, it only requires invariance of the first two moments:

$$\begin{cases} \mathbb{E}[X_t] = \mu \quad (\text{constant}) \quad \forall t \\ \mathrm{Var}(X_t) = \sigma^2 \quad (\text{constant}) \quad \forall t \\ \mathrm{Cov}(X_t, X_{t+h}) = \gamma(h) \quad (\text{depends only on } h, \text{ not on } t) \end{cases} \tag{6.9}$$

This is the definition used in most econometric models (ARIMA, GARCH) and in the Wiener–Khinchin spectral analysis (which we shall use later).

3. **Trend stationarity (deterministic stationarity)**: a non-stationary process that can be made stationary by removing deterministic components:

$$X_t = \alpha + \beta t + \epsilon_t, \tag{6.10}$$

where ϵ_t is stationary. The non-stationarity here comes exclusively from the temporal trend βt.

We shall use a hypothesis of strong stationarity (although weak stationarity often suffices). In this spirit, if $\tau = t - t'$ we have

$$\Gamma(\vec{r}', t, t') = \Gamma(\vec{r}', t - t') = \Gamma(\vec{r}', \tau). \tag{6.11}$$

As we can see, the mutual coherence at a point $\vec{r}'$ under a stationarity hypothesis gives rise to an explicit dependence on the time difference τ in the ensemble average.

It is worth asking: how necessary is it to analyse the ensemble average; why not take the temporal average? Answering this is **very complicated**, since it touches on the problem of *ergodicity*. Ergodicity, in the sense of ergodic theory, requires that a dynamical system satisfy that the temporal averages coincide with the spatial averages for almost every initial condition. Its rigorous proof is highly non-trivial, to the point that the problems for which ergodicity has been proven with mathematical rigour are few.

From a phenomenological standpoint we shall assume, from now on, an *ergodic hypothesis* in which electromagnetic radiation has a stationary and ergodic character. Under these conditions the mutual coherence function coincides with the temporal average

$$\Gamma(\vec{r}'; t, t') = \int_{\mathbb{R}} \mathcal{E}(\vec{r}', t') \mathcal{E}^*(\vec{r}', t - t')\, dt'. \tag{6.12}$$

The mutual coherence function has several properties, some of which are:

a) $\Gamma(\vec{r}', 0) \geq 0$. This follows from the definition, since $\Gamma(\vec{r}', 0) = \langle |\mathcal{E}(\vec{r}')|^2 \rangle$, i.e. it corresponds to the intensity of the electric field and can only be zero when trivially the field is zero at every instant.

b) $\Gamma(\vec{r}', -\tau) = \Gamma^*(\vec{r}', \tau)$. This is known as the *Hermitian property* of the mutual coherence function, and follows from the fact that $\mathcal{E}(\vec{r}', t)$ is stationary, which allows arbitrary temporal translations from the origin. In this spirit,

$$\Gamma(\vec{r}', \tau) = \langle \mathcal{E}(\vec{r}')\mathcal{E}^*(\vec{r}', t+\tau) \rangle = \langle \mathcal{E}(\vec{r}', t-\tau)\mathcal{E}^*(\vec{r}') \rangle = \Gamma^*(\vec{r}', -\tau). \tag{6.13}$$

c) $|\Gamma(\vec{r}', \tau)| \leq \Gamma(\vec{r}', 0)$. This property follows from the *Cauchy–Schwarz inequality*

$$|\Gamma(\vec{r}', \tau)|^2 = |\,\langle \mathcal{E}(\vec{r}')\mathcal{E}^*(\vec{r}', t+\tau) \rangle\,|^2 \leq \langle \mathcal{E}(\vec{r}', t)\mathcal{E}^*(\vec{r}', t) \rangle \, \langle \mathcal{E}(\vec{r}', t+\tau)\mathcal{E}^*(\vec{r}', t+\tau) \rangle = \Gamma^2(\vec{r}', 0), \tag{6.14}$$

which implies that $|\Gamma(\vec{r}', \tau)|$ can never exceed its initial value $\Gamma(\vec{r}', 0)$. (This property is, let us recall, exclusive to the mutual coherence function; when we consider the coherence between two sources we will see that it is not satisfied — on the contrary, it will increase.)

d) $\Gamma(\vec{r}', \tau)$ is non-negative. We can see this from the definition through the following, rather obvious, inequality

$$\left\langle \left| \int_{T_1}^{T_2} f(t)\mathcal{E}(t)\, dt \right|^2 \right\rangle \geq 0, \tag{6.15}$$

where $f(t)$ is an arbitrary function and T_1, T_2 are arbitrary times. In this way we obtain the inequality

$$\int_{T_1}^{T_2} \int_{T_1}^{T_2} f^*(\vec{r}', t) f(\vec{r}', t') \Gamma(\vec{r}', t'-t)\, dt\, dt' \geq 0. \tag{6.16}$$

e) If the electric field is random, stationary and differentiable, then we can find a velocity $\xi(\vec{r}', t) = \dot{\mathcal{E}}(\vec{r}', t)$, which is stationary, random and of zero mean value, and whose mutual coherence function is

$$\Gamma_\xi(\vec{r}', \tau) = -\ddot{\Gamma}_{\mathcal{E}}(\vec{r}', \tau). \tag{6.17}$$

f) $\Gamma(\vec{r}', \tau)$ is continuous. This holds even when there are jump discontinuities.

Finally, it is worth mentioning that there is a quantity known as the complex degree of coherence, given by

$$\gamma(\vec{r}', \tau) = \frac{\Gamma(\vec{r}', \tau)}{\Gamma(\vec{r}', 0)}, \tag{6.18}$$

and from the Cauchy–Schwarz inequality we have $0 \leq |\gamma(\vec{r}', \tau)| \leq 1$.

6.3.1 Manifestations of coherence: the Wiener–Khinchin theorem

The mutual coherence function and the power spectral density are two sides of the same coin in the study of light and wave signals. Mutual coherence describes how the fluctuations of a luminous field are correlated in space and time, acting as a map that reveals to what extent the variations at one point of space are synchronised with those at another, even with a certain temporal delay. The power spectral density, in turn, decomposes the energy of the signal into its frequency components, showing which tones or colours — in the case of light — dominate its behaviour. The intimate relation between the two concepts lies in the fact that, through mathematical transformations, it is possible to convert the detailed information on the space–time correlations captured by mutual coherence into a precise description of how the energy is distributed in the frequency domain, and vice versa. This link allows researchers to jump between complementary perspectives: analysing interference patterns in an interferometer or identifying spectral signatures in a material, all from the same fundamental principles.

We are going to find one of the most important theorems in coherence theory, the *Wiener–Khinchin theorem.* Note that stationary random signals do not have a Fourier transform; hence we may define the following truncated random signals of the electric field

$$
{}^{\omega}\mathcal{E}_T(\vec{r},t) = \begin{cases} {}^{\omega}\mathcal{E}(\vec{r},t) & -\frac{T}{2} \leq t \leq \frac{T}{2} \\ 0 & \text{otherwise} \end{cases} \tag{6.19}
$$

and we define the spectral power for a realisation ω as the squared magnitude of the Fourier transforms of the electric fields (6.19), that is,

$$
{}^{\omega}S_T(\vec{r},\nu) = \left| \int_{\mathbb{R}}^{\omega} \mathcal{E}_T(\vec{r},t) e^{i2\pi\nu t} dt \right|^2 = \left| \int_{-T/2}^{T/2} {}^{\omega}\mathcal{E}(\vec{r},t) e^{i2\pi\nu t} dt \right|^2 , \tag{6.20}
$$

and the *power spectral density of the event* ω we compute as

$$
\lim_{T\to\infty} \frac{{}^{\omega}S_T(\vec{r},\nu)}{T} = {}^{\omega}S(\vec{r},\nu). \tag{6.21}
$$

Since we already possess a tool to study, in each event or realisation, the power spectral density, then, through its ensemble average, we can find the total density, that is,

$$
S(\vec{r},\nu) = E\left\{ {}^{\omega}S(\vec{r},\nu) \right\}, \tag{6.22}
$$

introducing equations (6.20) and (6.21) we then have

$$S(\vec{r},\nu) = E\left\{\lim_{T\to\infty}\left|\frac{1}{T}\int_{-T/2}^{T/2} {}^{\omega}\mathcal{E}(\vec{r},t)e^{i2\pi\nu t}dt\right|^2\right\}, \tag{6.23}$$

$$S(\vec{r},\nu) = \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}\int_{-T/2}^{T/2} E\left\{\mathcal{E}(\vec{r},t)\mathcal{E}^*(\vec{r}',t')\right\}e^{i2\pi\nu(t-t')}dt\,dt', \tag{6.24}$$

$$S(\vec{r},\nu) = \int_{\mathbb{R}}\Gamma(\vec{r},\vec{r}';\tau)e^{i2\pi\nu\tau}d\tau, \tag{6.25}$$

$$\boxed{S(\vec{r},\nu) = \mathcal{F}\left\{\Gamma(\vec{r},\vec{r}',\tau)\right\}}. \tag{6.26}$$

Result (6.26) reads: the Fourier transform of the coherence function corresponds to the power spectral density, and is known as the Wiener–Khinchin theorem. Its use is fundamental, since it tells us that by methods such as spectrometry we can find the coherence function of any phenomenon; we can even go further, since this result really stems from something more general — the correlation function — meaning that every stochastic phenomenon has an associated spectrum, and from the measurements or estimates of that spectrum we can compute the correlations in the system.

If we make a simple observation, the optical range of radiation — the visible — is a band of very low frequencies relative to the whole spectrum; the visible runs roughly from 400 nm to 700 nm, and we may say that all of optics is *narrowband*. This obeys the inequality

$$\frac{\triangle\nu}{\nu} \ll 1, \tag{6.27}$$

and, in this narrowband regime, the coherence function takes the form

$$\Gamma(\vec{r};\tau) = \tilde{\Gamma}(\vec{r})e^{i2\pi\tilde{\nu}\tau}, \tag{6.28}$$

which we can represent graphically as shown in Figure 6.2, where one can see the behaviour of the coherence function in the narrowband regime.[3] There one sees an envelope (the observable) enveloping all the fundamental frequencies.

6.3.2 The wave equation of coherence

Let us step a little out of the comfort zone and understand how correlations propagate through space and time. Let $\mathcal{E}^{(r)}(\vec{r},t)$ be some **real** component of the electric field, viewed as a random signal. As is well known from classical electromagnetic theory, it obeys Maxwell's equations and therefore has an associated wave equation, namely

[3] One usually finds in the literature a term similar but not equal to that of narrowband, known as the quasi-monochromaticity condition, written as $\frac{\triangle\lambda}{\lambda} \ll 1$; but this bounds the wavelength so that it is as monochromatic as possible, while the narrowband condition admits several frequencies, provided they are narrow relative to the whole frequency spectrum.

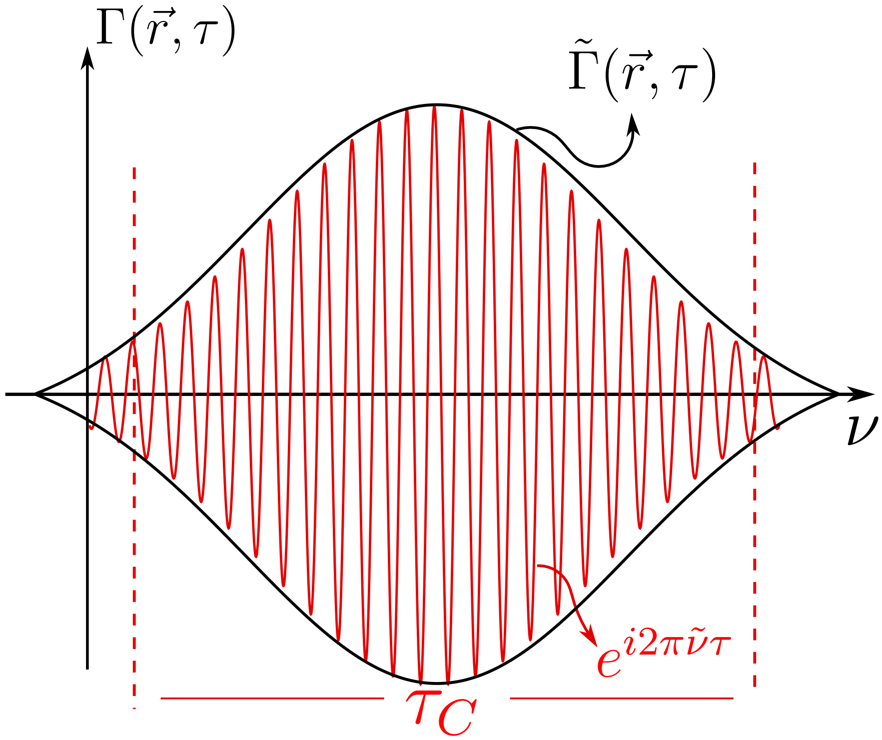


Figure 6.2: Graphical representation of the narrowband coherence function. In black, the envelope $\tilde{\Gamma}(\vec{r},\tau)$ is shown, and in red the oscillations $e^{i2\pi\tilde{\nu}\tau}$. These oscillations are confined within something known as the coherence time τ_C, which we shall see later.

$$\nabla^2 \mathcal{E}^{(r)}(\vec{r}, t) = \frac{1}{C^2} \frac{\partial^2 \mathcal{E}^{(r)}(\vec{r}, t)}{\partial t^2}, \tag{6.29}$$

where C is the speed of light in vacuum. We also have, by Fourier analysis,

$$\mathcal{E}(\vec{r}, t) = \int_0^\infty \tilde{\mathcal{E}}(\vec{r}, \nu) e^{-i2\pi\nu t} d\nu. \tag{6.30}$$

And if we apply the d'Alembertian operator $\Box = \nabla^2 - \frac{1}{C^2}\frac{\partial^2}{\partial t^2}$ to equation (6.30), we find that $\tilde{\mathcal{E}}(\vec{r}, \nu)$ has the form of a Helmholtz-type equation, that is,

$$\nabla^2 \mathcal{E}(\vec{r}, t) - \frac{1}{C^2} \frac{\partial^2 \mathcal{E}(\vec{r}, t)}{\partial t^2} = \int_0^\infty \left[\nabla^2 \tilde{\mathcal{E}}(\vec{r}, \nu) + k^2 \tilde{\mathcal{E}}(\vec{r}, \nu) \right] e^{-i2\pi\nu t} d\nu, \tag{6.31}$$

which leads us to the fact that the electric field, as an analytic and complex signal, also obeys a wave-type equation

$$\nabla^2 \mathcal{E}(\vec{r}, t) = \frac{1}{C^2} \frac{\partial^2 \mathcal{E}(\vec{r}, t)}{\partial t^2}. \tag{6.32}$$

Taking the complex conjugate of the previous equation we have

$$\nabla^2 \mathcal{E}^*(\vec{r}, t) = \frac{1}{C^2} \frac{\partial^2 \mathcal{E}^*(\vec{r}, t)}{\partial t^2}. \tag{6.33}$$

If we multiply both sides of equation (6.33) by $\mathcal{E}(\vec{r}', t')$, and since we have taken the d'Alembertian with respect to the unprimed variables, using some sign rules we have

$$\nabla^2 \left[\mathcal{E}^*(\vec{r}, t) \mathcal{E}(\vec{r}', t') \right] = \frac{1}{C^2} \frac{\partial^2}{\partial t^2} \left[\mathcal{E}^*(\vec{r}, t) \mathcal{E}(\vec{r}', t') \right]. \tag{6.34}$$

Taking the ensemble average of equation (6.34) over all realisations of the field, this wave equation becomes

$$\nabla^2 \Gamma(\vec{r}, \vec{r}'; t, t') = \frac{1}{C^2} \frac{\partial^2}{\partial t^2} \Gamma(\vec{r}, \vec{r}'; t, t'), \tag{6.35}$$

and, similarly, using $\Box' = \nabla'^2 - \frac{1}{C^2}\frac{\partial^2}{\partial t'^2}$ we have another wave equation given by

$$\nabla'^2 \Gamma(\vec{r}, \vec{r}'; t, t') = \frac{1}{C^2} \frac{\partial^2}{\partial t'^2} \Gamma(\vec{r}, \vec{r}'; t, t'). \tag{6.36}$$

Equations (6.35) and (6.36) tell us directly that the coherence functions between two space–time points evolve in time following a wave-type equation, separately, with respect to each pair of space–time coordinates — a result that is, without doubt, remarkable. We can take this result even further and study this propagation in the next section.

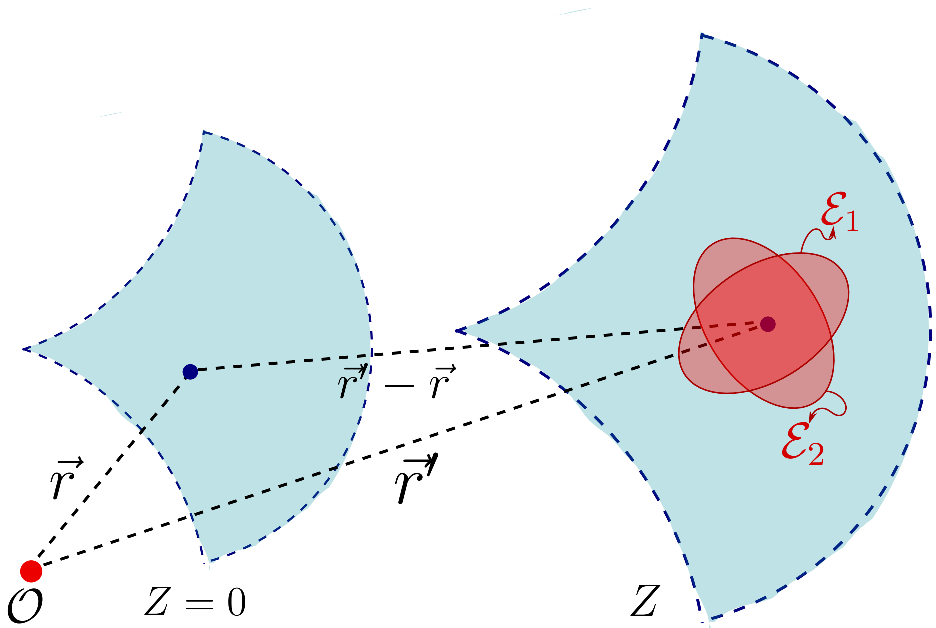


Figure 6.3: Propagation of the electric field from a plane $Z = 0$ to a plane Z, where each point of the wavefront carries its mean polarisation ellipse.

6.3.3 Propagation of coherence and the direction of time: the van Cittert–Zernike theorem

As we know, the electric field obeys a wave-type equation and the orientation of the fields is given by the polarisation of the field; in turn, as we saw in the diffraction chapter (see Section 5.2), at the classical level light fundamentally propagates as spherical waves, as illustrated in Figure 6.3, and at each point of the wavefront the components of the electric field oscillate randomly, but within the mean polarisation ellipse of their representative state. Mathematically we can represent this component as

$$\mathcal{E}_z(\vec{r}_1, \nu) = \frac{i}{\lambda} \int_\Sigma \mathcal{E}_0(\vec{r}, \nu) \frac{e^{ik\|\vec{r}_1 - \vec{r}\|}}{\|\vec{r}_1 - \vec{r}\|} \cos(\hat{z}; \vec{r}_1 - \vec{r})\, d\sigma. \tag{6.37}$$

Using equation (6.30) we have

$$\mathcal{E}_z(\vec{r}_1, t) = \frac{1}{i} \int_{\mathbb{R}^+} \frac{\nu}{V} \int_\Sigma \mathcal{E}_0(\vec{r}, \nu) \frac{\cos(\hat{z}; \vec{r}_1 - \vec{r})}{\|\vec{r}_1 - \vec{r}\|} e^{i2\pi\nu\left(t + \frac{\|\vec{r}_1 - \vec{r}\|}{V}\right)}\, d\sigma\, d\nu, \tag{6.38}$$

where V is the propagation velocity of the envelope. Rewriting a little more, we have

$$\mathcal{E}_z(\vec{r}_1, t) = \frac{1}{i} \int_\Sigma \frac{\cos(\hat{z}; \vec{r}_1 - \vec{r})}{\|\vec{r}_1 - \vec{r}\|} \int_{\mathbb{R}^+} \frac{\nu}{V} \mathcal{E}_0(\vec{r}, \nu) e^{i2\pi\nu\left(t + \frac{\|\vec{r}_1 - \vec{r}\|}{V}\right)}\, d\nu\, d\sigma, \tag{6.39}$$

and, taking the narrowband approximation, we have

$$\mathcal{E}_z(\vec{r}_1, t) = \frac{1}{i\tilde{\lambda}} \int_{\Sigma} \mathcal{E}_0 \left(\vec{r}, t + \frac{\|\vec{r}_1 - \vec{r}\|}{V} \right) \frac{\cos(\hat{z}; \vec{r}_1 - \vec{r})}{\|\vec{r}_1 - \vec{r}\|} d\sigma. \tag{6.40}$$

We can visualise graphically how we intend to propagate in Figure 6.4, where we observe that we are starting from the coherence function between two points $\vec{r}, \vec{r}'$ at instants t, t', respectively, and carrying it to a coherence function at the points $\vec{r}_1, \vec{r}_2$ at two instants t_1, t_2, respectively.

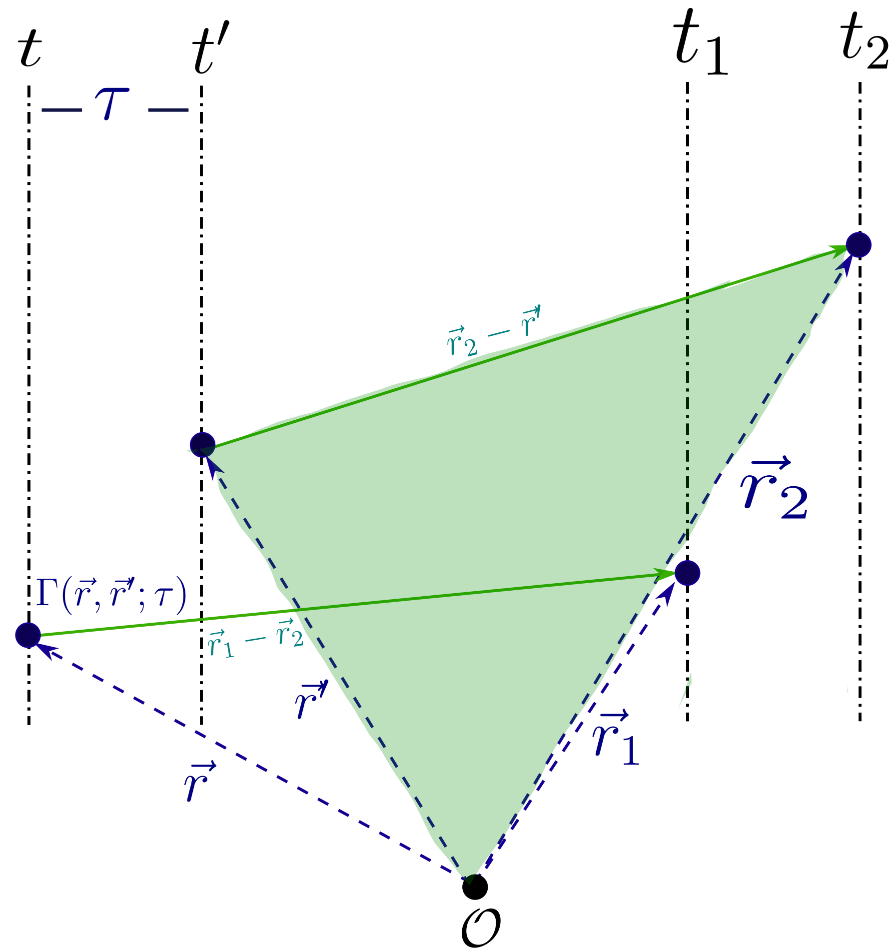


Figure 6.4: Propagation of the coherence function between two points $\vec{r}, \vec{r}'$ on the initial plane to the coherence between two points $\vec{r}_1, \vec{r}_2$ on the observation plane.

Starting from equation (6.40) we can compute the coherence function in its propagation as

$$\Gamma_z(\vec{r}_1, \vec{r}_2; \tau) = E\left\{\mathcal{E}_z(\vec{r}_1, t)\mathcal{E}_z^*(\vec{r}_2, t - \tau)\right\}, \tag{6.41}$$

$$\Gamma_z(\vec{r}_1, \vec{r}_2; \tau) = E\left\{\frac{1}{i\tilde{\lambda}}\int_\Sigma \mathcal{E}_0\left(\vec{r}, t + \frac{\|\vec{r}_1 - \vec{r}\|}{V}\right)\frac{\cos(\hat{z}; \vec{r}_1 - \vec{r})}{\|\vec{r}_1 - \vec{r}\|}d\sigma \right. \tag{6.42}$$

$$\left. \cdot\frac{1}{-i\tilde{\lambda}}\int_{\Sigma'} \mathcal{E}_0^*\left(\vec{r}', t - \tau + \frac{\|\vec{r}_2 - \vec{r}'\|}{V}\right)\frac{\cos(\hat{z}; \vec{r}_2 - \vec{r}')}{\|\vec{r}_2 - \vec{r}'\|}d\sigma'\right\}, \tag{6.43}$$

$$\Gamma_z(\vec{r}_1, \vec{r}_2; \tau) = \frac{1}{\tilde{\lambda}^2}\int_\Sigma\int_{\Sigma'} E\left\{\mathcal{E}_0\left(\vec{r}, t + \frac{\|\vec{r}_1 - \vec{r}\|}{V}\right)\mathcal{E}_0^*\left(\vec{r}', t - \tau + \frac{\|\vec{r}_2 - \vec{r}'\|}{V}\right)\right\}\frac{\cos(\hat{z}; \vec{r}_1 - \vec{r})\cos(\hat{z}; \vec{r}_2 - \vec{r}')}{\|\vec{r}_2 - \vec{r}'\|\cdot\|\vec{r}_1 - \vec{r}\|}d\sigma\, d\sigma', \tag{6.44}$$

$$\Gamma_z(\vec{r}_1, \vec{r}_2; \tau) = \frac{1}{\tilde{\lambda}^2}\int_\Sigma\int_{\Sigma'} \Gamma_0\left(\vec{r}, \vec{r}'; \tau + \frac{\|\vec{r}_1 - \vec{r}\| + \|\vec{r}_2 - \vec{r}'\|}{V}\right)\frac{\cos(\hat{z}; \vec{r}_1 - \vec{r})\cos(\hat{z}; \vec{r}_2 - \vec{r}')}{\|\vec{r}_2 - \vec{r}'\|\cdot\|\vec{r}_1 - \vec{r}\|}d\sigma\, d\sigma' \tag{6.45}$$

The propagation of the random oscillations across n realisations can be visualised in Figure 6.5. We are going to impose the following condition on the coherence function

$$\Gamma(\vec{r}, \vec{r}'; \tau) = \Gamma(\vec{r}'; \tau)\delta(\vec{r} - \vec{r}'), \tag{6.46}$$

which, in a few words, means that, spatially speaking, the distribution representing the coherence function is point-localised and weighted by the mutual coherence at the point $\vec{r}'$. Imposing this on equation (6.45) we have

$$\Gamma_z(\vec{r}_1, \vec{r}_2; \tau) = \frac{1}{\tilde{\lambda}^2}\int_\Sigma \Gamma_0\left(\vec{r}; \tau + \frac{\|\vec{r} - \vec{r}_1\| - \|\vec{r} - \vec{r}_2\|}{V}\right)\frac{\cos(\hat{z}; \vec{r}_1 - \vec{r})\cos(\hat{z}; \vec{r}_2 - \vec{r})}{\|\vec{r}_2 - \vec{r}\|\cdot\|\vec{r}_1 - \vec{r}\|}d\sigma. \tag{6.47}$$

In nature we will not always have narrowband sources; however, we can resort to another argument, namely that most instruments do not have an infinite or broad range but a short one — or, in other words, measurement instruments measure a narrow part of the spectrum. Therefore, including the narrowband condition we have

$$\Gamma_z(\vec{r}_1, \vec{r}_2)e^{i2\pi\tilde{\nu}\tau} = \frac{1}{\tilde{\lambda}^2}\int_\Sigma \Gamma_0(\vec{r})e^{i2\pi\tilde{\nu}\left(\tau + \frac{\|\vec{r} - \vec{r}_1\| - \|\vec{r} - \vec{r}_2\|}{V}\right)}\frac{\cos(\hat{z}; \vec{r}_1 - \vec{r})\cos(\hat{z}; \vec{r}_2 - \vec{r})}{\|\vec{r}_2 - \vec{r}\|\cdot\|\vec{r}_1 - \vec{r}\|}d\sigma. \tag{6.48}$$

Given the coherence time defined by $\tau_C = \int_{\mathbb{R}^+}|\gamma(\vec{r}, \tau)|^2 d\tau$, and if we impose a condition of very small times, smaller than a coherence time $\tau \ll \tau_C$, we have

$$\Gamma_z(\vec{r}_1, \vec{r}_2) = \frac{1}{\tilde{\lambda}^2}\int_\Sigma I_0(\vec{r})e^{i2\pi\left(\frac{\|\vec{r} - \vec{r}_1\| - \|\vec{r} - \vec{r}_2\|}{V/\tilde{\nu}}\right)}\frac{\cos(\hat{z}; \vec{r}_1 - \vec{r})\cos(\hat{z}; \vec{r}_2 - \vec{r})}{\|\vec{r}_2 - \vec{r}\|\cdot\|\vec{r}_1 - \vec{r}\|}d\sigma. \tag{6.49}$$

Taking the far-field approximation we can simplify the problem as follows

$$\Gamma_z(\vec{r}_1, \vec{r}_2) = \frac{1}{\tilde{\lambda}^2 z^2}\int_\Sigma I_0(\vec{r})e^{i2\pi\left(\frac{\|\vec{r} - \vec{r}_1\| - \|\vec{r} - \vec{r}_2\|}{V/\tilde{\nu}}\right)}d\sigma. \tag{6.50}$$

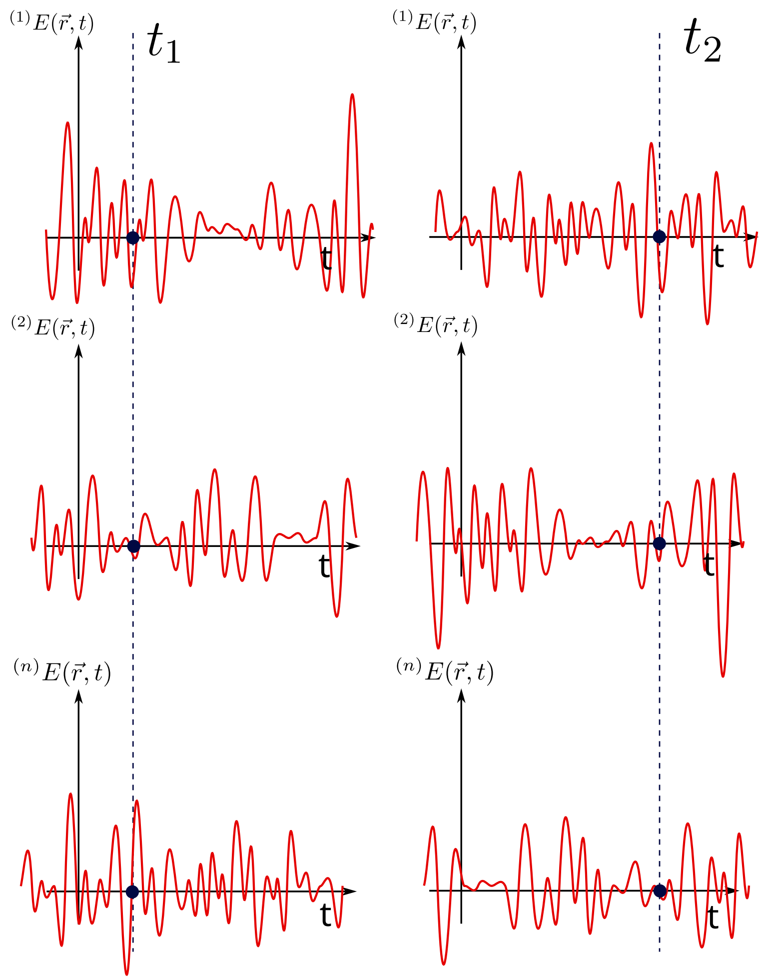


Figure 6.5: Evolution of the realisations upon propagating the electric field.

We may now make a Taylor-series expansion, the same one performed in the diffraction chapter. In this spirit the following terms become

$$\|\vec{r}-\vec{r}_1\| = \left(z^2+(x-x_1)^2+(y-y_1)^2\right)^{1/2} = z\left(1+\frac{(x-x_1)^2+(y-y_1)^2}{z^2}\right)^{1/2} \approx z+\frac{(x-x_1)^2+(y-y_1)^2}{2z}, \tag{6.51}$$

$$\|\vec{r}-\vec{r}_2\| \approx z+\frac{(x-x_2)^2+(y-y_2)^2}{2z}. \tag{6.52}$$

Introducing this into integral (6.50) we obtain

$$\Gamma_z(\vec{r}_1,\vec{r}_2) = \frac{1}{\tilde{\lambda}^2 z^2}\int_{\mathbb{R}^2} I_0(x,y)e^{-\frac{i2\pi}{\tilde{\lambda}z}(xx_1+yy_1)}e^{\frac{i2\pi}{\tilde{\lambda}z}(xx_2+yy_2)}e^{-\frac{i\pi}{\tilde{\lambda}z}(x_1^2+y_1^2)}dx\,dy. \tag{6.53}$$

Let us define the vectors $\vec{S}_1=(x_1,y_1)$, $\vec{S}_2=(x_2,y_2)$ and $\vec{S}=(x,y)$. Then

$$\Gamma_z(\vec{S}_1,\vec{S}_2) = \frac{e^{i\frac{\pi}{\tilde{\lambda}z}(\|\vec{S}_1\|^2-\|\vec{S}_2\|^2)}}{\tilde{\lambda}^2 z^2}\int_{\mathbb{R}^2} I_0(\vec{S})e^{-i\frac{2\pi}{\tilde{\lambda}z}\vec{S}\cdot(\vec{S}_1-\vec{S}_2)}dS. \tag{6.54}$$

If we set $\vec{S}_2=\vec{0}$ we arrive at the Fourier integral

$$\boxed{\Gamma_z(\vec{S}_1) = \frac{e^{\frac{i\pi}{\tilde{\lambda}z}\|\vec{S}_1\|^2}}{\tilde{\lambda}^2 z^2}\int_{\mathbb{R}^2}\Gamma_0(\vec{S})e^{-\frac{i2\pi}{\tilde{\lambda}z}\vec{S}\cdot\vec{S}_1}dS}, \tag{6.55}$$

which, basically, means that the correlations, under all the approximations we have made, propagate by means of a Fourier integral; or, what is the same, the van Cittert–Zernike theorem tells us that coherence diffracts — it propagates naturally as a diffraction integral (see Figure 6.6).

How can we tell in which direction time flows? This may sound like a question rather out of context for these notes, but if we could travel back to the golden age of Boltzmann, we would find a heated debate of the time on how statistical physics gives a direction to time, an *arrow*. Paraphrasing Boltzmann, it is said that *time advances in the direction in which correlations increase*. If we look at the van Cittert–Zernike theorem, we can start from two completely uncorrelated sources but, as they propagate, they gradually become correlated; time grows and so does coherence, until a point is reached at which the radiation of those sources is coherent. This is, without doubt, one of the strongest and most beautiful results we can find in statistical physics.

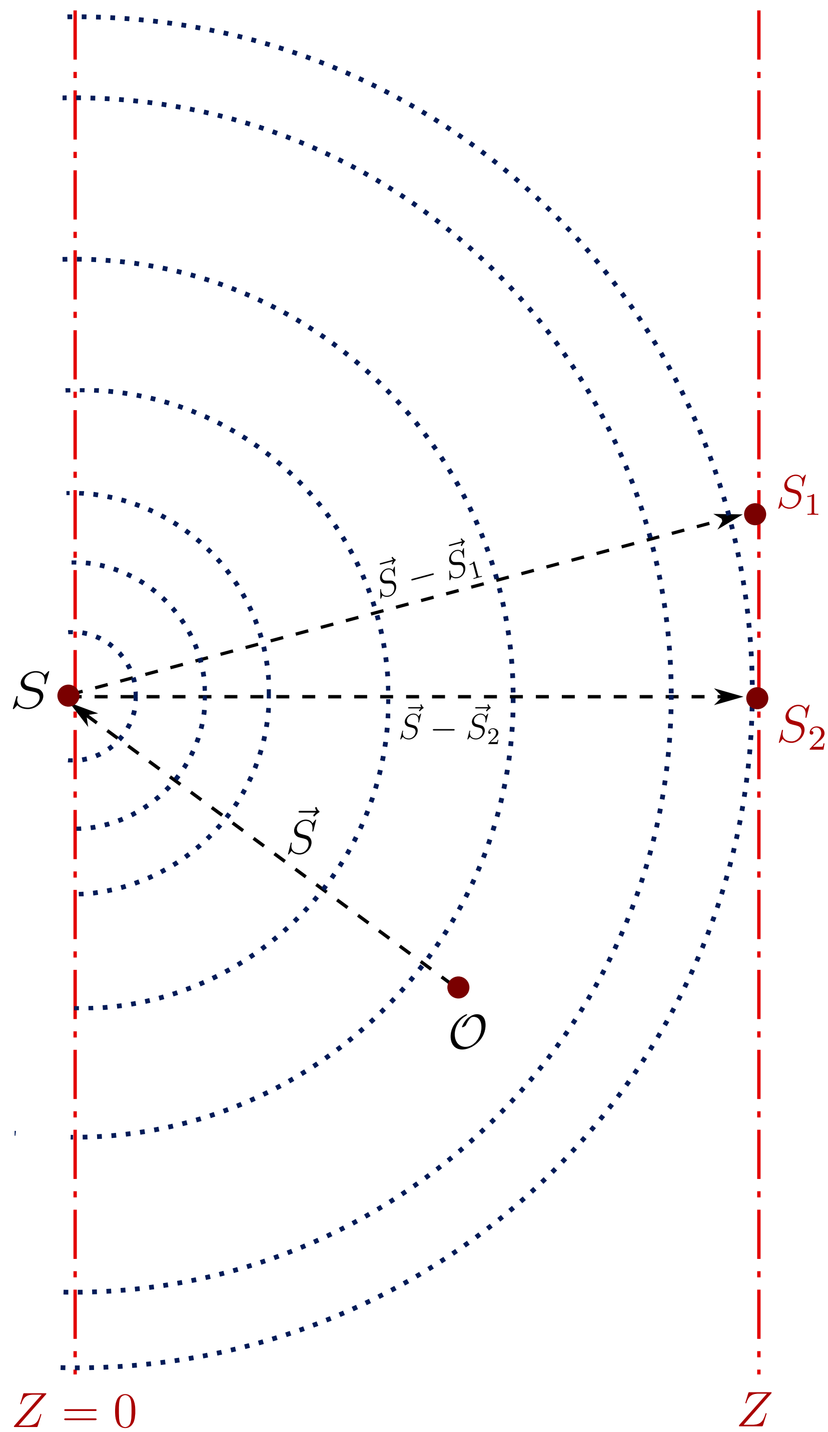


Figure 6.6: The mutual coherence propagated to the far field is the Fourier transform of the intensity distribution $\Gamma_0(\vec{S})$ of the source.

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